Alkyl & Aryl Halides

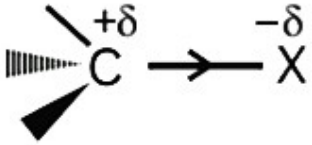
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Introduction about Halogen Derivatives

If we replace one or more than one hydrogen from hydrocarbon and attach halogen atom/s will named as halogen derivatives.
Generally named with respective their functional group like alkane , alkene and arenes as haloalkane , haloalkenes and haloalkynes.

Structure of Alkyl Halides:



More electronegative than carbon atom ,therefore carbon halogen bond will act as dipole . Carbon atom is slightly positive and halogen atom carry partial negative charge.

Body of Halogen Derivatives:

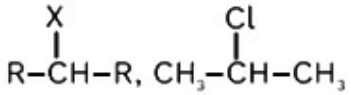
(i) Primary (1 Degree) halogen derivative:

If the halogen carrying carbon is attached to single carbon atom or with zero carbon atom directy

Example : CH3 – X

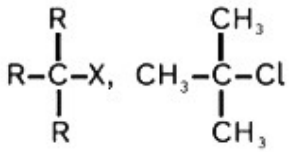
(ii) Sec (2 degree) Halogen Derivative::

If two carbon atoms are directly attached to the halogen bearing carbon.

Example : 

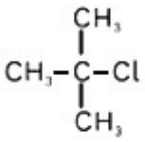
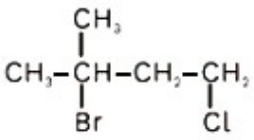
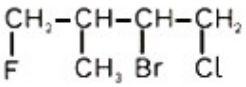
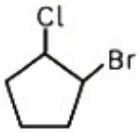
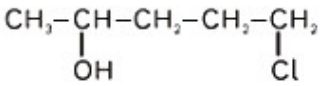
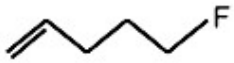
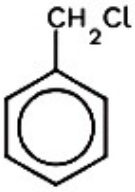
(iii) Tertiary (3 degree) Halogen Derivatives :

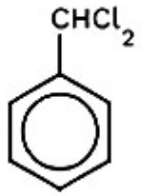
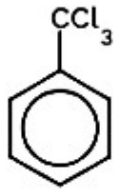
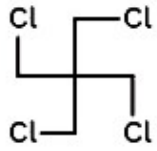
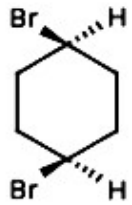
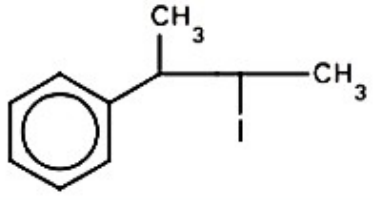
Three other carbon atom attached to the halogen bearing carbon atom.

Example : 

Haloalkanes can be classified into following three categories.
(i) Monohaloalkanes
(ii) Dihaloalkanes
(iii) Polyhaloalkanes

IUPAC Nomenclature of Alkyl Halides

S.N.	Compound	IUPAC Name
1.		2 – Chloro-2-methylpropane
2.		3-Bromo-1-chlorobutane
3.		2-Bromo-1-chloro-4-fluoro-3-methylbutane
4.		2-Bromo-1-chloro-3-iodocyclopentane
5.		5-chloropentan-2-ol
6.		5-Fluoropent-1-ene
7.		Chlorophenylmethane

S.N.	Compound	IUPAC Name
8.		Dichlorophenylmethane
9.		Trichlorophenylmethane
10.		2,2-Bis(chloromethyl)-1,3-dichloropropane
11.		Cis-1,4-dibromocyclohexane
12.		2-iodo-3-phenylbutane

Bonding in Alkyl Halide:

Carbon halogen bond lengths

Bond	Bond length (Å)
$CH_3 - F$	1.39
$CH_3 - Cl$	1.78
$CH_3 - Br$	1.93
$CH_3 - I$	2.14

Physical Properties of Alkyl Halide:

Halo derivatives of alkane are colorless bromides and iodides mature colour when exposed to light. Some volatile halogen compounds have sweet smell.

1. Theory of dipole in Halogen Derivatives:

$\mu = 4.8 \times \delta \times d$

Here δ is the charge and d is the distance between to atoms.

These two factors charge and distance clash with each other, with the bigger halogens having longer bond but weaker electronegativity. The general result is that the bond dipole moment grow in the following manner

C – I < C – Br < C – F < C – Cl (carbon and respective halogen atom)

μ : 1.29 D 1.48 D 1.51 D 1.56 D

The electronegativities of the halogen increase in the order:

I < Br < Cl < F

X	CH_3X	CH_2X_2	CHX_3	CX_4
F	1.82 D	1.97 D	1. 65 D	0
Cl	1.94 D	1.60 D	1.03 D	0
Br	1.79 D	1.45 D	1.02 D	0
I	1.64 D	1.11 D		0

Table : Molecular dipole moments of methyl halides

Melting and Boiling Points Respective to Halogen Derivatives

As we know due to larger polarity as well as advanced molecular weight as related to the root hydrocarbon and stronger intermolecular forces of attraction (dipole-dipole and van der Waals) in the haloalkanes ,boiling points of chlorides, bromides and iodides are considerably higher than those of the hydrocarbons of similar molecular weight.

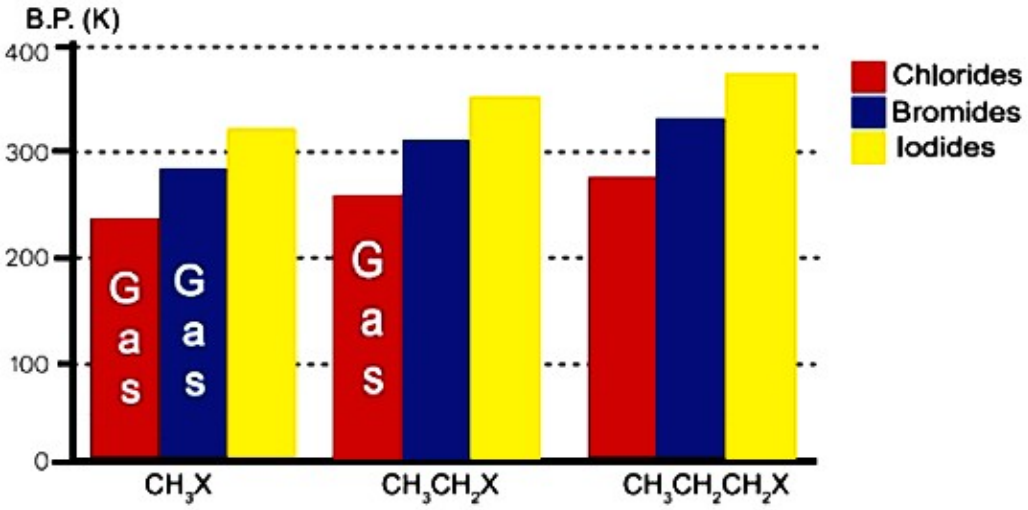
When size of molecules get bigger, attractions get stronger and having more number of electrons.

The boiling points values of alkyl halides decline in the direction:

R-I > R-Br > R-Cl > R-F.

This is because with the growth in size and mass of halogen atom, the magnitude of van der Waal forces increases.

Comparison of boiling points of some alkyl halides



Comparison of boiling points of some alkyl halides

Boiling points of some alkyl halide in °C (1 atm)

Formula	X = F	X = Cl	X = Br	X = I
$CH_3 - X$	– 78	– 24	3	42
$CH_3 - CH_2X$	– 32	12	38	72
$CH_3 - CH_2 - CH_2X$	– 3	47	71	103

$CH_3 - (CH_2)_3 - CH_2X$	65	108	129	157

Fluorine is exclusive among the halogens is that increasing the number of fluorines does not lead to higher and higher boiling point.
The boiling points value of the chlorinated derivatives of CH_4 increase with the number of chlorine atoms as of an increase in the induced dipole/dipole attractive forces

Compound	CH_3Cl	CH_2Cl_2	$CHCl_3$	CCl_4
B.P.	24°C	40°C	61°C	77°C

	$CH_3 - CH_2F$	$CH_3 - CHF_2$	$CH_3 - CF_3$	$CF_3 - CF_3$
B.P.	–32°C	–25°C	–47°C	–78°C

Density :
Alkyl fluorides and chlorides are less dense and alkyl bromides and iodides denser, than water.

	$CH_3–(CH_2)_6–CH_2F$	$CH_3–(CH_2)_6–CH_2Cl$	$CH_3–(CH_2)_6–CH_2Br$	$CH_3–(CH_2)_6–CH_2I$
Density (20°C)	0.80 g/mL	0.89 g/mL	1.12 g/mL	1.34 g/mL

Reason

Alkyl halides are insoluble in nature (in water) ,as according to observations mixture of an alkyl halide and water splits into two films.

If alkyl derivatives are made from fluorine or chlorine , upper layer contain halo derivatives and water is the lower. Will observe reversed situation when the alkyl halide is a bromide or an iodide. In these cases the alkyl halide is in the lower layer.

Solubility Concept in Alkyl Halo Derivatives

The haloalkanes are very little soluble in water ,to dissolve haloalkane in water some kind of energy is obligatory to overcome the pulls between the haloalkane molecules and disrupt the hydrogen bonds formed among H_2O molecules.

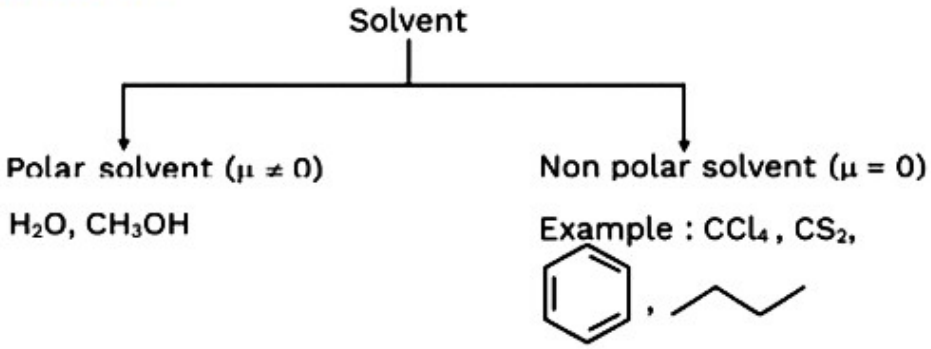
Not as much of energy is released when new attraction forces are set up among the alkylhalide and the H_2O molecules as these are not as sturdy as the original hydrogen bonds in H_2O. So here we can say the solubility of alkylhalides in water is little low.

1. **Solvent**
2. **Effect of Solvent at Nucleophilicity**
3. **Details about Reaction Intermediates**
4. **Leaving group**

Solvent :

Substantial medium that dissolves another substance to form a solution named as solvent

Classification of Solvent :



Solubility Principle:

"Like dissolve like"

Polar compound are dissolved in polar solvent.

Non polar compound are dissolved in non-polar solvent.

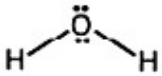
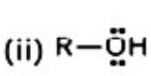
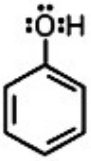
Polar Solvents: (μ ≠ 0)

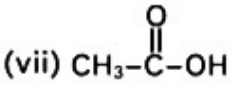
Solvents having large dipole movements. They contain bonds between atoms with very different electronegativities.

(a) Polar Protic Solvent (PPS) :

Solvent which is polar in nature and have acidic hydrogen. H-bonding is possible in polar protic solvent.

Example :

(i)  (ii)  (iii)  (iv) NH_3

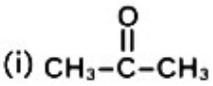
(V) (vi) R–NH–R (vii)  (viii) HF

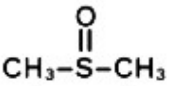
(ix) R–SH (x) H_3PO_4 (xi) H_2SO_4

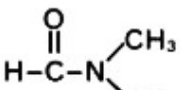
(b) Polar Aprotic Solvent (PAS) :

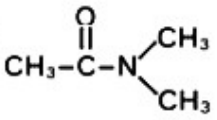
Solvent which is polar in nature but do not have acidic hydrogen. H-bonding is not possible in polar aprotic solvent.

Example:

(i) 
(Acetone)

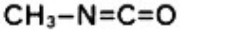
(ii) 
(DMSO)

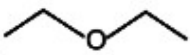
(iii) 
(DMF)

(iv) 
(DMA)

(v) 
(THF)

(vi) CH_3CN

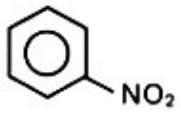
(vii) 
(Methylisocynate)

(viii) 

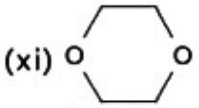
(ix) 
(Furan)

Points to remember

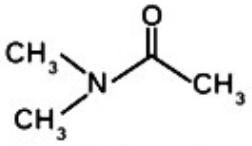
Polar protic solvent can dissolve cation as well as anion part of reagent.

(x) 

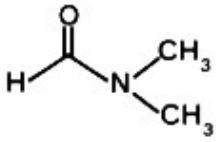
(Nitrobenzene)

(xi) 

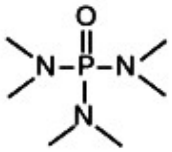
(1,4-dioxane)

(xii) 

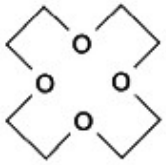
(Dimethyl acetanide)

(xiii) 

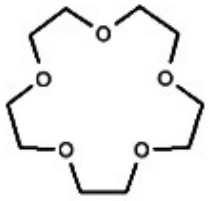
(Dimethyl formamide)

(xiv) 

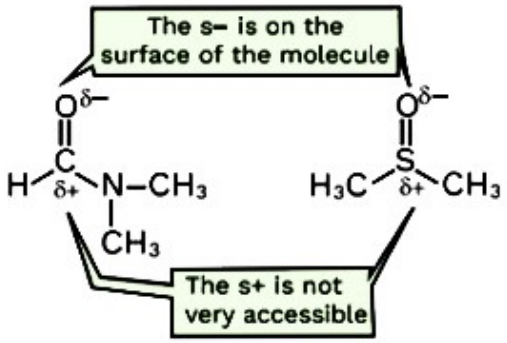
(Hexamethyl phosphoramide)

(xv) 

(12-Crown-4-ether)

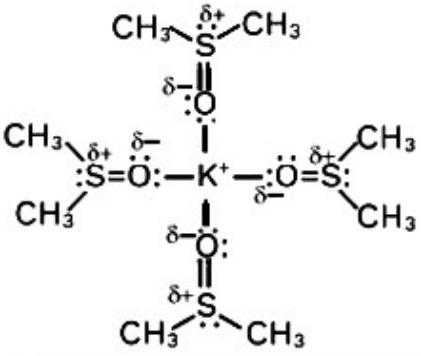
(xvi) 

(15-crown-5-ether)



N,N-dimethylformamide Dimethyl sulfoxide

DMF **DMSO**



DMSO can solvate a cation better than it can solvate an anion

(C) Non Polar Solvent's :

Solvents that do not have dipole moment and contains bonds between atoms with similar electronegativities

Example Hexane, Pentane.

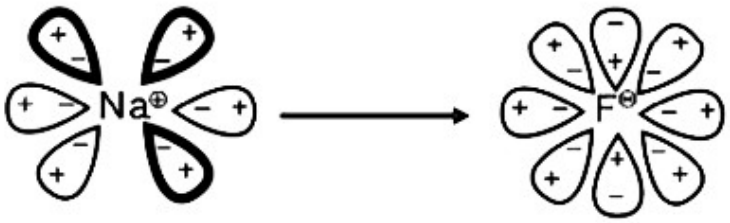
Points to remember

Polar aprotic solvent dissolve only cationic part of reagent.

Effect of Solvent at Nucleophilicity

a. In a Polar Protic Solvent

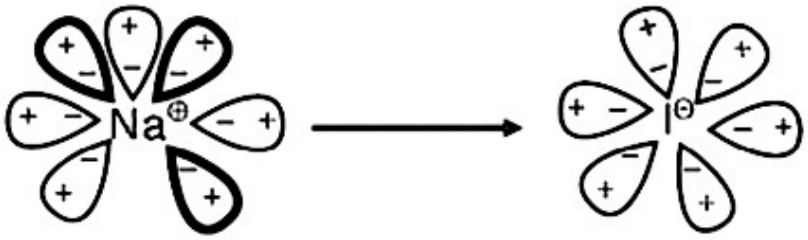
$F^{\ominus}$ $Cl^{\ominus}$ $Br^{\ominus}$ $I^{\ominus}$



$\Rightarrow$ Hydration α charge density

$$\alpha \frac{1}{\text{Size}}$$

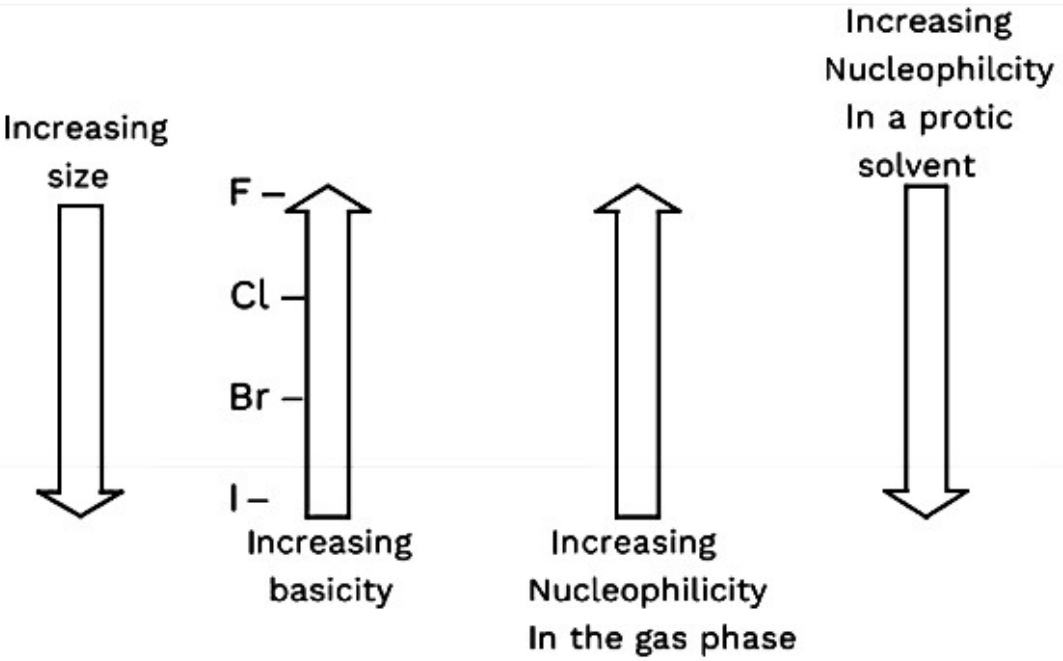
Size of $F^{\ominus}$ is small
Hydration ↑
Ionic mobility ↓
Nucleophilicity ↓



Size of $I^{\ominus}$ is large
Hydration↓
Hydrated size↓
Ionic mobility
Nucleophilicity

Points to remember

Nucleophilicity : $F^{\ominus} < Cl^{\ominus} < Br^{\ominus} < I^{\ominus}$
Basic Strength : $F^{\ominus} > Cl^{\ominus} > Br^{\ominus} > I^{\ominus}$



$\Rightarrow$ Relative Nucleophilicity Toward CH_3I in Methanol

Increasing nucleophillicity

(b) In Polar Aprotic Solvent :

In polar aprotic solvent, hydrolysis of only cationic part of reagent takes place.
Hydration of anion does not takes place.
As size ↑ionic mobility↓ Nucleophilicity↓
Nucleophilicity : $F^{\ominus} > Cl^{\ominus} > Br^{\ominus} > I^{\ominus}$
Basic Strength : $F^{\ominus} > Cl^{\ominus} > Br^{\ominus} > I^{\ominus}$

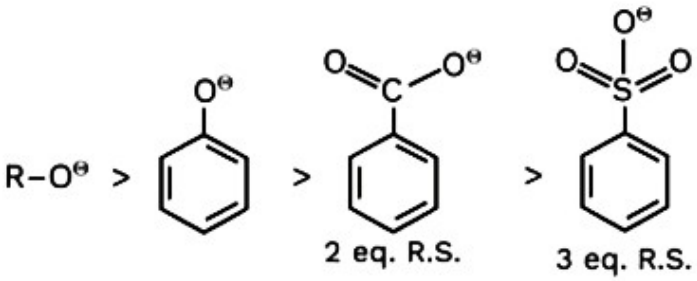
Example:

In Polar Protic Solvent :	$F^- < Cl^-$	$OH^- < SH^-$	$\ddot{N}H_3 < \ddot{P}H_3$
In Polar Aprotic Solvent :	$F^- > Cl^-$	$OH^- > SH^-$	$\ddot{N}H_3 > \ddot{P}H_3$

(c) Magnitude of negative charge :

Magnitude of negative charge ↑Nucleophilicity

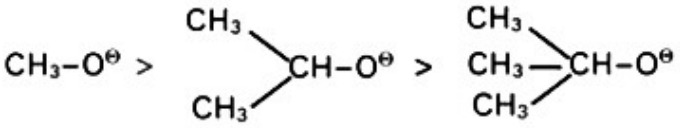
Example :



(d) Steric hinderance :

Steric hinderance ↑Nucleophilicity↓

Example :



(e) Number of Donor Atom :

Number of donor atom ↑ Nucleophilicity ↑

Example :

(i) $H_2\ddot{O} < H-\ddot{O}-\ddot{O}-H$ (ii) $\ddot{N}H_3 < \ddot{N}H_2-\ddot{N}H_2$

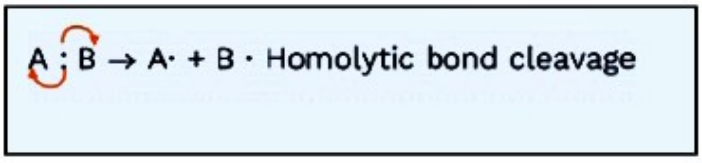
3. Reaction Intermediates :

Reactive, short lived, high energy, unstable species, those are formed in the course of organic reactions are called reaction intermediates. Reaction intermediates generally formed in between reaction. A covalent bond can get cleaved either by:

(A) Heterolytic cleavage **or** **(B) Homolytic cleavage**

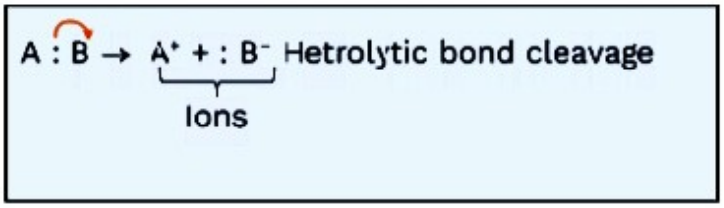
(A) Homolytic fission of covalent bonds :

Bond breaking in such way that each fragment gets one unpaired electron.

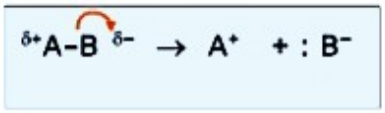


(B) Heterolytic fission of covalent bonds :

The bond may part in such a way that one portion takes away electrons of the bond leaving the other portion with vacant orbital. This kind of breakage called heterolysis, produces charged portions or ions..



Heterolysis of a bond normally requires that the bond should be polarized.



Polarization of a bond usually result from different electronegativity of the atoms joined by the bond. The larger the change in electronegativity, the greater the polarization. In the given case, atom B is more electronegative than A.

1. Carbanion :

Intermediate species which contain three bond pair and a negative charge on it, is called carbanion.

Note Hybridisation of carbanion could be sp^3, sp^2 and sp.

	Hybridisation	**Example**
(i)	sp^3	$\overset{\ominus}{C}H_3$, $CH_3-\overset{\ominus}{C}H_2$, $CH_3\overset{\ominus}{C}HCH_3$, $(CH_3)_3\overset{\ominus}{C}$,
(ii)	sp^2	$CH_2=\overset{\ominus}{C}H$, $CH_2=CH=\overset{\ominus}{C}H_2$
(iii)	sp	$HC\equiv\overset{\ominus}{C}$

Stability of Carbanion :

Carbanions are stabilized by electron withdrawing effect as :

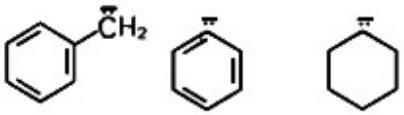
(i) –I effect **(ii) –M effect** **(iii) Delocalisation of charge**

Carbanions are Lewis bases. In their reactions they seek a proton or some other positive centre.

Example :

(a) $\overset{\ominus}{C}H_3, \overset{\ominus}{C}H_2-CH_3, CH_3-\overset{\ominus}{C}H-CH_3$ $H_3C-\overset{\ominus}{\underset{CH_3}{\underset{|}{C}}}-CH_3$ (Stability order : 1 > 2 > 3 > 4)

(b) $CH\equiv\overset{\ominus}{C}, CH_2=\overset{\ominus}{C}H, CH_3-\overset{\ominus}{C}H_2$ (Stability order : 1 > 2 > 3)

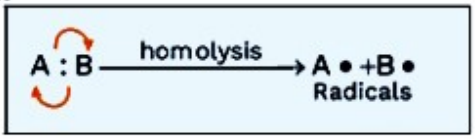
(c)  (Stability order : 1 > 2 > 3)

Points to remember

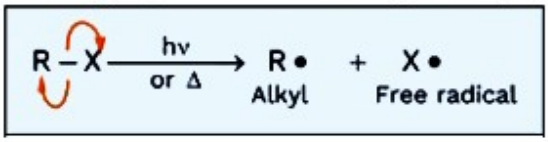
Rearrangement: Generally carbanions do not undergo rearrangement.

Free Radicals :

Homolysis of covalent bond results into free radical intermediates possess the unpaired electrons.



It is generated in presence of Sun light, Peroxides or High temperature



An uncharged intermediate which has three bond pair and an unpaired electron on carbon.

Note :

(i) It is Neutral species with odd e^-

(ii) It is paramagnetic in nature due to odd e^-

(iii) No rearrangement is observed generally.

(iv) Carbon atom having odd electron is in sp^2 hybridised state.

(v) Any reaction if it is carried out in the presence of sunlight, peroxide or high temperature it generally proceeds via free radical intermediate.

$(H_3C)\dot{C} > H_3C-\dot{C}H-CH_3 > H_3C-\dot{C}H_2 > \dot{C}H_3$ (Stability order)

$\dot{C}H_3 < CH_3-CH_2 < CH_3-\dot{C}H-CH_3 < CH_3-\underset{CH_3}{\underset{|}{\dot{C}}}-CH_3$ (Stability order)

(due to resultant of inductive effect and hyperconjugation, both operates in same direction)

2. Carbocation :

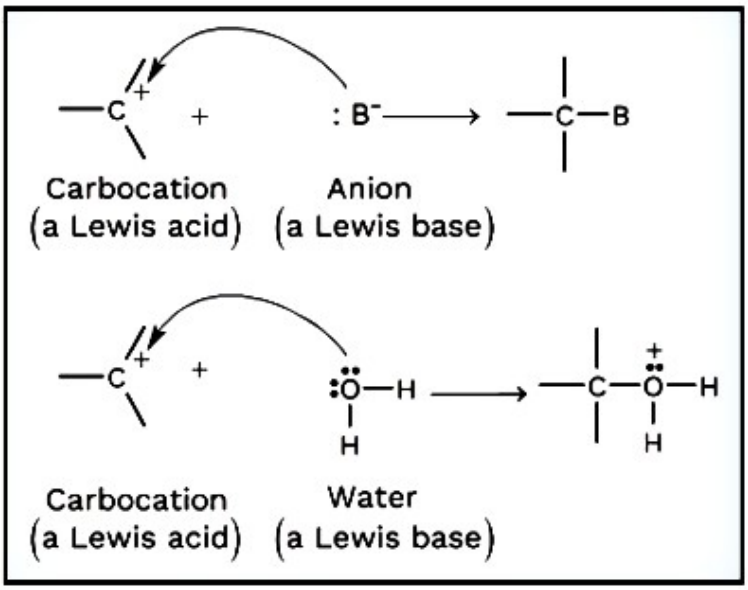
A carbon intermediate which contain three bond pair and a positive charge on it is called carbocation.

Hybridization: Carbocation may be sp^2 and sp hybridized

Example:

sp^2 : $\overset{\oplus}{C}H_3, CH_3\overset{\oplus}{C}H_2, CH_3\overset{\oplus}{C}HCH_3, (CH_3)_3\overset{\oplus}{C}$

sp : $H_2C = \overset{\oplus}{C}H, HC \equiv \overset{\oplus}{C}$



	Carbon free radical	Carbocation	Carbanion
			
	Trigonal planar sp^2 7	Trigonal planar sp^2 6	Pyramidal sp^3 8

3. Carbenes (Divalent Carbon Intermediates) :

Intermediate carbon species which forms only two bonds, divalent neutral species. Most carbenes are highly unstable that are capable of only fleeting existence. Soon after carbenes are formed, they usually react with another molecules.

Types of carbene	Singlet 	Triplet –Ċ–
Shape	Bent	Linear
Hybridisation	sp^2	sp
Nature of reaction	Stereospecific	None
State	Excited state	Ground state
Magnetic	diamagnetic	Paramagnetic
Nature	Paired electrons	Diradical

4. Nitrenes :

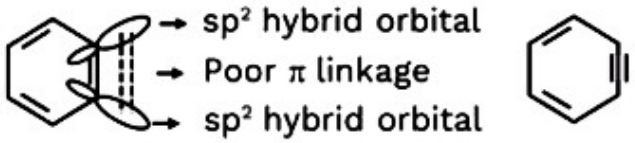
Octet of nitrenes is incomplete so very much reactive .In nitrenes only one valences of N are satisfied.





5. Benzyne :

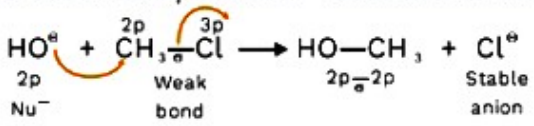
The benzene ring has one extra C–C π bond in benzyne



It is also significant to note that hybridization of each carbon involved in 'Benzylic bond' is sp^2 since the overlap between these sp^2 hybrid orbitals is not so much effective.

Leaving Group/Nucleofuge

A group or atom which leave from compound or reactant is known as leaving group.



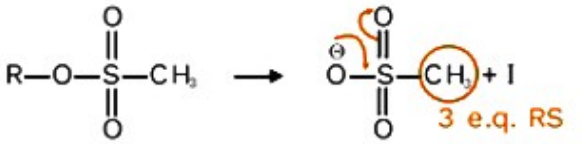
Types of Leaving Groups based on ease of elimination

(A) Super Leaving Group

E.g. : N_2, CO_2, SO_3

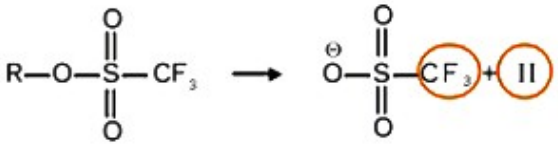
(B) Good Leaving Group :

(i)



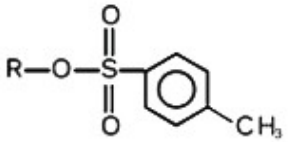
Mesylate ion/OMS

(ii)



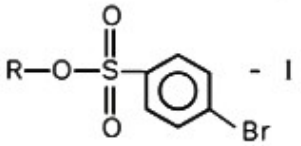
Triflate ion

(iii)



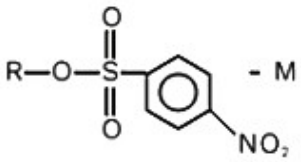
Tosylate ion (OTS)

(iv)



Brosylate ion (OBS)

(v)



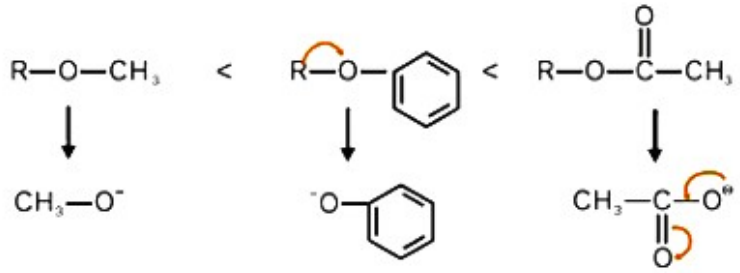
Nosylate (ONS)

(C) Moderate Leaving Group :

R—F < R—Cl < R—Br < R—I

2p 3p 4p 5p





(D) Bad Leaving Group – Strong Bond

$R-CH_3$

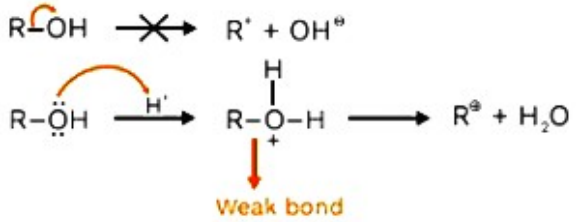
R–OH

$R-NH_2$

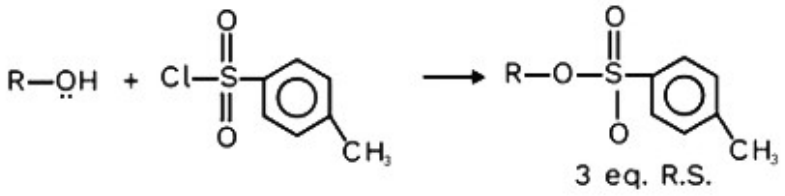
R–F

Conversion :

Method – 1



Method –2



Substutition Reaction (S_N2)

How alkyl halide reacts?

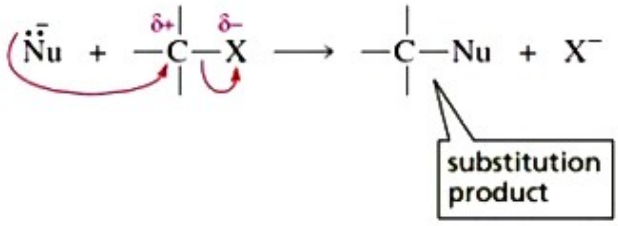
A halogen is more electronegative than carbon.
The two atoms do not part their bonding electrons equally

$\overset{\delta+}{R}CH_2-\overset{\delta-}{X}$ X = F, Cl, Br, I

There are two important mechanisms for the substitution reaction:

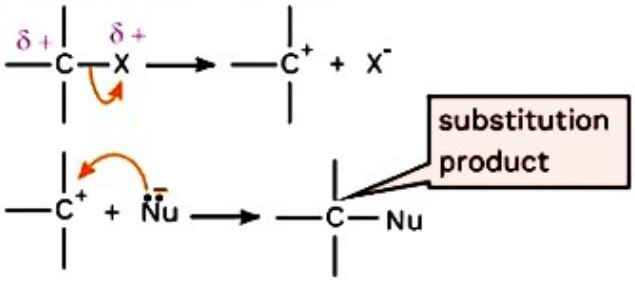
1. A nucleophile is attracted to the partially positively charged carbon (an electrophile).
2. Carbon halogen bond is formed during hetrolytic bond breaking and forming .

Example S_N2 Reaction



3. The carbon–halogen bond breaks heterolytically without any assistance from the nucleophile, forming a carbocation. The carbocation (an electrophile) then reacts with the nucleophile to form the substitution product.

Example S_N1 Reaction



Nucleophilic Substitution Reaction

Irrespective of the mechanism by which a substitution reaction occurs, it is called Nucleophilic Substitution Reaction because a nucleophile substitutes for the leaving group (mostly Halogen atom).

Factors involved in rate of reaction

1. the structure of the alkyl halide
2. the reactivity of the nucleophile
3. the concentration of the nucleophile
4. the solvent in which the reaction is carried out

S_N2 Reaction

Reaction

$HO^- + CH_3\text{-}Cl \rightarrow CH_3OH + Cl^-$

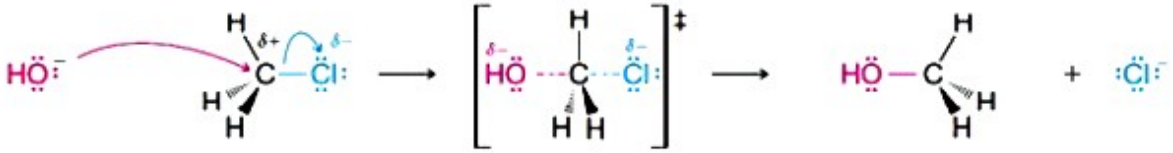
where "S" stands for substitution, "N" for nucleophilic, and "2" for bimolecular. Bimolecular means that two molecules are involved in the rate-determining step.

S_N2 Reaction Mechanism

Reaction

$$HO^- + CH_3Cl \longrightarrow CH_3OH + Cl^-$$

Mechanism

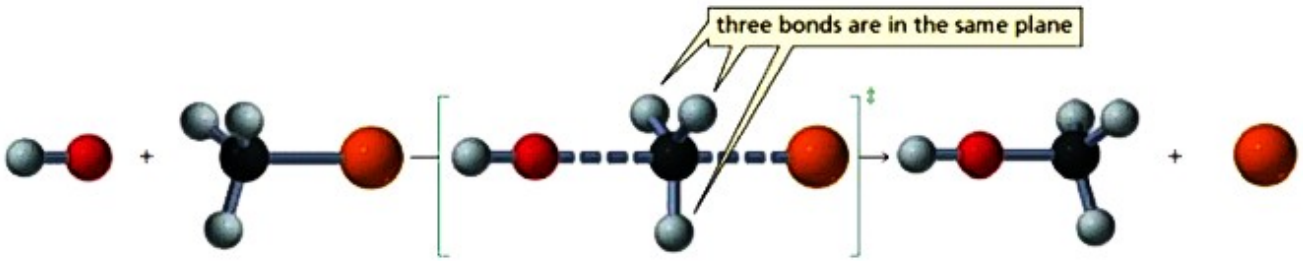


The negative hydroxide ion brings a pair of electrons to the partially positive carbon from the back side with respect to the leaving group. The chlorine begins to move away with the pair of electrons that bonded it to the carbon.

In the transition state, a bond between oxygen and carbon is partially formed and the bond between carbon and chlorine is partially broken. The configuration of the carbon atom begins to invert.

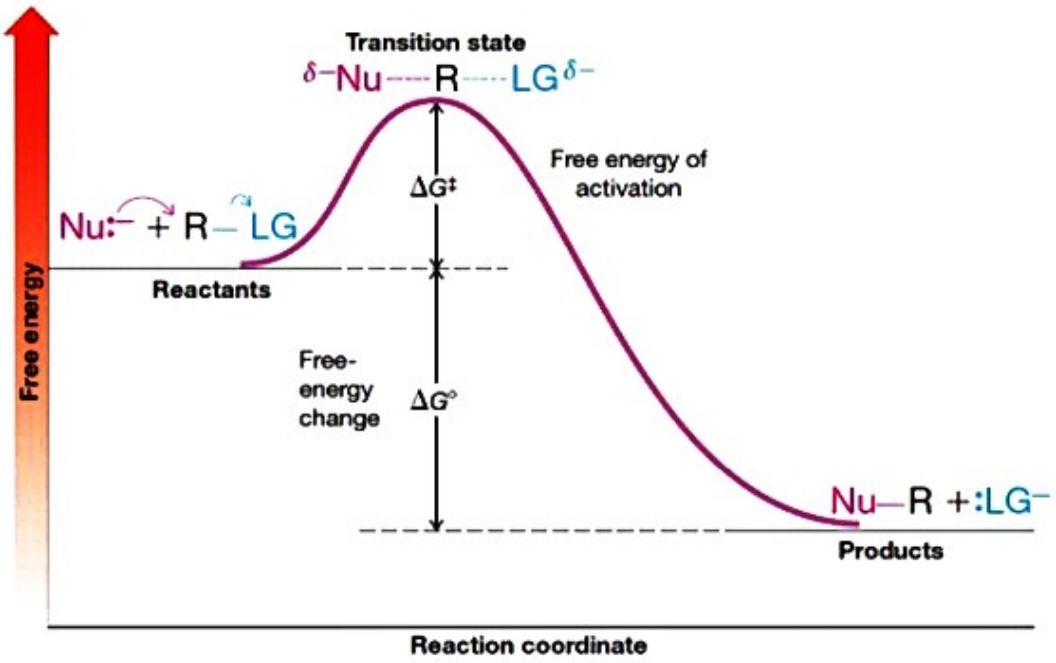
Now the bond between the oxygen and carbon has formed and the chloride ion has departed. The configuration of the carbon has inverted.

Molecular Diagram



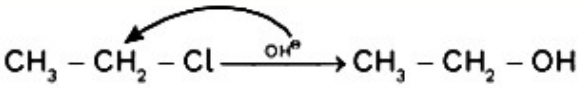
Some facts

1. S_N2 is bimolecular reaction because in slow step reagent and compound both are involved.
2. In S_N2 reaction nucleophile attacks at the anti-bonding orbital of leaving group that's why Walden inversion takes place.
3. In S_N2 reaction pentavalent carbon form is T.S. which is highly unstable.
4. Hybridization is sp^2 not sp^3d (does not have d-orbital)
5. In S_N2 reaction no rearrangement because no intermediate formed.
6. No relation in reactant and product because they are not isomers of each other.

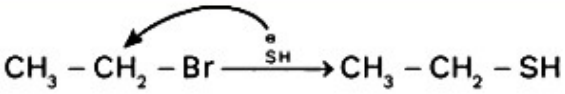


Identify the product in each reaction

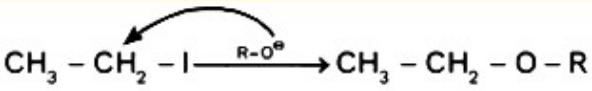
Q1 $CH_3CH_2Cl + HO^- \longrightarrow$

A1 

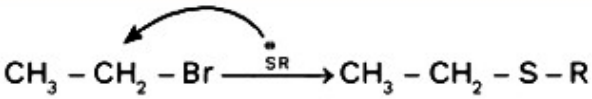
Q2 $CH_3CH_2Br + HS^- \longrightarrow$

A2 

Q3 $CH_3CH_2I + RO^- \longrightarrow$

A3 

Q4 $CH_3CH_2Br + RS^- \longrightarrow$

A4 

Q5 $CH_3CH_2Cl + {}^-NH_2 \longrightarrow$

A5 $CH_3 - CH_2 - NH_2$

Q6 $CH_3CH_2Br + {}^-C \equiv CR \longrightarrow$

A6 $CH_3 - CH_2 - C \equiv C - R$

Q7 $CH_3CH_2I + {}^-C \equiv N \longrightarrow$

A7 $CH_3 - CH_2 - C \equiv N$

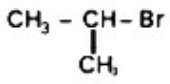
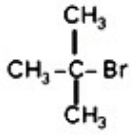
Factors Affecting S_N2 reactions

(a) Degree of alkyl halide

most reactive — methyl halide > 1° alkyl halide > 2° alkyl halide > 3° alkyl halide — least reactive

Comparative Rates of S_N2 Reactions for Several Alkyl Halides

$$R - Br + Cl^- \xrightarrow{S_N2} R - Cl + Br^-$$

Alkyl halide	Class of alkyl halide	Relative rate
CH_3–Br	methyl	1200
CH_3CH_2–Br	primary	40
$CH_3CH_2CH_2$–Br	primary	16
	secondary	1
	tertiary	too slow to measure

(b) The Leaving Group

Relative leaving abilities of the halide ions

best leaving group — $I^- > Br^- > Cl^- > F^-$ — worst leaving group

	relative rates of reaction
$HO^- + RCH_2I \rightarrow RCH_2OH + I^-$	30,000
$HO^- + RCH_2Br \rightarrow RCH_2OH + Br^-$	10,000
$HO^- + RCH_2Cl \rightarrow RCH_2OH + Cl^-$	200
$HO^- + RCH_2F \rightarrow RCH_2OH + F^-$	1

Note: For each of the subsequent pairs of SN2 reactions, specify which reaction occurs faster:

Q8 $CH_3CH_2Br + H_2O$ **or** $CH_3CH_2Br + HO^-$

A8 $\overset{\ominus}{O}H > H_2O$ (Nucleophilicity) order

S_N2 reaction occurs fast in present of better nucleophile if leaving group is same.

Q9 $CH_3\underset{\displaystyle CH_3}{\underset{|}{C}}HCH_2Br + HO^-$ or $CH_3CH_2\underset{\displaystyle CH_3}{\underset{|}{C}}HBr + HO^-$

A9 1° > 2° (Rate of S_N2)

Q10 $CH_3CH_2Cl + CH_3O^-$ or $CH_3CH_2Cl + CH_3S^-$ (in ethanol)

A10 In polar protic solvent ethanol
$CH_3S^{\ominus}$ is better nucleophile then $CH_3O^{\ominus}$
Bigger sized $S^{\ominus}$ ion is less solvated.

Q11 $CH_3CH_2Cl + I^-$ or $CH_3CH_2Br + I^-$

A11 $Br^{\ominus} > Cl^{\ominus}$ (leaving group ability)

(c) Role of the solvent

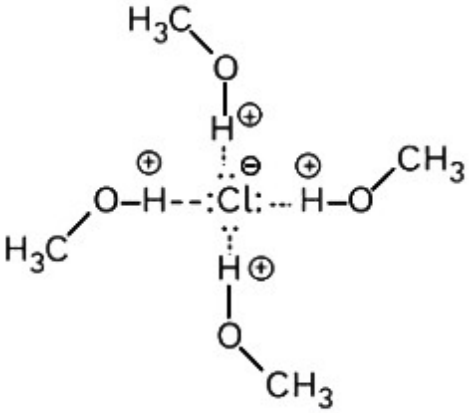
$CH_3–Br + Nu: \longrightarrow CH_3–Nu + Br^-$

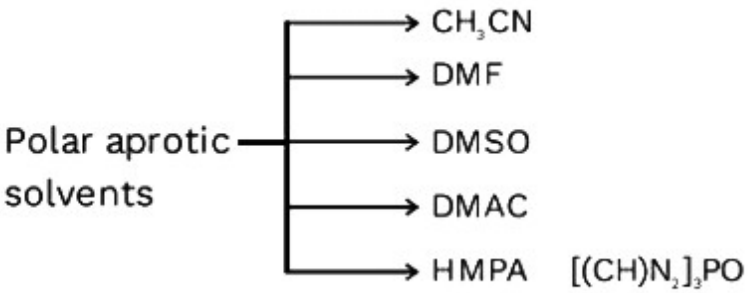
Relative Reactivity

The rates of S_N2 are affected by the solvent protic solvents which contain –OH or –NH groups are generally worst solvent for S_N2 reaction.

Polar aprotic solvents which have strong dipoles but do not have –OH or –NH groups, are the best.

Protic solvents such as MeOH and EtOH slow down S_N2 reactions by clustering around the reactant nucleophile, called solvation. because of H-bond with Nu: and forms like a cage which lowers down its reactivity





They solvate metal cation rather than nucleophilic anion. As a result unsolvated anion have a greater nucleophilicity, hence S_N2 reaction becomes faster.

Example

Rate increases by 2,00,000 times by changing from MeOH to HMPA in the reaction of N_3^- with 1 bromobutane

$RBr + N_3^- \longrightarrow R–N_3 + Br^-$

Solvent	MeOH	H_2O	DMSO	DMF	CH_2CN	HMPA
Relative reactivity	1	7	1300	2800	5000	2,00,000

Which reaction in each of the following parts will take place more rapidly?

Q12 **(a)** $CH_3Br + HO^- \longrightarrow CH_3OH + Br^-$

(b) $CH_3Br + H_2O \longrightarrow CH_3OH + HBr$

A12 (a) H_2O act as nucleophile and polar protic solvent.
(b) $OH^{\ominus} > H_2O$ (Better Nucleophile)

Q13 $CH_3I + HO^- \longrightarrow CH_3OH + I^-$

$CH_3Cl + HO^- \longrightarrow CH_3OH + Cl^-$

A13 $I^{\ominus} > Cl^{\ominus}$ (leaving group ability)

Q14 $CH_3Br + NH_3 \longrightarrow CH_3\overset{+}{N}H_3 + Br^-$

$CH_3Br + H_2O \longrightarrow CH_3OH + HBr$

A14 $NH_3 > H_2O$ (Nucleophilicity order)
Better Nucleophile, more rate of reaction.

Q15 $CH_3Br + HO^- \xrightarrow{DMSO} CH_3OH + Br^-$

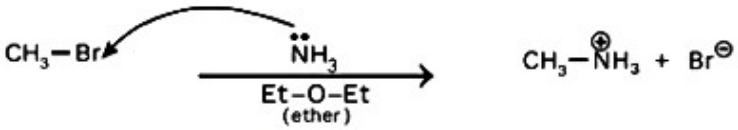
$CH_3Br + HO^- \xrightarrow{EtOH} CH_3OH + Br^-$

A15 $HO^{\ominus}$ D.M.S.O polar aprotic
H_2O EtOH polar protic
Better nucleophile and polar a protic solution suitable for S_N2 reaction.

Q16 $CH_3Br + NH_3 \xrightarrow{Et_2O} CH_3\overset{+}{N}H_3 + Br^-$

$CH_3Br + NH_3 \xrightarrow{EtOH} CH_3\overset{+}{N}H_3 + Br^-$

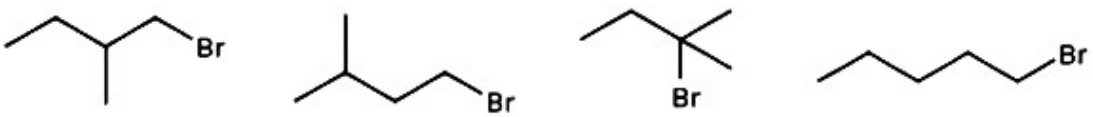
A16



Polar aprotic solvent

Q17 **Position the following alkyl bromides in order of declining reactivity in an S_N2 reaction:**
1-bromo-2-methylbutane, 1-bromo-3methylbutane, 2-bromo-2-methylbutane, and 1-bromopentane.

A17



(a) (b) (c) (d)

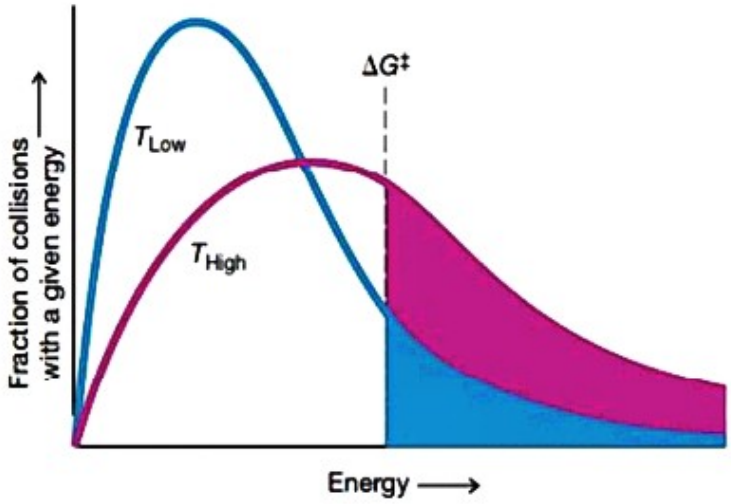
Rate of reaction directly depends upon steric crowding at carbon having alkyl halide (L.G.).

d > b > a > c

↓ ↓

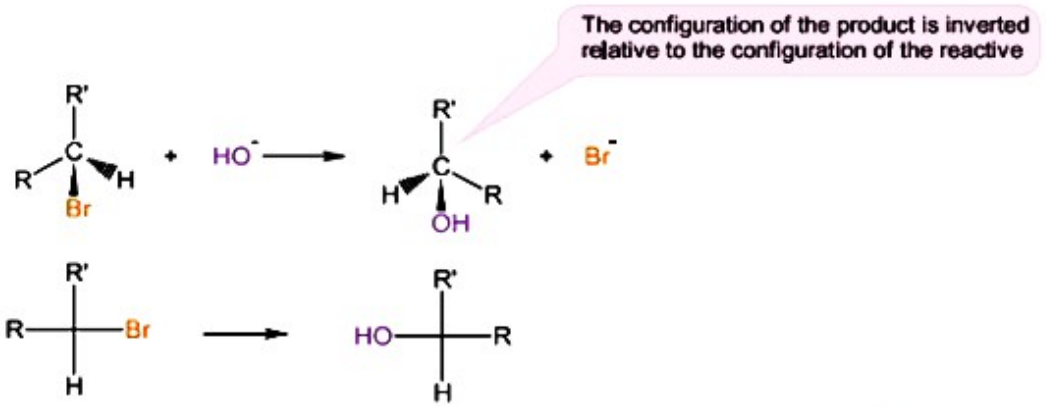
Least crowded Most crowded

Temperature and Reaction Rate



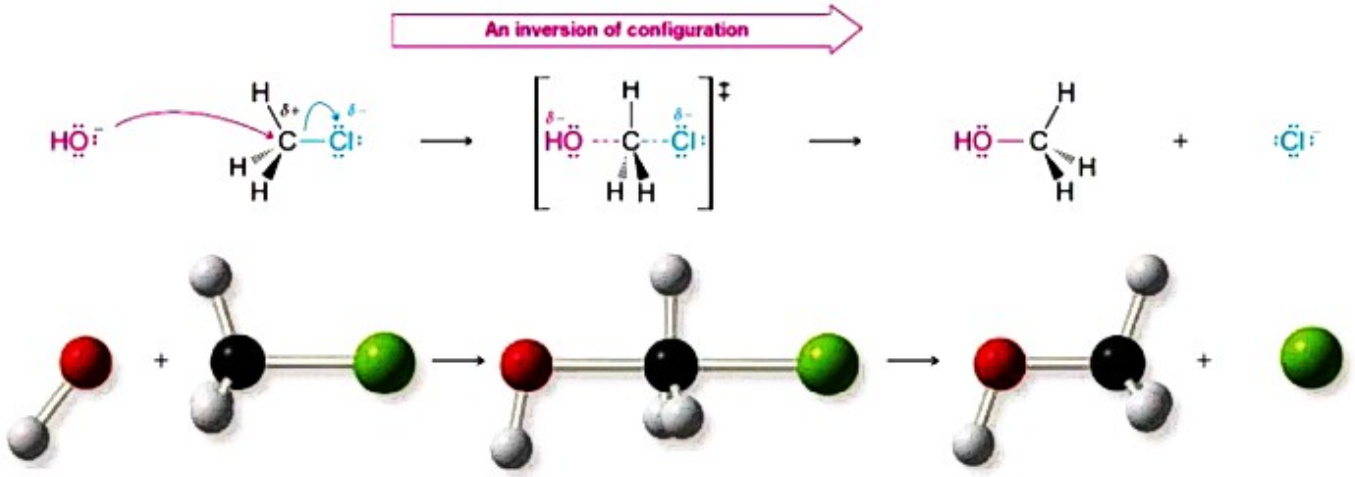
The Stereochemistry of S_N2 Reactions

Stereochemistry in S_N2 reaction is defined on the basis of attack of nucleophile in anti bonding of compound:
The nucleophile approaches the substrate carbon from the back side with respect to the leaving group.
In other words, the bond to the nucleophile that is forming is opposite (at 180°) to the bond to the leaving group that is breaking.
Nucleophilic displacement of the leaving group in an S_N2 reaction causes inversion of configuration at the substrate carbon.

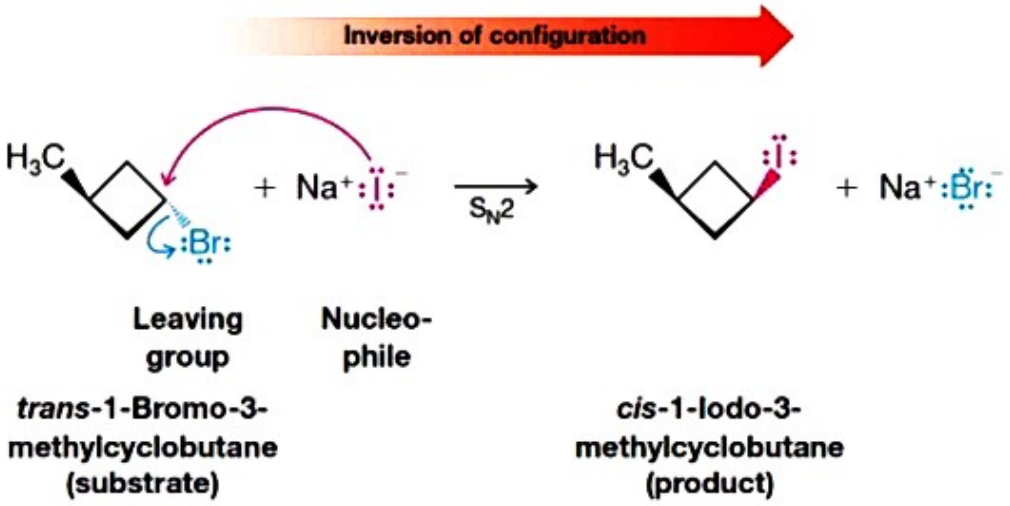


Carbon at which substitutions occur, get inverted in configurations , just As an umbrella has tendency to invert in a storm. It is called walden inversion.
Paul Walden, firstly discovered that the configuration of a compound inverts in an S_N2 substitution.
Only one substitution product is formed .

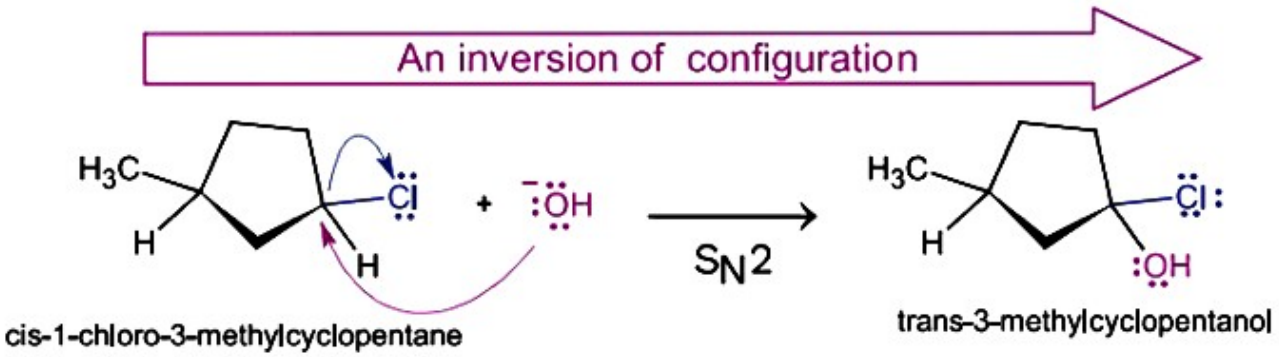
An inversion of configuration



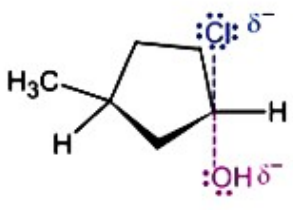
Inversion of configuration in Ring Compound



Example



Apparently, the transition state for this reaction is like that shown here.



Leaving group departs from the top side.

Nucleophile attacks from the bottom side.

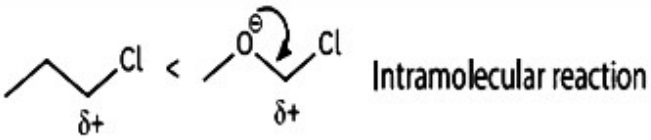
Which alkyl halide would you expect to be more reactive in an S_N2 reaction with a given nucleophile?

Q18 $CH_3CH_2CH_2Br$ **or** $CH_3CH_2CH_2I$

A18 b > a (Leaving Ability)

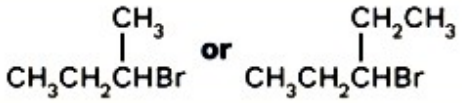
Q19 $CH_3CH_2CH_2Cl$ **or** CH_3OCH_2Cl

A19



Intramolecular reaction

Q20



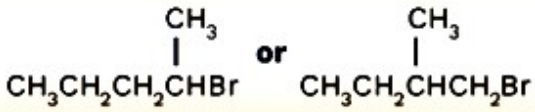
A20



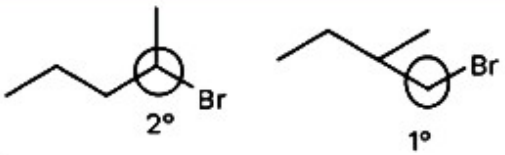
more bulkyl group

a > b

Q21

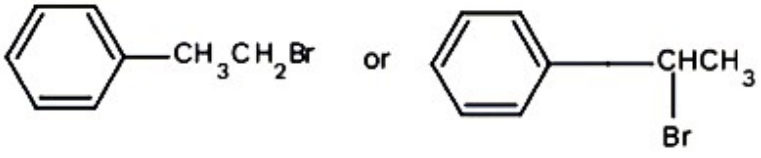


A21

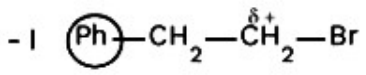


b > a

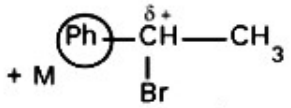
Q22



A22

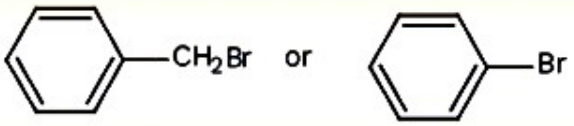


b > a

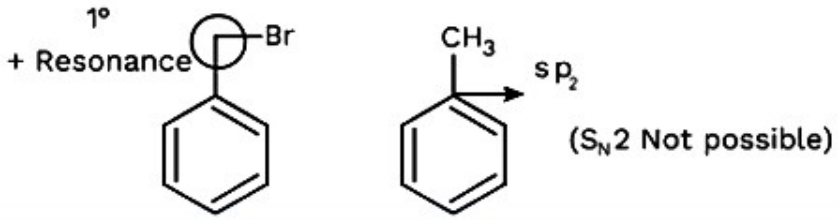


(more stable δ+)

Q23

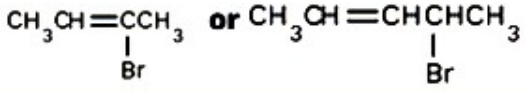


A23

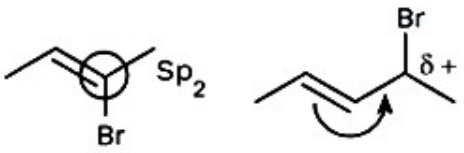


(S_N2 Not possible)

Q24



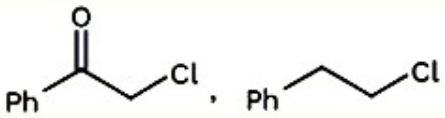
A24



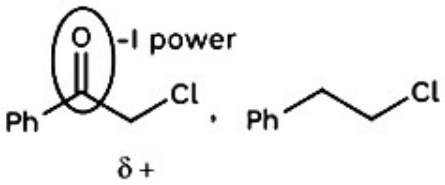
More stable transition state in case of b.

b > a

Q25



A25

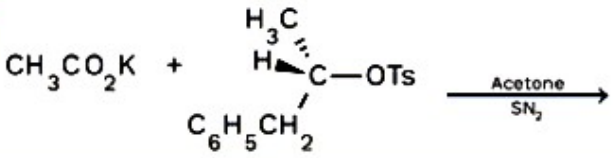


More stable transition state.

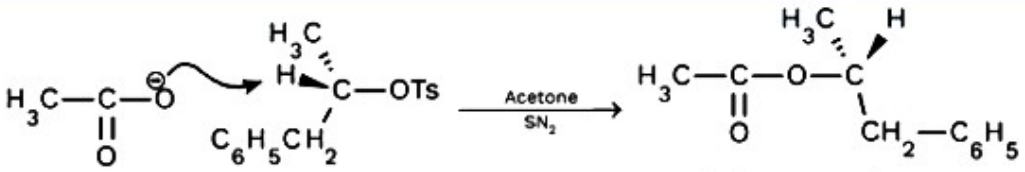
a > b

Identify the product

Q26

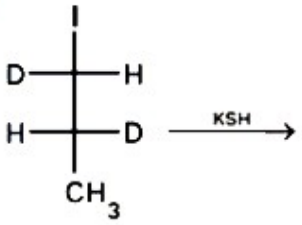


A26



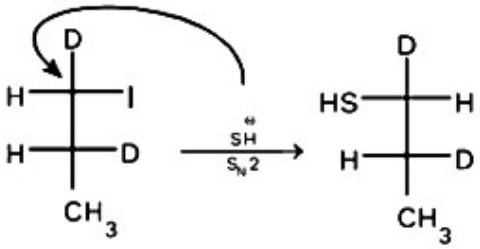
Walden Invertion

Q27

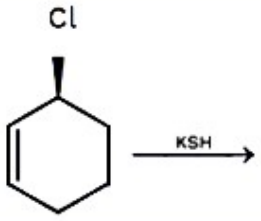


(Hint: First draw perfect Fischer projection)

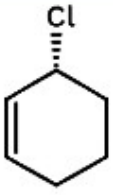
A27



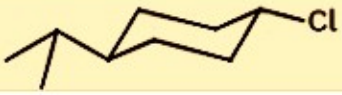
Q28



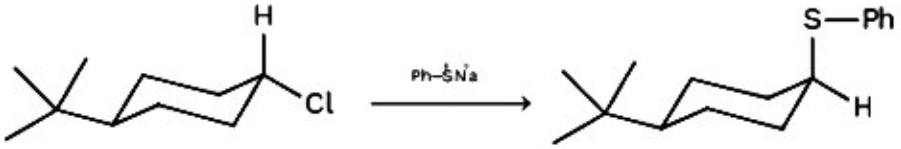
A28



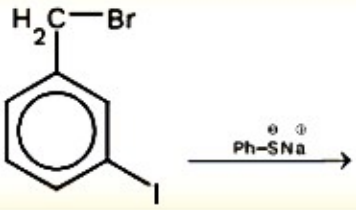
Q29



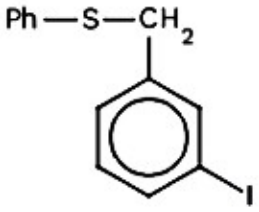
A29



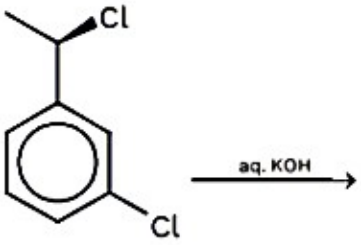
Q30



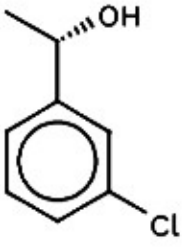
A30



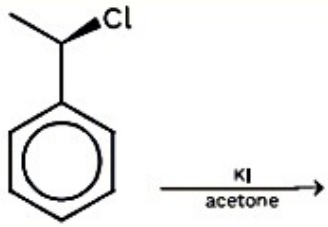
Q31



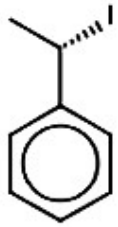
A31



Q32



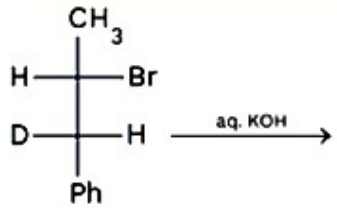
A32



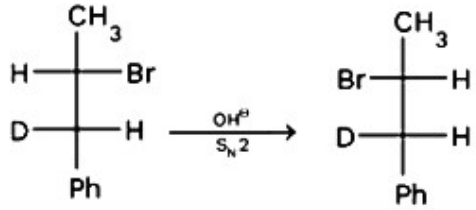
Q33 $Cl - CH = CH - CH_2 - Cl \xrightarrow{\text{aq. KOH}}$

A33 $Cl - CH = CH - CH_2 - OH$

Q34

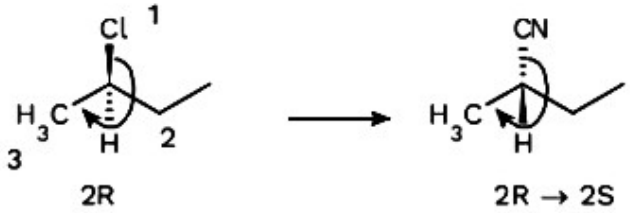


A34



Q35 **2R-chlorobutane** $\xrightarrow{NaCN}$

A35 2R-chlorobotane $\xrightarrow[S_N2]{CN^{\ominus}}$ 2S-cyanobutane



The Williamson Ether Synthesis

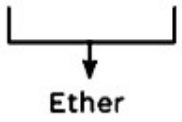
Substitution by alkoxy group:

Alkyl halides on treatment with metal alkoxides ($R—\bar{O}Na^+$) produce ethers. This method of preparation of ethers is often referred to as **Williamson reaction.**

Great way to synthesis unsymmetrical ethers via nucleophilic substitution reaction known as the williamson's ether synthesis..

The Williamson ether synthesis involves of an S_N2 reaction of a sodium alkoxide stereotypically with an alkyl halide or alkyl sulfonate

Alkonide + alkyl Halide



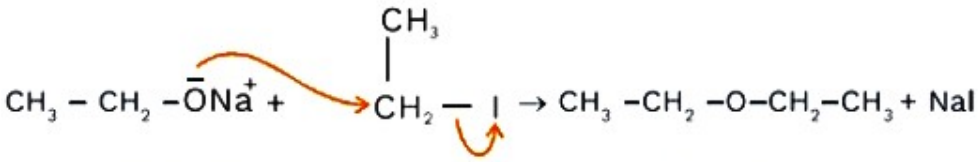
Ether

The alkoxide ion (RO^-) for the Williamson ether synthesis is obtained by using sodium metal or sodium hydride (NaH) to eliminate a proton from an alcohol.

$$ROH + Na \rightarrow RO^- + Na^+ + \frac{1}{2}H_2$$

$$ROH + NaH \rightarrow RO^- + Na^+ + H_2$$

Mechanism of the Reaction



Sodium Ethoxide DiethylEther

The alkoxide ion reacts with the alkyl halide in an S_N2 reaction, with the subsequent development of functional group ether.

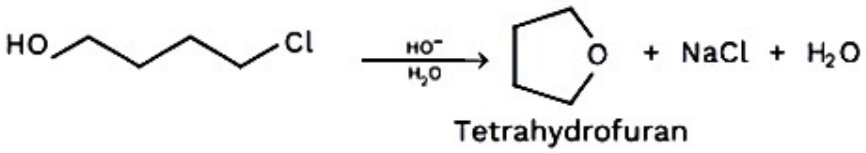
The substrate must be unrestricted and bear a decent migratory group. Representative substrates are 1° or 2° alkyl halides and alkyl sulfonates:

The usual limits of S_N2 reactions applicable in reactions.

Finest results are attained when the alkyl halide or sulfonate is principle or else CH_4.

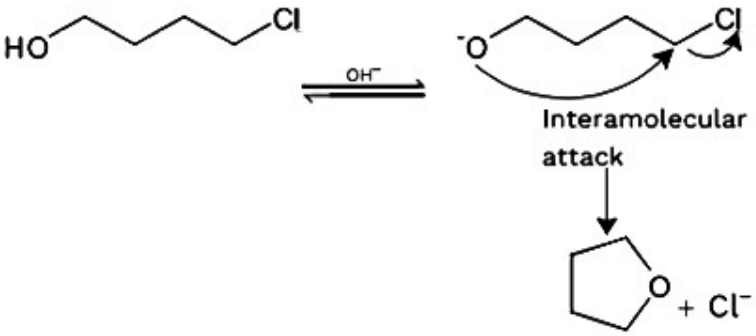
If the substrate is 3 degree elimination reaction is the limited result. Substitution is also favoured over elimination at lower temperatures.

Q36 **The cyclic ether tetrahydrofuran (THF) can be synthesized by treating 4-chloro-1-butanol with aqueous NaOH (see below). Propose a mechanism for this reaction.**



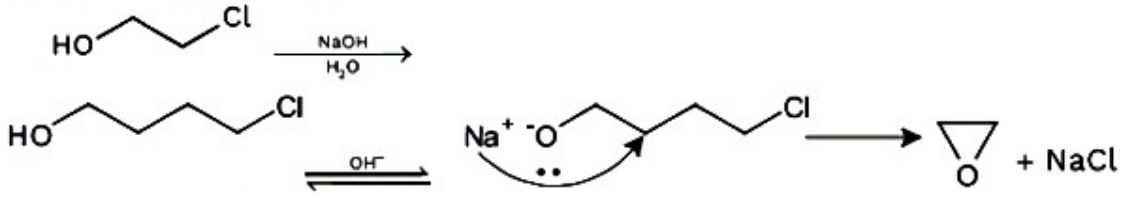
Tetrahydrofuran

A36 The cyclic ether.....

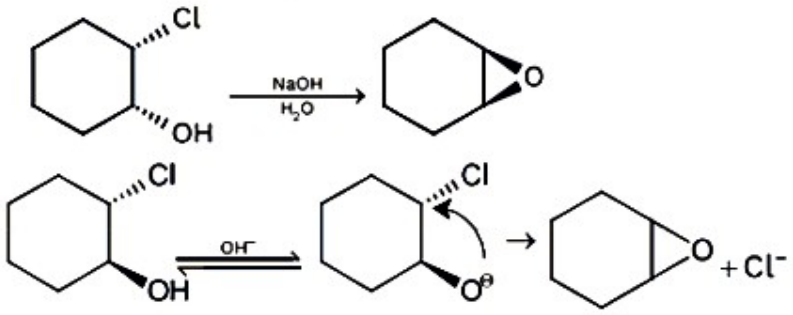


Q37 **Epoxides can synthesized by treating halohydrins with aqueous base. Propose a mechanism for reactions (a) and (b), and explain why no epoxide formation is observed in (c).**

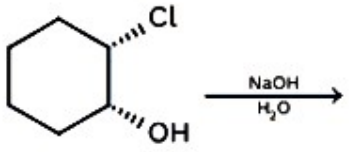
(a)



(b)

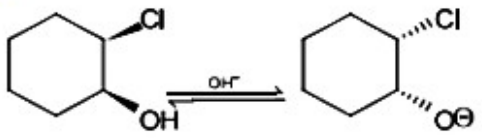


(c)



No epoxide observed

A37



As both are on same side back side attack is not possible

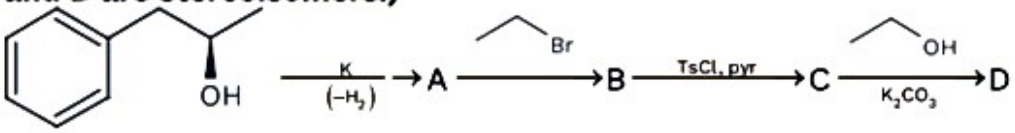
Let's understand

A modification of this method involves the reaction of an alkyl halide and alcohol in presence of dry silver oxide. This method is known as **Purdie method of alkylation** and is quite useful in sugar chemistry.

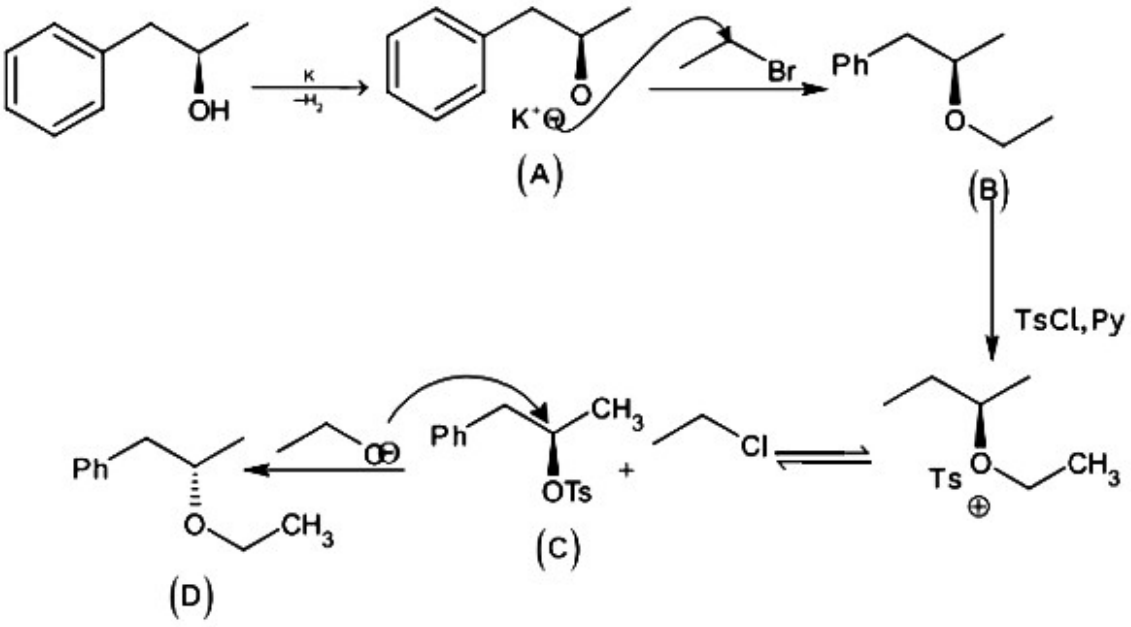
$$2CH_3OH + 2CH_3I + Ag_2O \rightarrow 2CH_3{-}O{-}CH_3 + 2AgI + H_2O$$

dimethyl ether

Q38 **Write structures for products A, B, C and D, shown stereochemistry. (Hint: B and D are stereoisomers.)**



A38

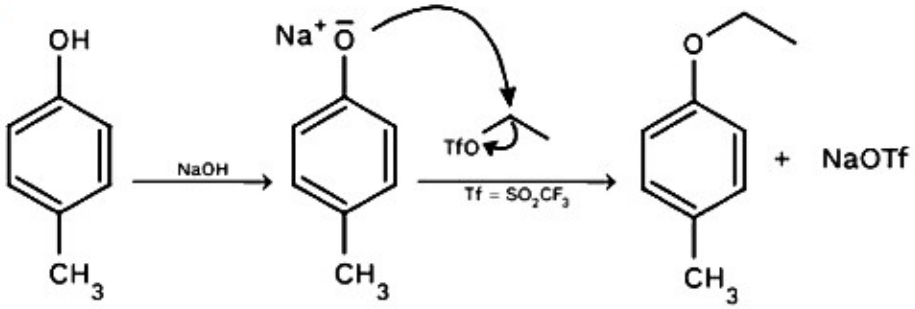


[Enantiomer of B]

Preparation of phenol using williamson's Ether Synthesis

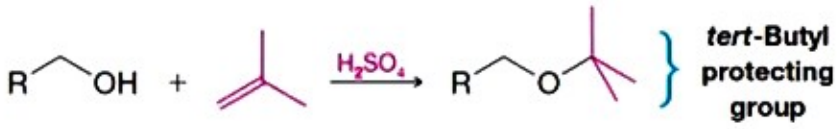
Phenols can also be converted to ethers through the Williamson synthesis.
Phenols are more acidic than alcohols, they can be converted to phenoxides by reaction with weaker bases such as sodium hydroxide, or by use of stronger bases such as NaH.
Leaving groups on the alkyl reactant can be halides or solfonates (e.g., OTs, Oms, or OTf).

Example.



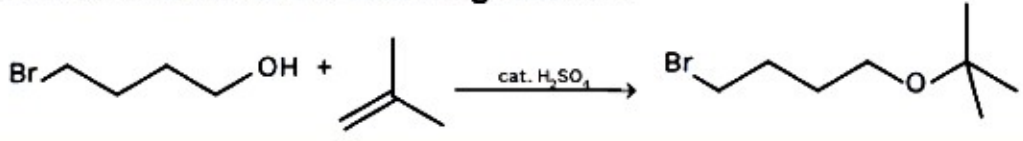
Tert-Butyl Ethers by Alkylation of Alcohols:

Primary alcohols can be converted to tert-butyl ethers by dissolving them in a strong acid such as sulphuring acid and then adding isobutylene to the mixture. (This procedure minimizes dimerization and polymerization of the isobutylene.)

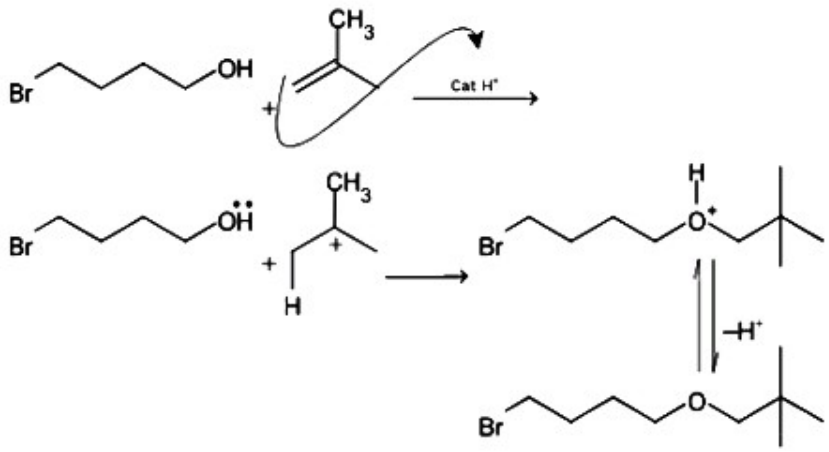


A tert-butyl ether can be cast-off to "protect" the OH group of a 1 degree alcohol, while another reaction is supported out on some other fragment of the compound.
A tert-butyl protective group can be removed easily by treating the ether with dilute aqueous acid.

Q39 **Propose mechanisms for the following reaction.**

(a) 

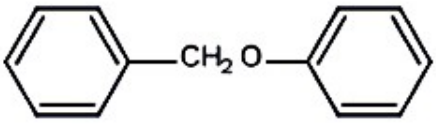
A39

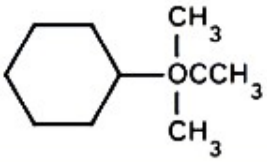


Q40

How could the following ethers be prepared, using an alkyl halide and an alcohol?

(A) $CH_3CH_2\underset{\displaystyle CH_3}{\underset{|}{C}}HOCH_2CH_2CH_3$

(B) 

(C) 

(D) $CH_3CH_2OCH_2\underset{\displaystyle CH_3}{\underset{|}{C}}HCH_2CH_2CH_3$

A40

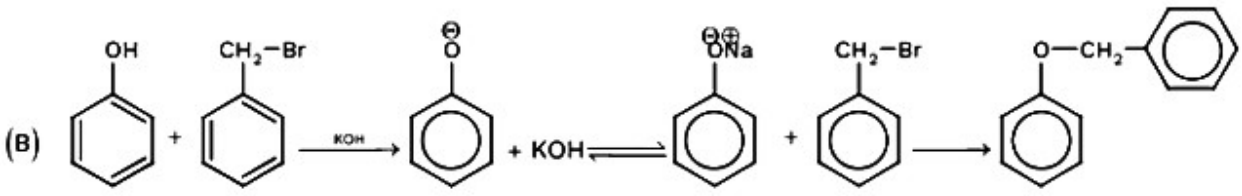
(a) $CH_3-CH_3-\underset{\displaystyle CH_3}{\underset{|}{C}}H-OH + CH_3-CH_2-Br \xrightarrow{NaOH}$

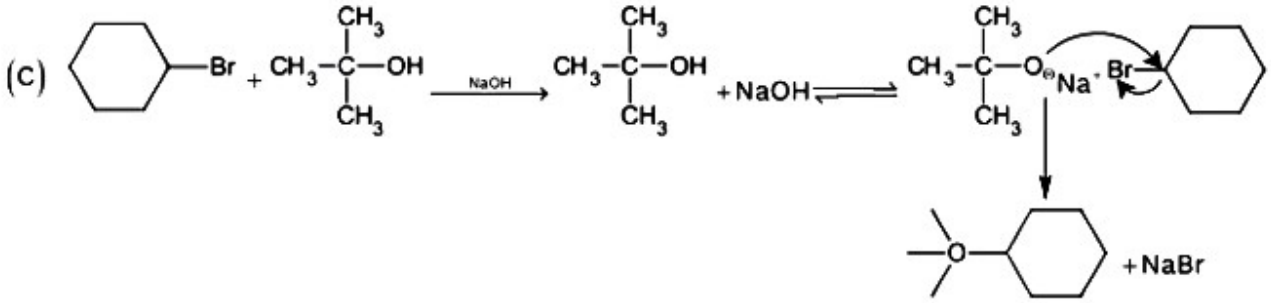
$$CH_3-CH_2-\underset{\displaystyle CH_3}{\underset{|}{C}}H-OH + NaOH \rightleftharpoons CH_3-CH_2-\underset{\displaystyle CH_3}{\underset{|}{C}}H-O^{\ominus}Na^{\oplus} + H_2O$$

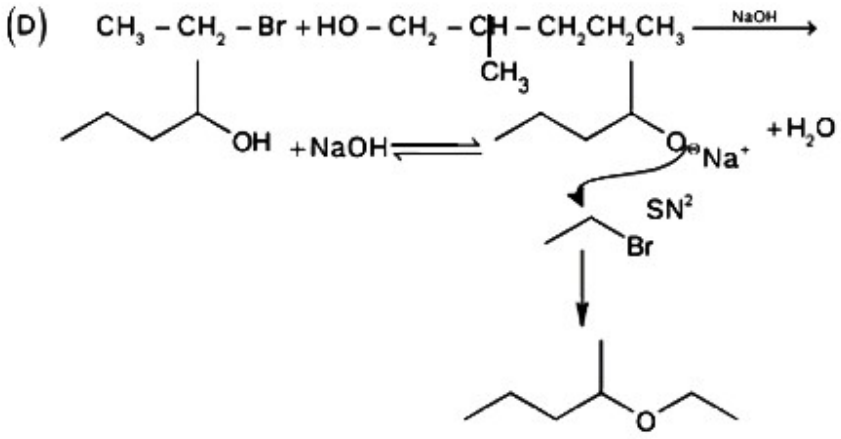
SN^2

$$CH_3-CH_2-CH_2-Br$$

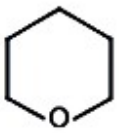
$$CH_3-CH_2-\underset{\displaystyle CH_3}{\underset{|}{C}}H-O-CH_2-CH_2-CH_3$$



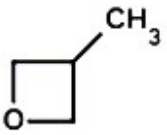


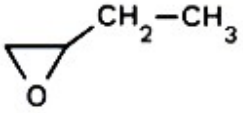


Q41 **4-chloro-1-butanol + NaOH → (B) Product (B) of the above reaction is:**

(A) 

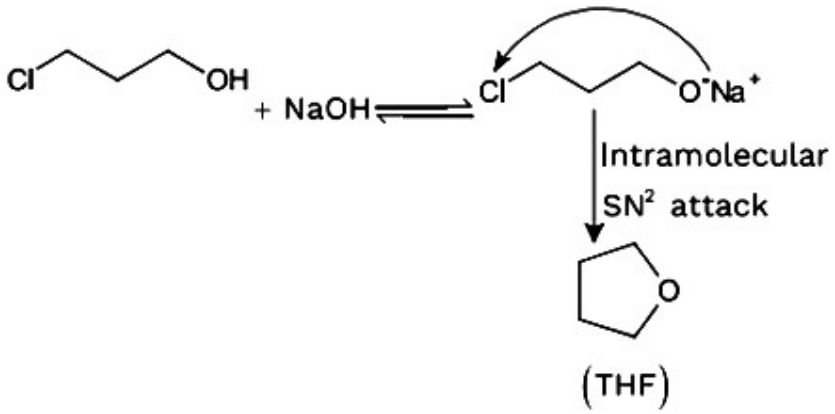
(B) 

(C) 

(D) 

A41 B

4-choloro-1-butanol + NaOH ⟶ ?



⇒ Tetrahydrofuran will be formed

Q42 **In the given couple of reaction in which pair the second reaction is more reactive than first toward S_N2 reaction?**

(A) $CH_3 - CH_2 - Cl + CH_3 - CH_2 - O^- \rightarrow Et - O - Et$ **(or)**

$CH_2 - CH_2 - Cl + CH_3 - CH_2 - OH \rightarrow Et - O - Et$

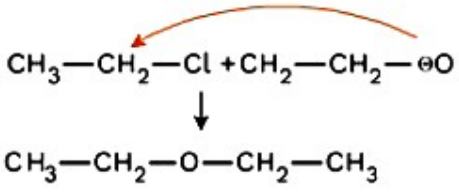
(B) $CH_3 - CH_2 - Cl + EtO^- \rightarrow Et - O - Et$ **(or)**

$CH_3 - CH_2 - Cl + EtS^- \rightarrow CH_3 - CH_2 - S - Et$

(C) $\underset{(1m)}{Et} - Cl + \underset{(2m)}{CH_3O^-} \rightarrow Et - O - CH_3$ **(or)** $\underset{(2m)}{Et} - Cl + \underset{(1m)}{CH_3O^-} \rightarrow Et - O - CH_3$

(D) $Et - Br + Ph_3P \rightarrow Et - \overset{\oplus}{P}Ph_3$ **(or)** $Et - Br + Ph_3N \rightarrow E + -\overset{\oplus}{N}Ph_3$

A42 (a)

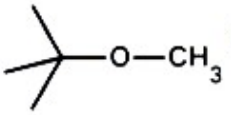


(Nucleophilicity order)

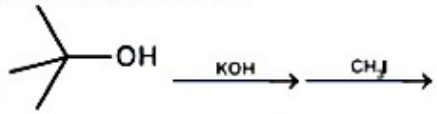
(b) Assume polar a protic solvent EtO^{Θ} (nucleophilicity order)

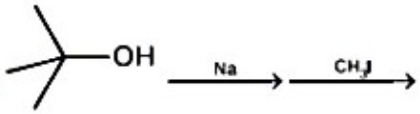
(c) Concentration of substrate and nucleophile equally important.

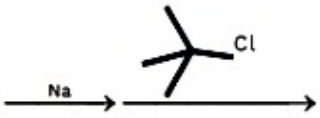
(d) $Ph_3P < Ph_3N$ (order nucleophilicity)
Assume (polar aprotic solvent)

Q43 **Best method to prepare**  **is:**

The combination of

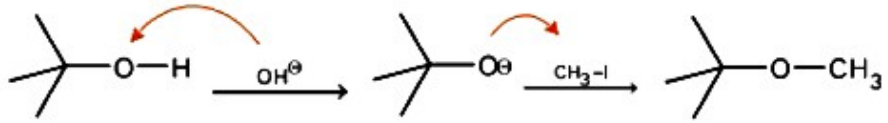
(A) 

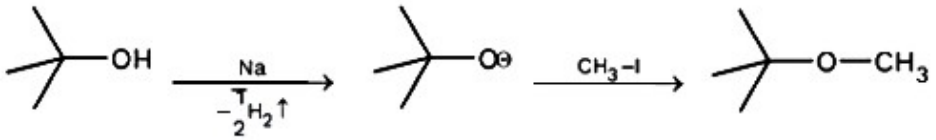
(B) 

(C) MeOH 

(D) (A) and (B) both

A43 (A, B)





Q44 **The best combination for the preparation of ether by Williamson's synthesis are**

(A) $(CH_3)CBr$ **and** CH_3ONa

(B) $(CH_3)CBr$ **and** CH_3OH

(C) CH_3Br **and** $(CH_3)_3CONa$

(D) CH_3Br **and** $(CH_3)_3COH$

A44 (C)

1° Alkyl halide + 3° alkoxie

Darzen's Method ($SOCl_2$)

Thionyl chloride reacts with alcohols without rearrangement. In the presence or absence of pyridine

$CH_3-CH_2-OH + SOCl_2 \xrightarrow{SN^1} CH_3-CH_2-Cl + \mathbf{SO_2 + HCl}$

$CH_3-CH_2-OH + SOCl_2 \xrightarrow{SN^2} CH_3-CH_2-Cl + \mathbf{SO_2 + HCl}$

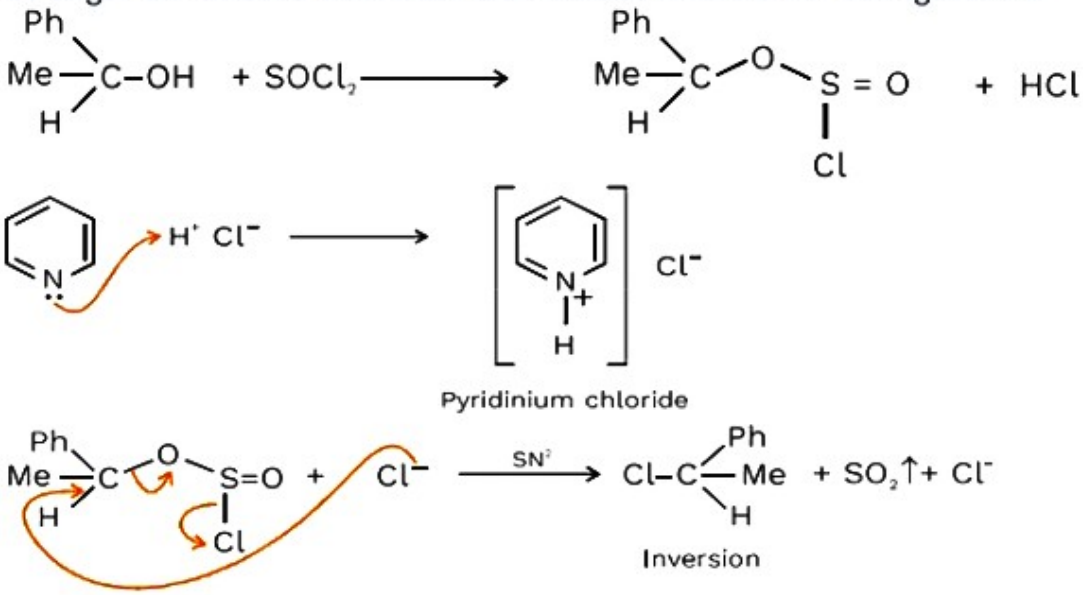
This is the one of the best methods to prepare haloalkanes from alcohols because both the by products are gaseous in nature.

$SOBr_2$ is less stable and SOI_2 does not exist

Role of Pyridine

When alcohol and $SOCl_2$ combine chlorosulphate and HCl form pyridine, prevents the formation of HCl (which is weak Nu– supplier) producing pyridinium chloride which is ionic in nature and Cl– is more free thus better nucleophilic.

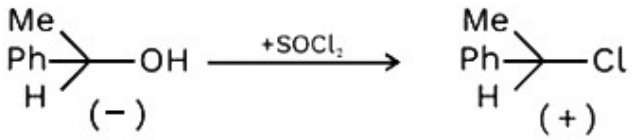
Stronger Cl– attacks from back side causes inversion of configuration.



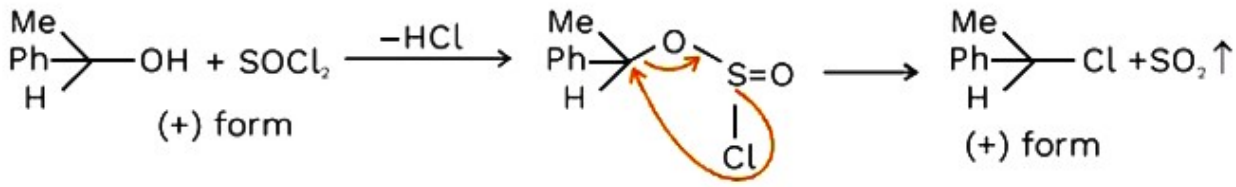
S_Ni (Internal Nucleophilic Substitution)

We know that the S_N2 reactions are very often accompanied by inversion of configuration. But a few S_N2 reactions proceed with actual retention of configuration instead of usual inversion such reaction are called "Nucleophilic internal substitution reactions or simply S_Ni reactions. Therefore, in S_Ni reactions, reactant and product have the same configuration

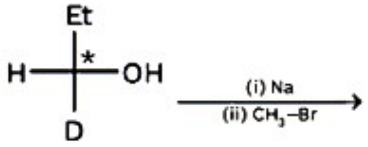
Example

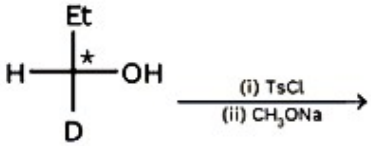


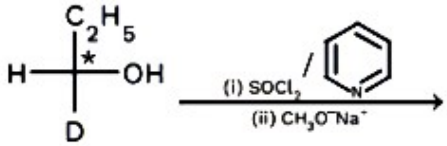
- This reaction is believed to proceed through choro sulphite intermediate in a concentrated mechanism.
- The Cl^- ion attacks from the front side due to which the net result is the retention of configuration.

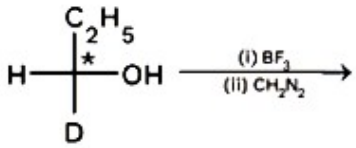


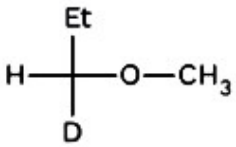
Q45 **In which of the following cases, the configuration of chiral C is retained in product?**

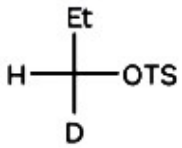
(A) 

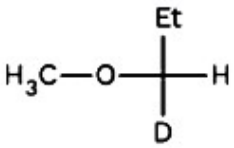
(B) 

(C) 

(D) 

A45 (A) 

(B) 

(C) 

Halide Exchange Reactions

1. Swartz Reaction (Metal fluoride with Alkyl Bromide 1°)

C–F bond formation

$$AgF + CH_3 - Br \xrightarrow[S_N2]{\Delta} CH_3 - F + AgBr$$

Or Hg_2F_2
Or CoF_3

Points to remember

The best way to prepare alkyl fluoride from an alkyl chloride or bromide is Heated in the presence of metallic fluoride, such as AgF, Hg_2F_2, CoF_3 or SbF_3 to give alkyl fluoride.

2. Finkelstein Reaction

Of preparative importance is the substitution of chloride or bromide with iodide.

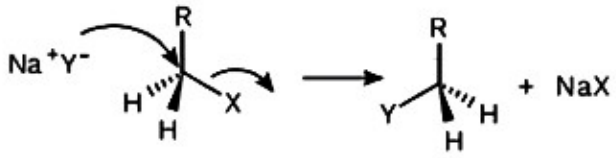
The more reactive alkyl iodides are better substrates for further transformations.

Alkyl iodides often are difficult to prepare directly, which is why the conversion of readily accessible chlorides or bromides via a Finkelstein reaction is often preferred.

$$R{-}X \underset{-X^-}{\overset{+Y^-}{\rightleftharpoons}} R{-}Y$$

Concept behind alkyl Iodide substitutions

For preparative use it is necessary to shift the equilibrium in favour of the desired product. This may for example be achieved by taking advantage of different solubilities of the reactants.



(a) With primary alkyl halides the Finkelstein reaction proceeds by a S_N2- mechanism. An alkali halide is used to deliver the nucleophilic halide anion.

(b) Sodium iodide is much more soluble in acetone than are sodium chloride or sodium bromide.

(c) Upon treatment of an alkyl chloride or bromide with sodium iodide in acetone, the newly formed sodium chloride or bromide precipitates from the solution and is thus removed from equilibrium.

(d) Alkyl iodides can be conveniently prepared in good yields by this route. Alkyl bromides are more reactive as the corresponding chlorides.

$$NaI + \overset{2p}{CH_3} - \overset{3p}{Cl} \underset{}{\overset{\text{acetone}}{\rightleftharpoons}} \overset{2p}{CH_3} - \overset{4p}{I} + \underset{\text{(Insoluble)}}{\overset{\oplus\ \ominus}{NaCl}}$$

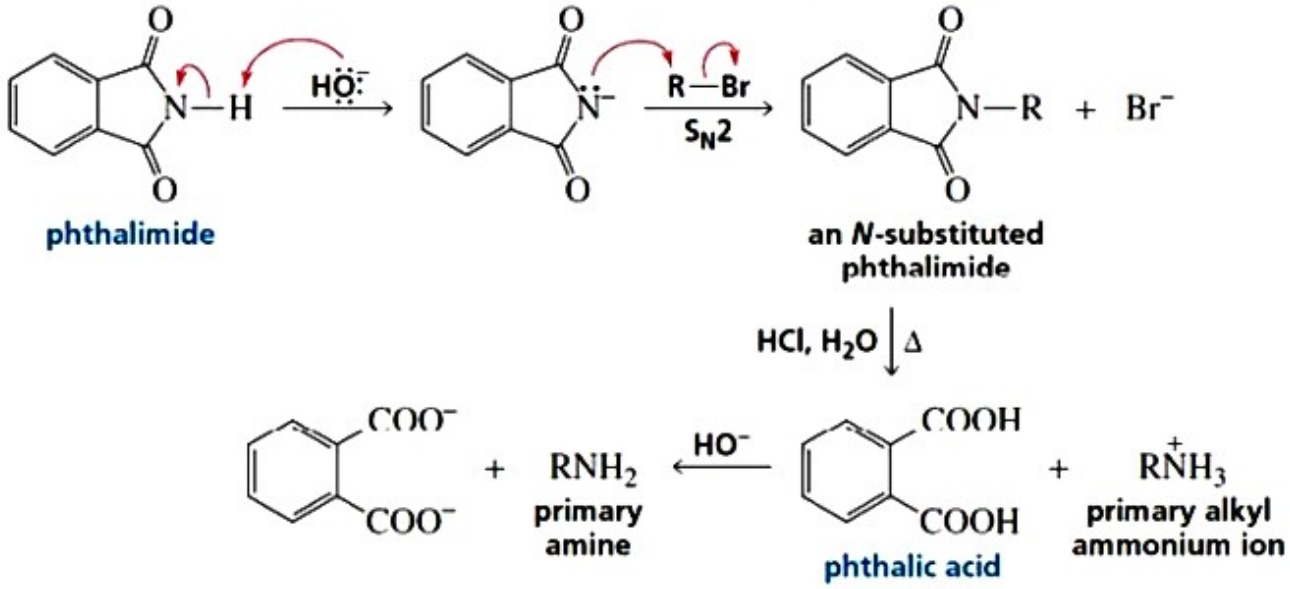
Hydrolysis of an Imide: The Gabriel Synthesis

An imide is a molecule with two acyl groups bonded to a nitrogen. The Gabriel synthesis, which converts alkyl halides into primary amines, involves the hydrolysis of an imide.

Points to remember

Le-Chatelier's Principles follow :

- Reactant must be in Excess (NaI)
- Product is Consumed in Equilibrium.



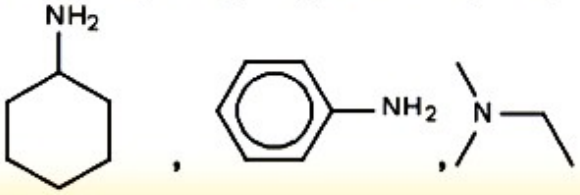
Only one alkyl group can be placed on the nitrogen because there is only one hydrogen bonded to the nitrogen of phthalimide. This means that the Gabriel synthesis can be used only for the preparation of primary amines.

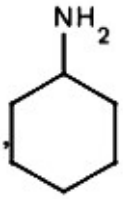
Gabriel synthesis of primary amines

$$RCH_2Br \xrightarrow[\substack{2.\ HCl, H_2O, \Delta \\ 3.\ HO^-}]{1.\ \text{phthalimide, } HO^-} RCH_2NH_2$$

Which of following will be given with Gabriel Pthalamide Reaction systhesis-

Q46 $CH_3 - CH_2 - NH_2$, $CH_2 = CH - NH_2$, $CH_3 - NH - CH_3$,



A46 $CH_3 - CH_2 - NH_2$, $CH_2 = CH - NH_2$,  only

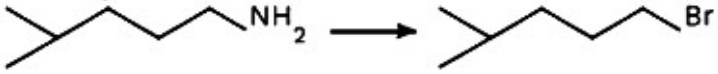
2 degree and 3 degree amine synthesis not possible

Q47 **What alkyl bromide would you use in a Gabriel synthesis to prepare each of the following amines?**

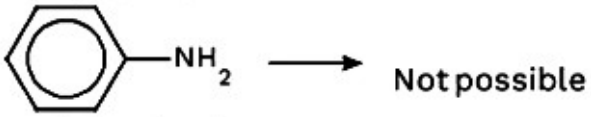
A47 (A) Pentylamine



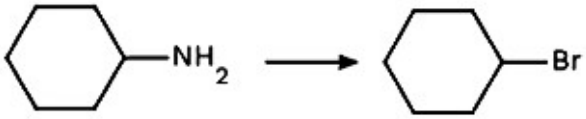
(B) Isohexylamine



(C) Benzylamine

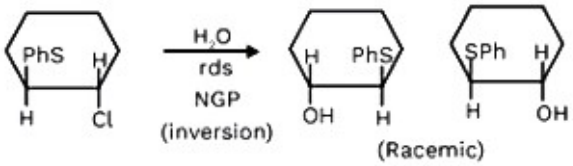


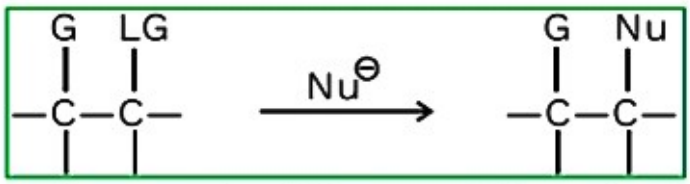
(D) Cyclohexylamine



Neighbouring Group Participation: NGP

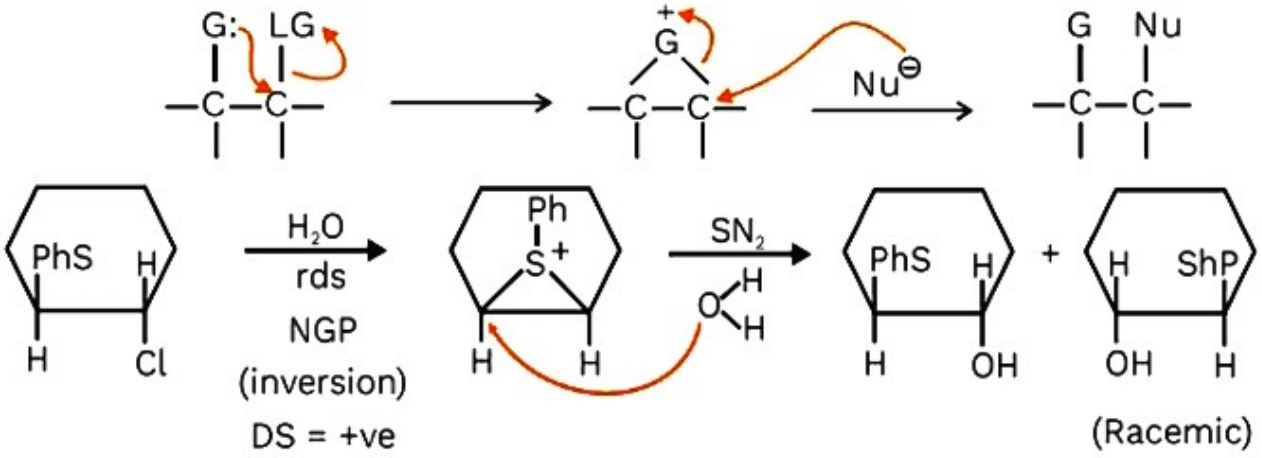
- The reaction is thus a combination of two S_N2 reactions.
- Double inversion is retention.





- In many substitution reactions, a nucleophilic group within a molecule, close to the reaction

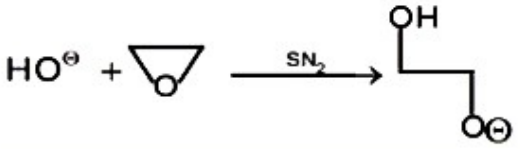
centre, participates temporarily in the reaction to control both rate and stereochemistry. Such involvement of a group within the same molecule is known as neighbouring group participation (NGP).



Points to remember

- **NGP > SN_1 > SN_2 {fast reaction}**

$HO^{\ominus} + R - Cl \xrightarrow{SN_2} HO - R + Cl^{\ominus}$ External leaving group



Internal leaving group

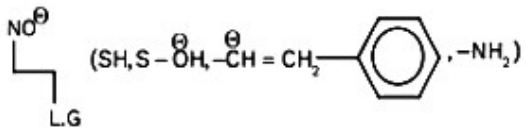
Characteristics :

(i) Rate $\propto$ concentration of halide

(ii) NGP is inversion step

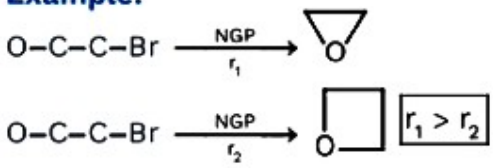
(iii) Order of reaction = 1

(iv) Anti [leaving group and neucleophile must be anti]



(v) NGP is fast reaction because of energy increase and formation of non-classical carbocation.

Example:

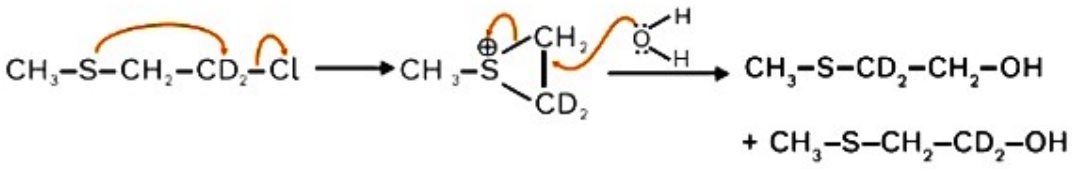


3 membered rings will be formed, more fast because of minimum entropy loss of conformer.

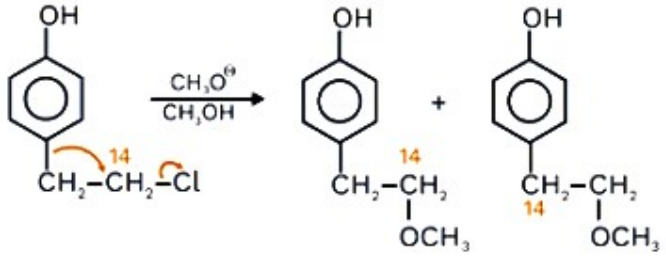
(vi) Conformation of non classical carbocation is rate determining step.

Q48 $CH_3–S–CH_2–CD_2–Cl \xrightarrow{H_2O} CH_3–S–CH_2–CD_2–OH + CH_3–S–CH_2–CD_2–CH_2–OH$

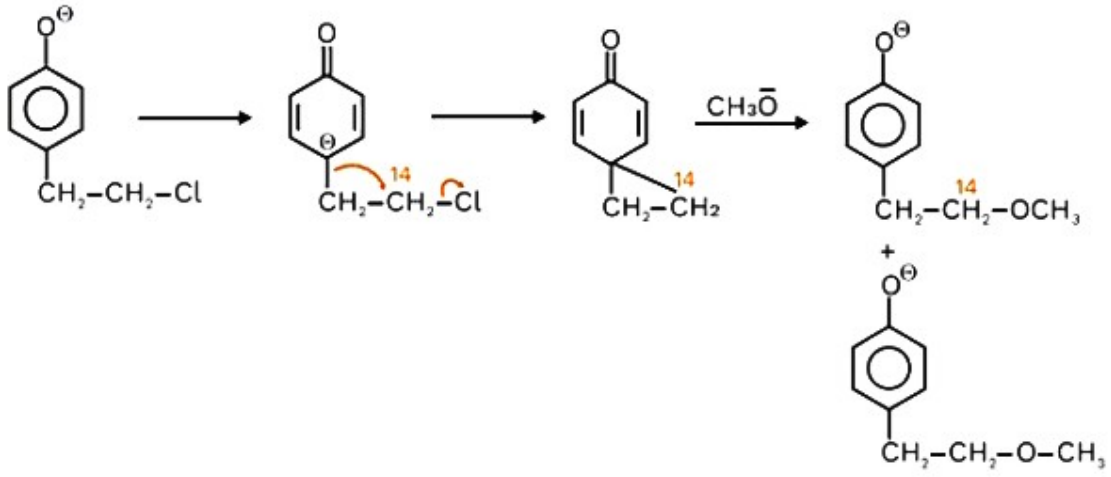
A48



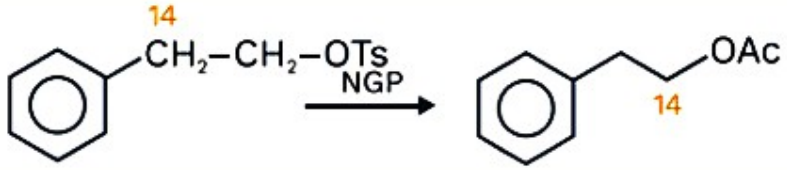
Q49



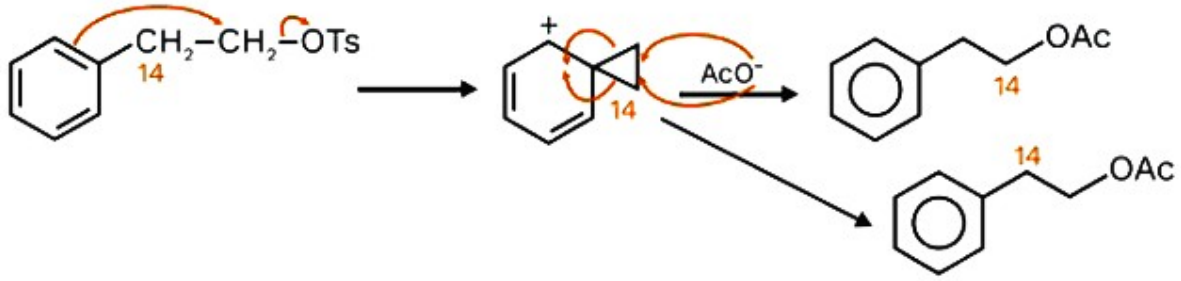
A49



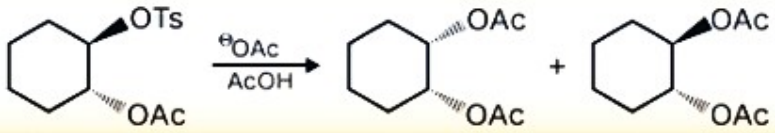
Q50



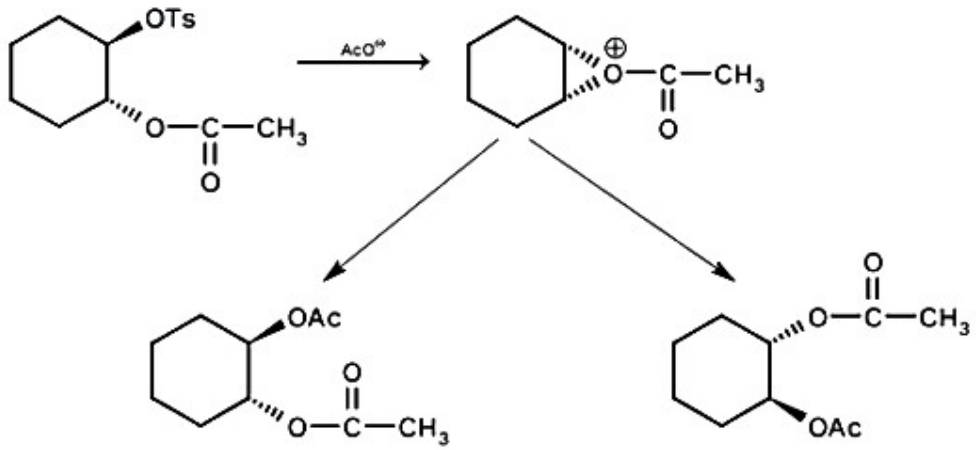
A50



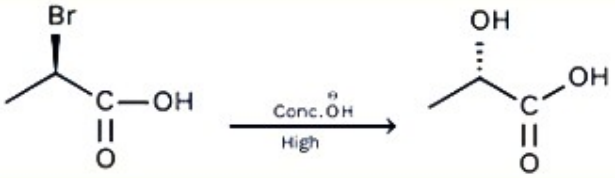
Q51



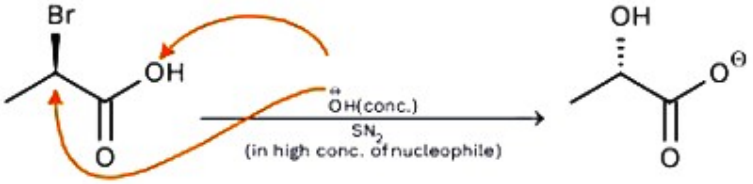
A51



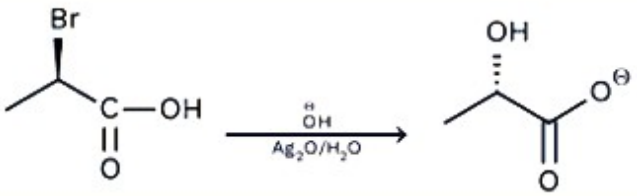
Q52



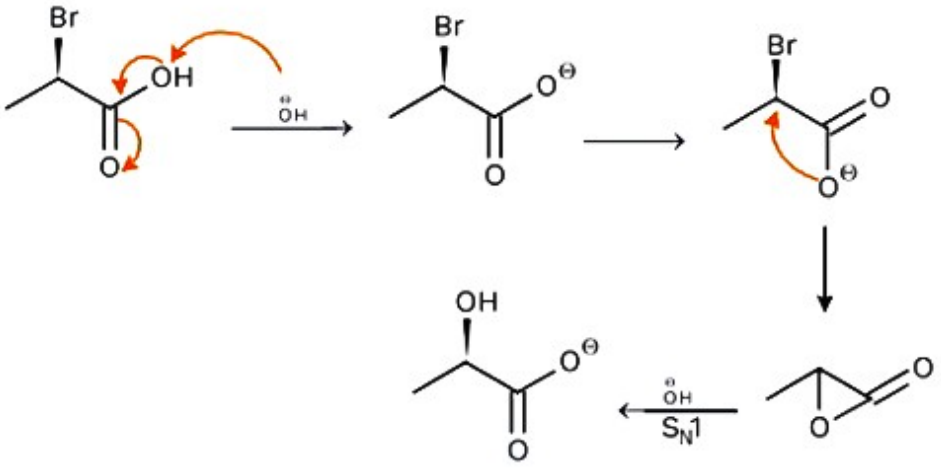
A52



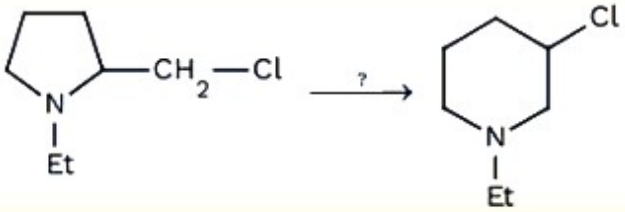
Q53



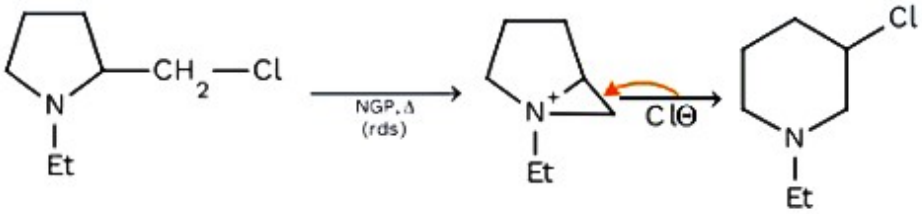
A53



Q54

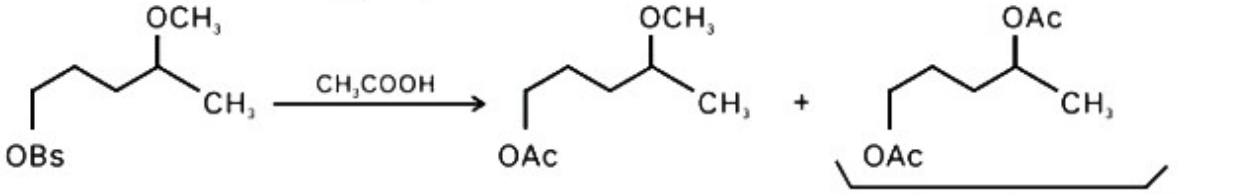


A54



Alkoxy group as a neighbouring group

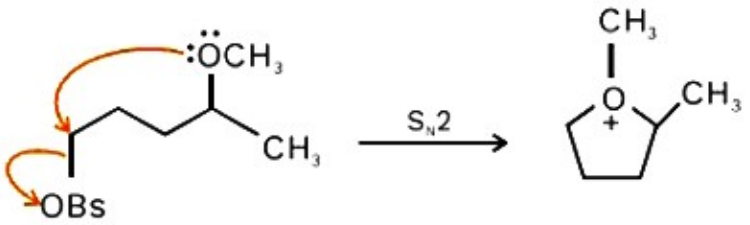
–OR is a neighbouring group.



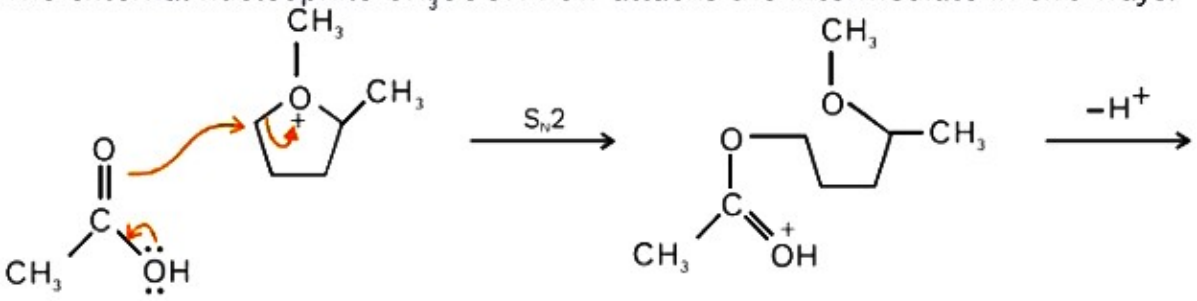
where does this come from?

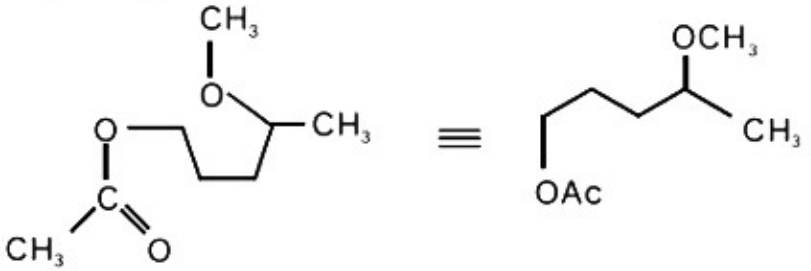
OBs = Brosylate anion

The $-OCH_3$ group acts as neighbouring group to remove the leaving group in a S_N2 reaction.

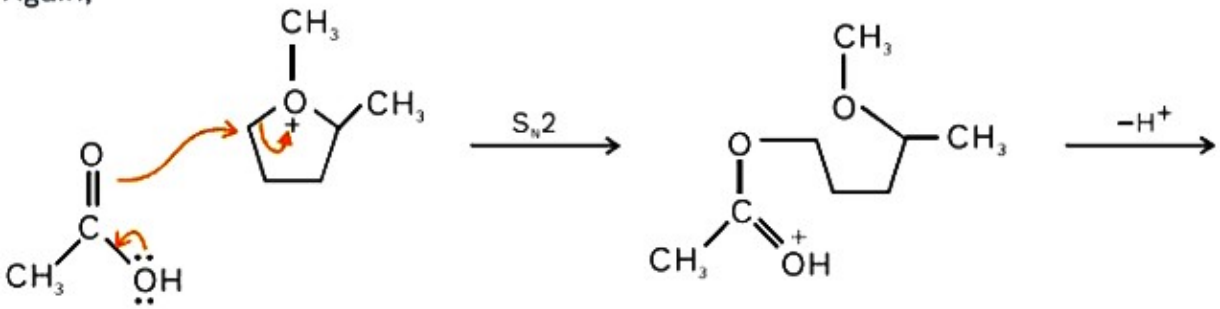


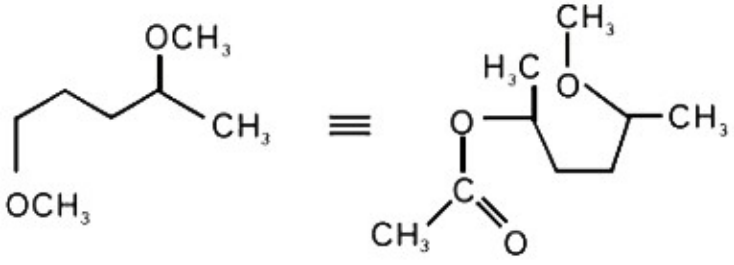
The external nucleophile CH_3COOH now attacks the intermediate in two ways.





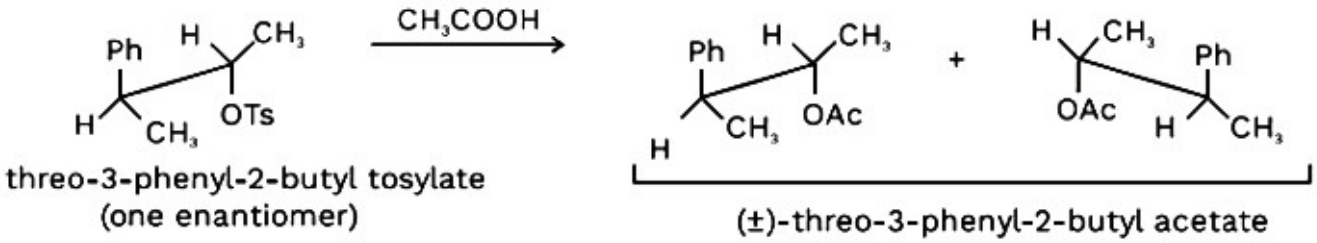
Again,





Aromatic ring as a neighbouring group

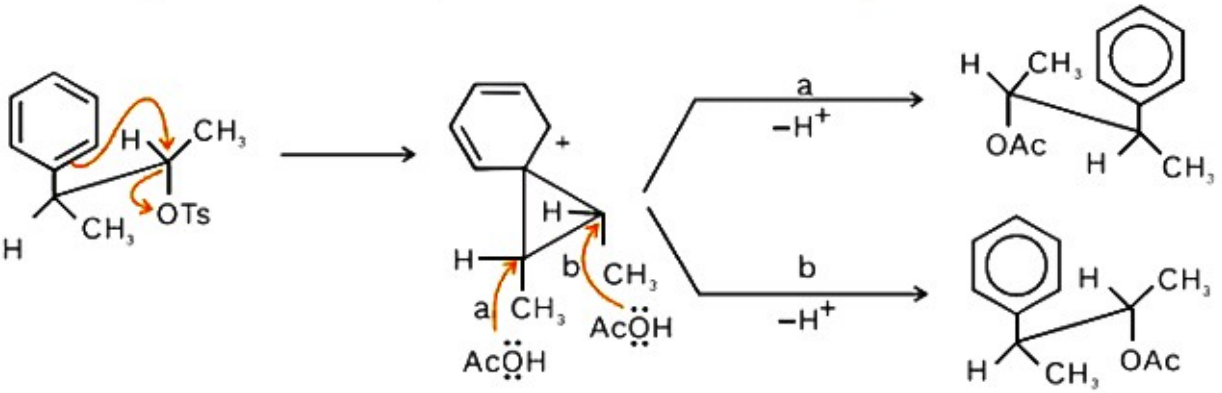
Optically active threo-3-phenyl-2-butyl tosylate gives a racemic mixture of corresponding acetate on acetolysis.



threo-3-phenyl-2-butyl tosylate (one enantiomer)

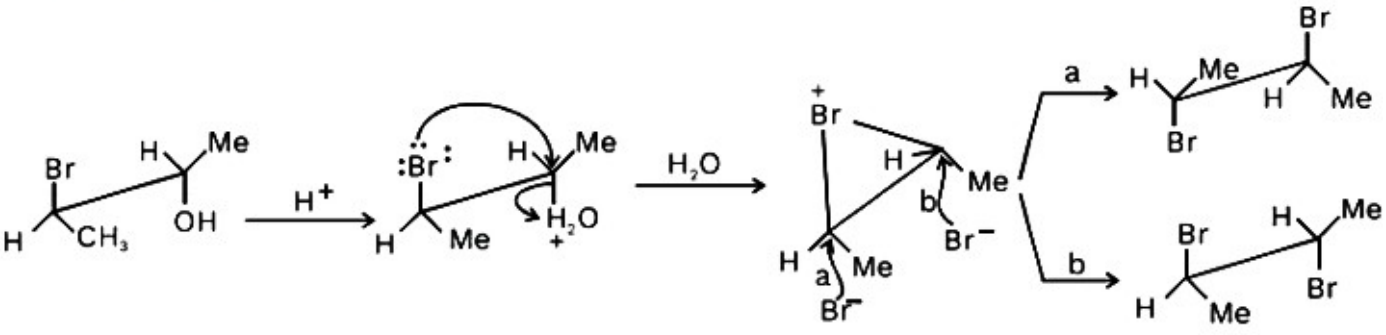
(±)-threo-3-phenyl-2-butyl acetate

The aromatic ring as a neighbouring group to remove the tosylate in a S_N2 reaction. Acetic acid finally breaks down the phenonium ion in another S_N2 reaction.

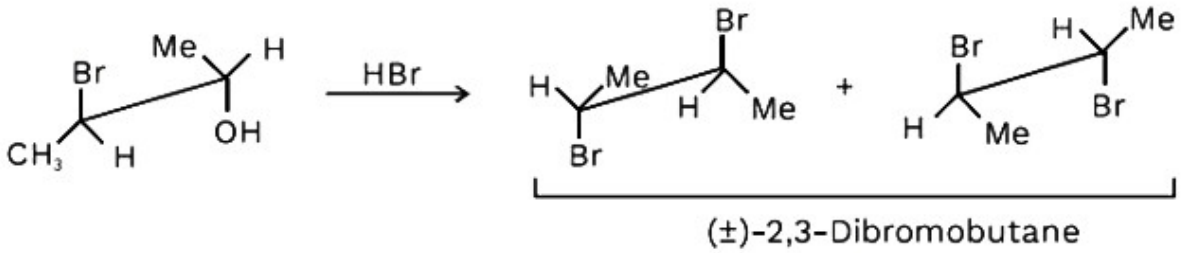


Halogen atom as a neighbouring group

Active threo-3-bromo-2-butanol upon treatment with HBr gives rise to (±)-2,3-dibromobutane. Had it been a pure SN2, meso product would have been obtained. Again, if it were a pure SN1, a mixture of meso and active forms would be the product. Formation of only (±)-2,3-dibromobutane can be explained with the help of bromine atom acting as neighbouring groups.

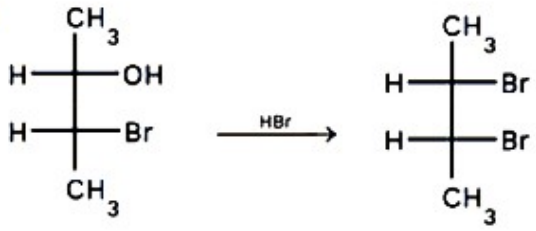


The other enantiomer of threo-3-bromo-2-butanol would also give the same (±)-2,3-dibromobutane.

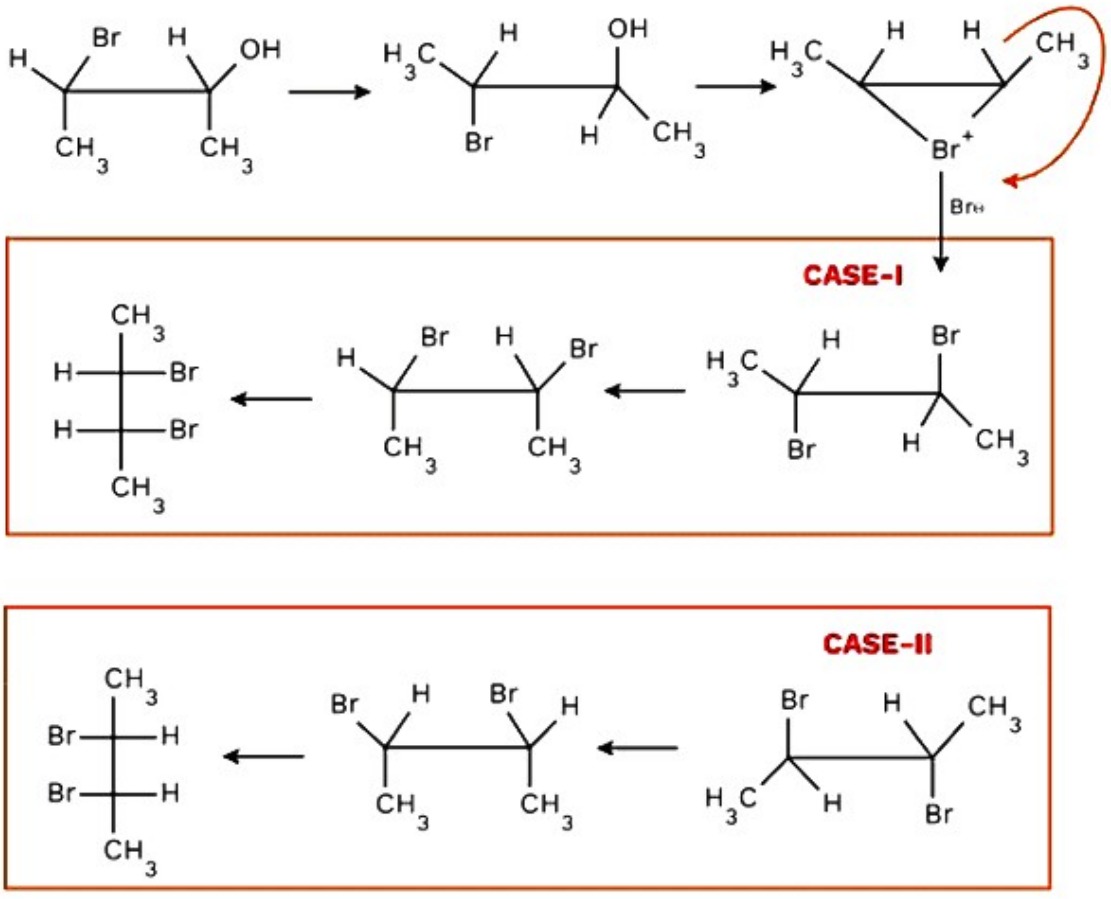


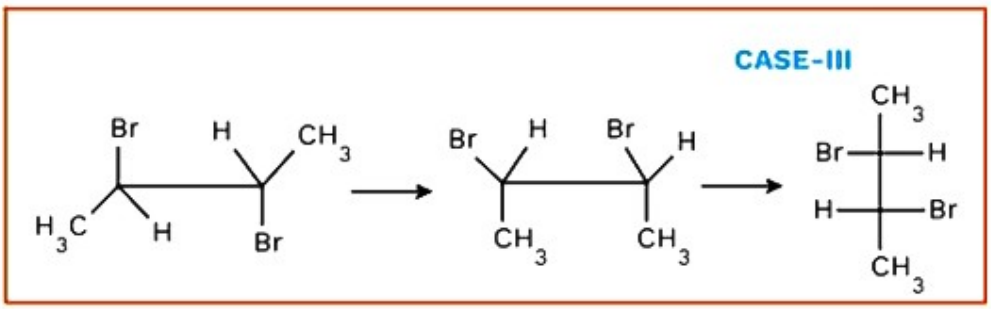
(±)-2,3-Dibromobutane

Q55



A55





Substitution by hydroxyl group

When treated with aqueous potash or silver oxide suspended in boiling water, alkyl halides are hydrolysed to form alcohols. Here the halogen is substituted by the nucleophile –OH.

$CH_3 - CH_2 - I$ + aq. KOH → $CH_3 - CH_2 - OH$ + KI

ethyl iodide ethyl alcohol

$CH_3 - Br + AgOH \rightarrow CH_3 - OH + AgBr$

methyl bromide methyl alcohol

Substitution by alkynyl group (R—C≡C⁻)

Alkyeact with sodium acetylides to form higher alkynes. Thus,

$R - C \equiv C^- Na^+ + CH_3 - Br \longrightarrow R - C \equiv C - CH_3 + NaBr$

Sodium alkynide Methylalkyne

Example :

$H - C \equiv C^- Na^+ + CH_3 - Br \longrightarrow H - C \equiv C - CH_3 + NaBr$

Sodium acetylide Propyne

This reaction occurs by S_N2 mechanism with primary alkyl halides and by S_N1 mechanism with tertiary alkyl halides.

Substitution by thiol group (SH):

Alkyeact with ethanolic potassium hydrogen sulphide to form thioalcohols. Thus,

$K^+ \overset{\ominus}{S}H + CH_3 - CH_2 - I \longrightarrow CH_3 - CH_2 - SH + KI$

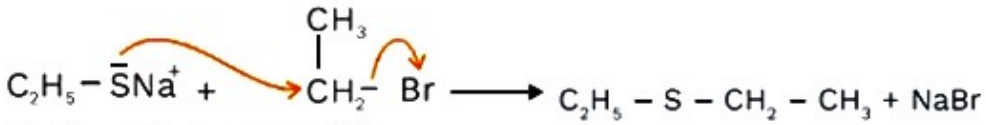
Ethanethiol

Substitution by thioether group (—S—):

Thioethers are produced when alkyeact with potassium sulphide or hot ethanolic solution of a mercaptide.

$2C_2H_5 - I + K_2S \rightarrow C_2H_5 - S - C_2H_5 + 2KI$

diethyl thioether



Sodium ethyl mercaptide

Hoffmann Ammonolysis

Alkyl halides when treated with ethanolic ammonia under pressure give primary amines.

$$C_2H_5 - Br + :NH_3 \longrightarrow C_2H_5 - \overset{+}{N}H_3 \xrightarrow{-H^+} C_2H_5 - NH_2 + HBr$$

If excess of alkyl halide is introduced, the reaction further form secondary amine, **tertiary** amine and finally tetra alkyl ammonium halide (quaternary salt).

$C_2H_5Br + C_2H_5NH_2 \longrightarrow (C_2H_5)_2NH + HBr$

(second molecule)diethyl amine

$C_2H_5Br + (C_2H_5)NH \longrightarrow (C_2H_5)_3N + HBr$

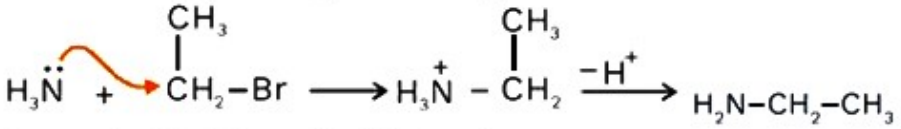
(third molecule)triethylamine

$C_2H_5Br + (C_2H_5)_3N \longrightarrow (C_2H_5)_4\overset{+}{N}\overset{-}{Br}$

(second molecule)tetraethylammonium bromide (quaternary salt)

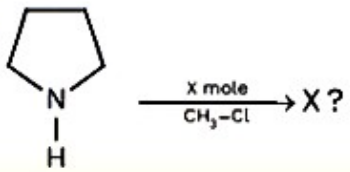
Primary alkyl halides give a good yield of amines. Secondary (except isopropyl halide) and tertiary halides preferentially undergo elimination to form alkenes.

This reaction probably follows S_N2 mechanism.

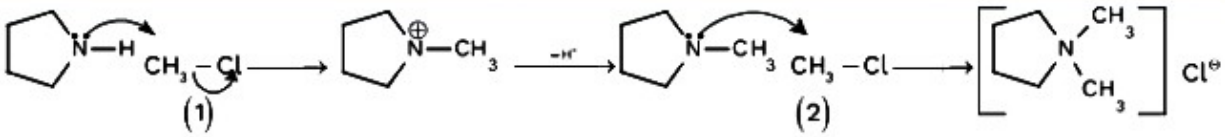


AmmoniaEthyl bromideEthylamine

Q56

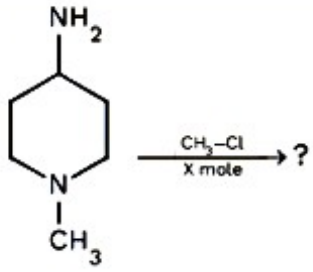


A56

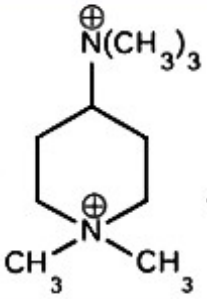


$\Rightarrow$ x = 2 moles

Q57

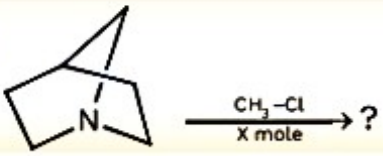


A57 X = 4 moles. As final product will be



.2Cl$^{\ominus}$

Q58

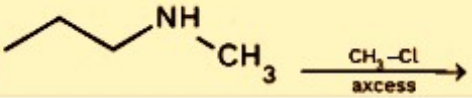


A58 X = 1 mole

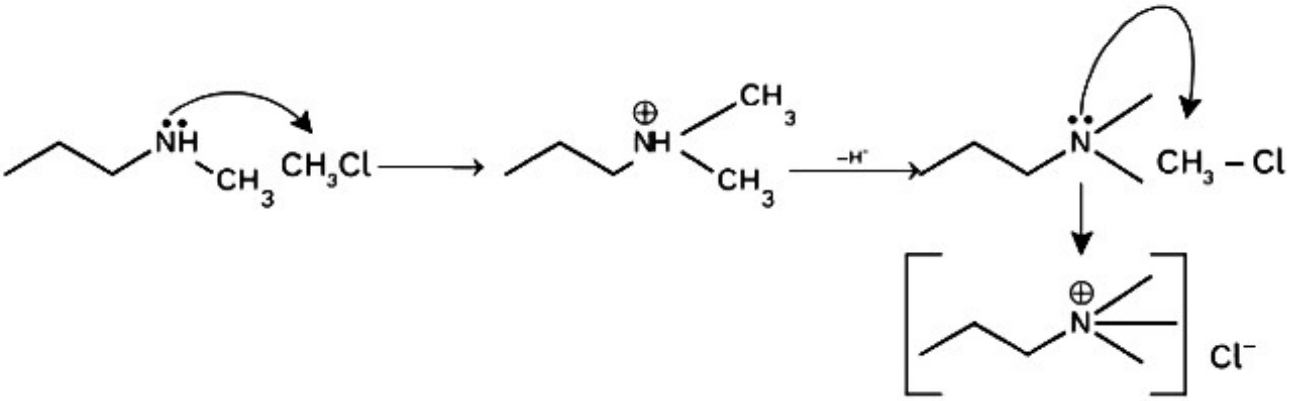


Cl (Final product)

Q59

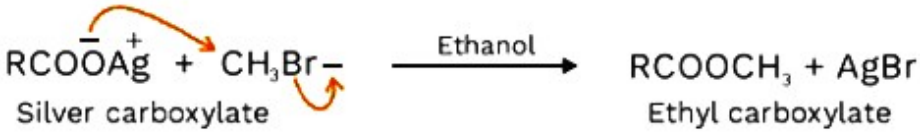


A59



Substitution by an ester (RCOO—).

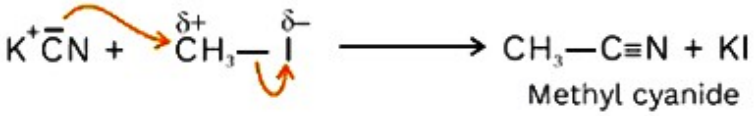
When heated with ethanolic solution of silver salt of carboxylic acids, alkyl halides yield esters.





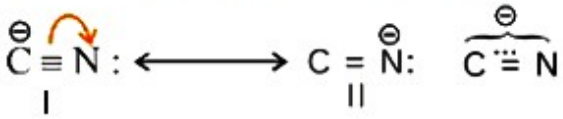
Substitution by a Cyano or Isocyano groups:

With ethanolic potassium cyanide, an alkyl halide gives corresponding alkyl cyanide (or nitrile) as the chief product along with a small amount of alkyl isocyanide. This involves the nucleophilic attack of the cyanide ion on the alyl halide.

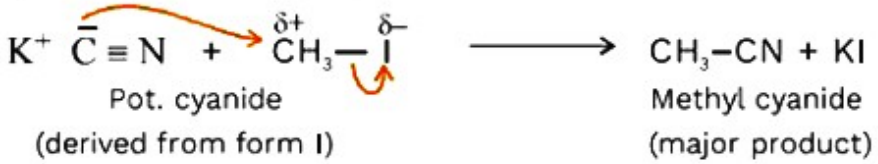


The reaction proceeds either by S_N1 or S_N2 mechanism.

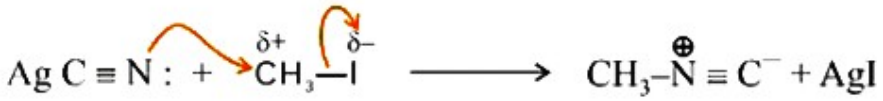
Cyanide is an amibident ion has two nuclophilic sites as shown below:



Therefore, cyanides can attack the positive carbon of the alkyl group either through carbon (form I) or nitrogen (form II) to form alkyl cyanides or alkyl isocyanides respectively. Thus, with alkali metal cyanides (KCN) it is the form I that reacts with alkyl halides to give alkyl cyanides as the major product.

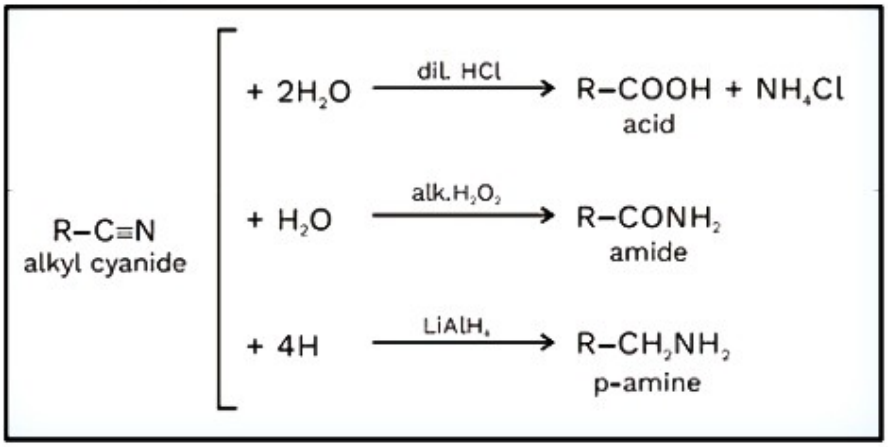


Unlike alkli metal cyanides, which are ionic, silver cyanide is predominantly covalent and hence reacts with the alkyl halides through form II to produce alkyl isocyanides as the major product:



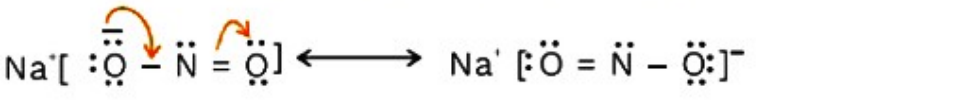
Silver cyanide (derived from form II) — Methyl isocyanide (major product)

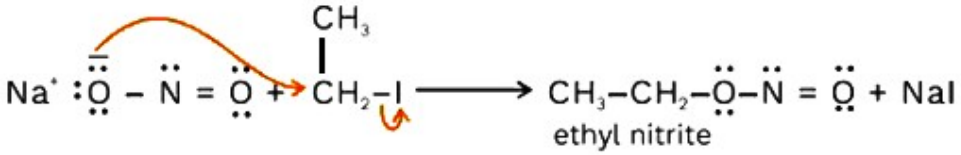
Alkyl cyanides are very useful synthetic reagents since they can be conveniently converted into acids, amides and amines.



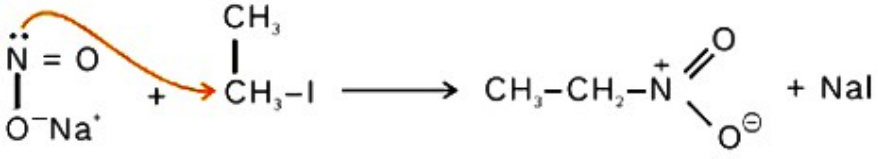
Substitution by a nitrite or a nitro groups.

The nitrite ion, like the cyanide ion, is an ambident ion and its nucleophilic attack on an alkyl halide can yield either an alkyl nitrite or a nitroalkane according as the attack on alkyl halides takes place through oxygen or nitrogen.





ethyl nitrite



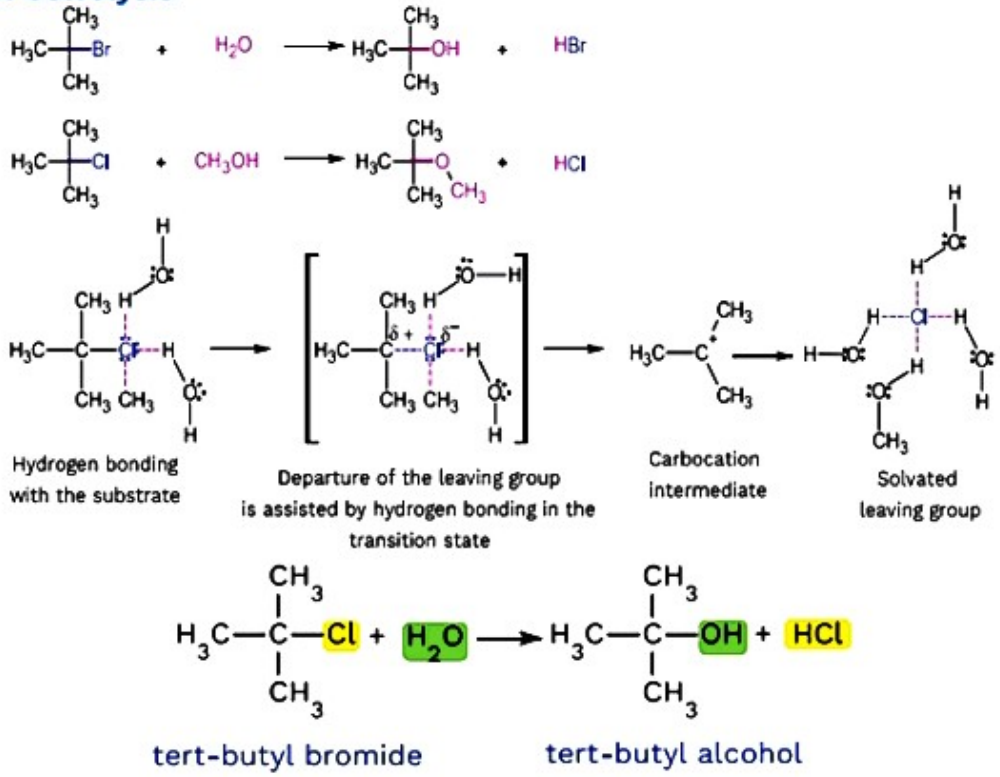
Thus with $\overset{+}{N}aN\overset{-}{O}_2$ or $\overset{+}{K}N\overset{-}{O}_2$ alkyl nitrites are obtained while with $AgNO_2$ nitroalkaknes are produced.

Solvolysis

A solvolysis reaction is a nucleophilic replacement in which the nucleophile is a molecule of the solvent (solvent + lysis: cleavage by the solvent).

1. The S_N1 reaction of an alkyl halide with water is an example of solvolysis. When the solvent is water, we could also call the reaction a hydrolysis.
2. If the reaction had taken place in methanol, we would call it a methanolysis.

Examples of Solvolysis

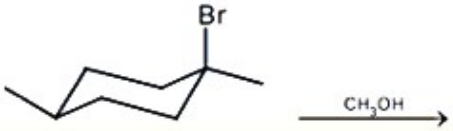


tert-butyl bromide tert-butyl alcohol

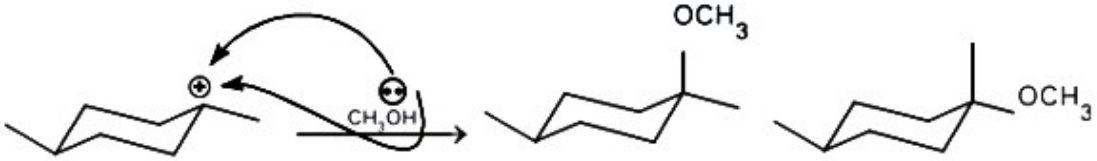
rate = k[alkyl halide]

The rate of the reaction depends on the concentration of only one reactant, the reaction is a first-order reaction.

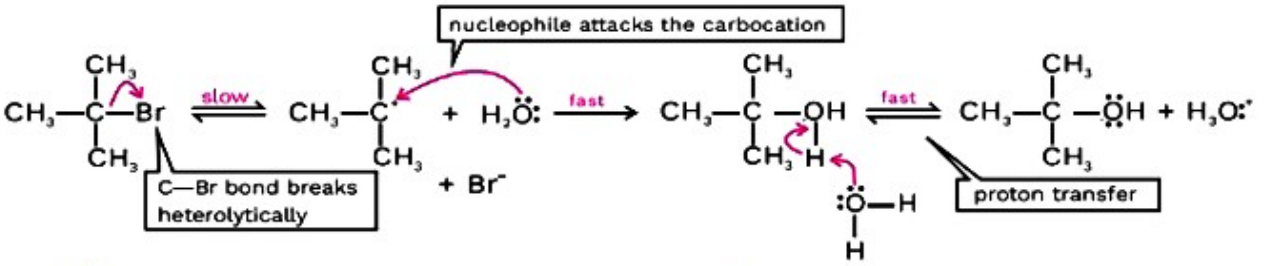
Q60 **What product(s) would you expect from the following solvolysis?**

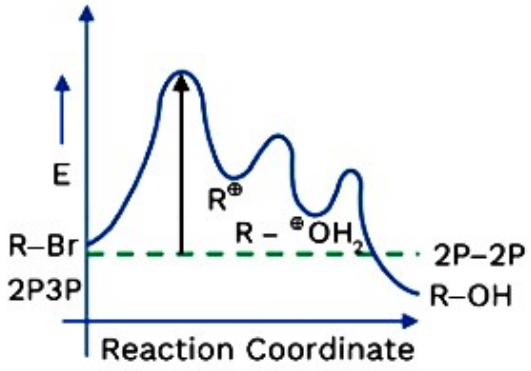


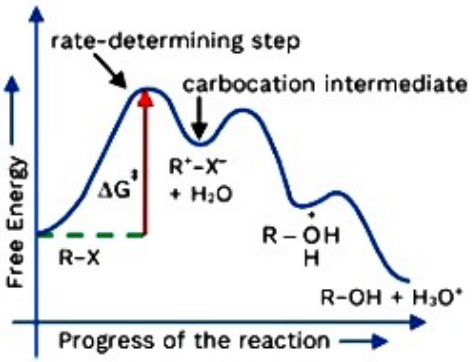
A60



Mechanism of the S_N1 reaction







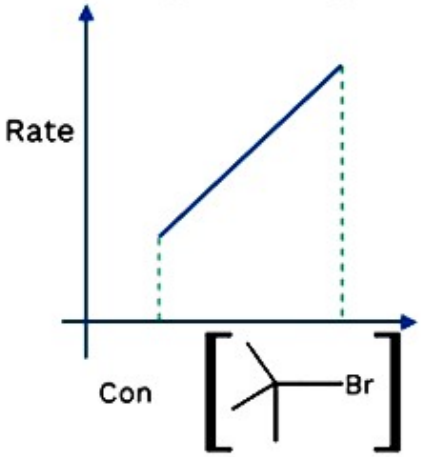
In SN^1 Reaction rate depends only on substrate not upon nucleophile, that's why it is unimolecular reaction

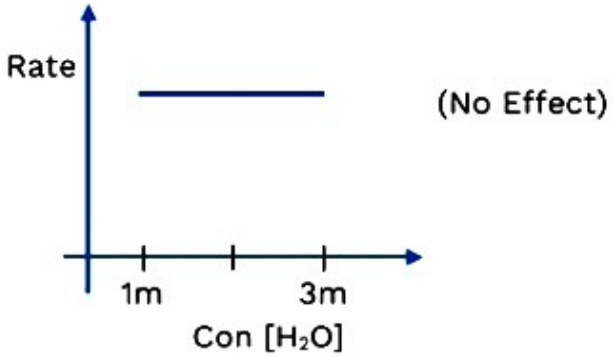
Rate α [Br] [H_2O]

Excess Pseudo

1st Order Reaction

Rate α [Br]



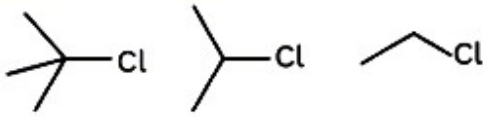


Points to remember

1st step is slowest in which carbocation is formed, reversible reaction.
In SN^1 reaction carbocation formation is R.D.S, it means more stable carbocation, more will be rate of reaction.

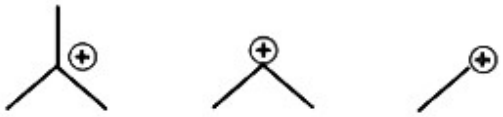
Identify the rate of reaction.

Q61



(a) **(b)** **(c)**

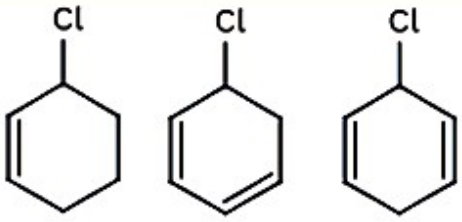
A61 (a > b > c)



9α-H (a) 6α-H (b) 3α-H (C)

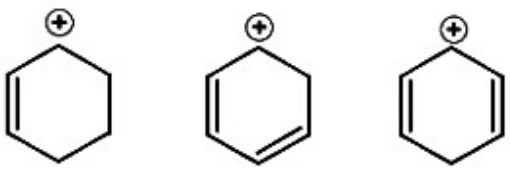
→ Rate α stability of carbocation

Q62



(a) **(b)** **(c)**

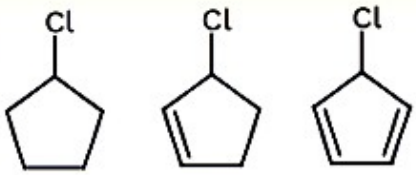
A62 (b > c > a)



Linear conjugation Cross conjugation

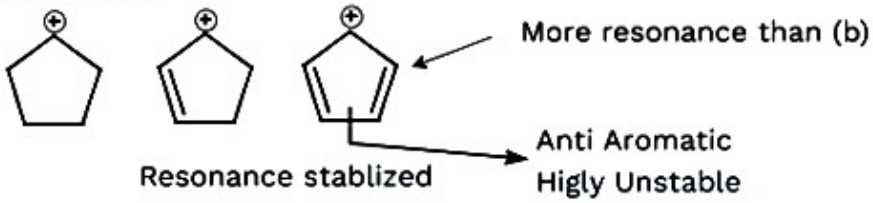
→ Linear conjugation provides more stability than cross conjugation

Q63

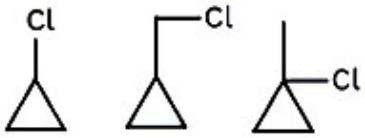


(a) **(b)** **(c)**

A63 (b > a > c)

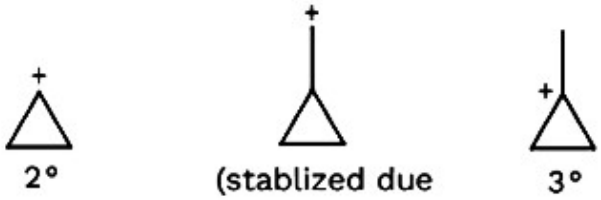


Q64



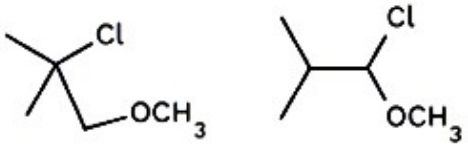
(a) **(b)** **(c)**

A64 (b > c > a)



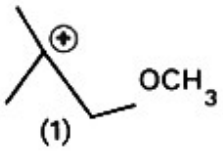
2° (stablized due to dancing resonance) 3°

Q65

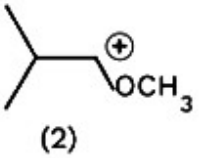


(a) **(b)**

A65 (b > a)



(1)

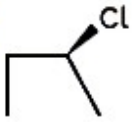


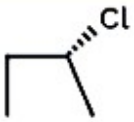
(2)

(8α-H)

(Resonance stabilized)

Q66





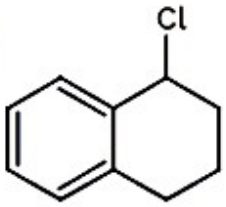
(a) **(b)**

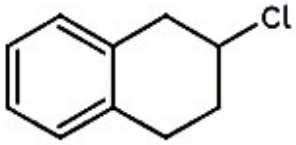
A66 (a = b)



→ Both will form same carbocation hence, rate will be equal.

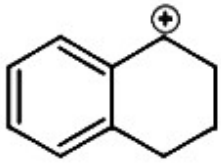
Q67

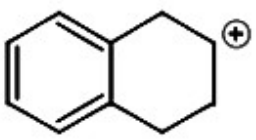




(a) **(b)**

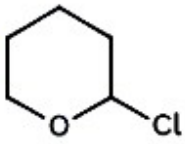
A67 (a > b)

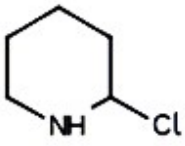


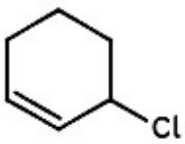


(Resonance stabilized carbocation) (2° carbocation)

Q68

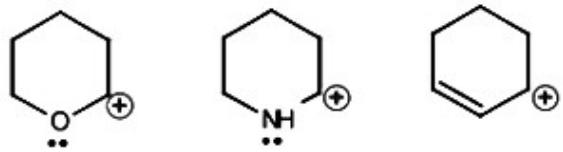






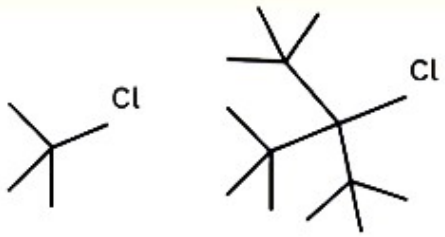
(a) **(b)** **(c)**

A68 (b > a > c)



(+m) (+m) (Resonance)
(Back bonding)
→ N-donors have more +m effect than)-donars.

Q69



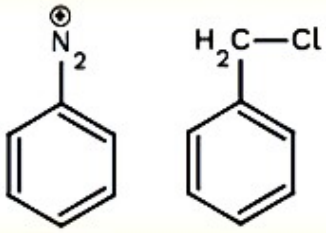
(a) **(b)**

A69 (b >a)



9α-H oα-H
More stable
[shift b > a due to faster formation of carbocation].

Q70



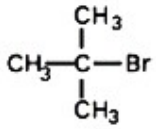
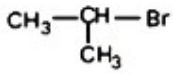
A70 (B>A) [rate of reaction].

Polar Protic Solvents Favour S_N1 Reactions

A protic solvent has at least one hydrogen atom capable of participating in a Hydrogen bond. Protic solvents, such as water, EtOH, and MeOH, facilitate formation of a carbocation by forming hydrogen bonds with the leaving group as it departs, thereby lowering the energy of the transition state leading to a carbocation.

Relative reactivities of alkyl halides in an SN1 reaction

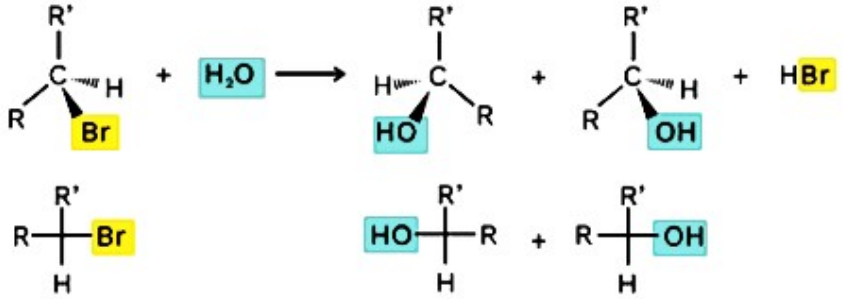
most reactive 3° alkyl halide > 2° alkyl halide > 1° alkyl halide **least reactive**

Alkyl bromide	Class of alkyl bromide	Relative rate
	tertiary	1,200,000
	secondary	11.6
CH_3-CH_2-Br	primary	1.0*
CH_3-Br	methyl	1.05*

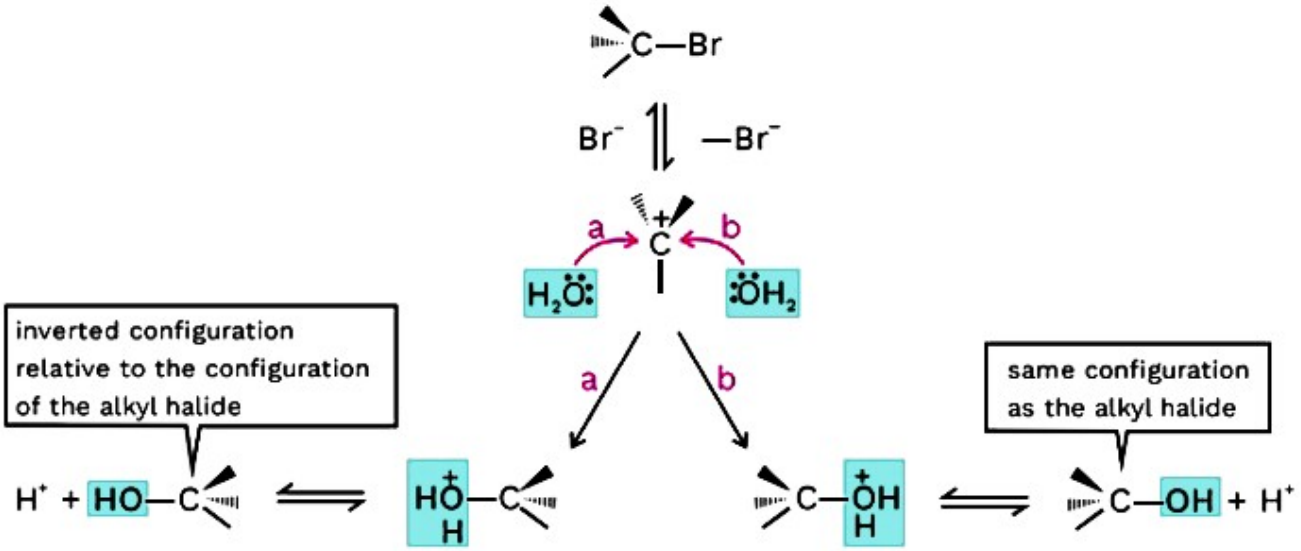
An S_N1 reaction of an alkyl halide in which the leaving group is attached to asymmetric carbon forms two stereoisomers because attack of the nucleophile on one side of the planar carbocation forms one stereoisomer and attack on the other side produces the other stereoisomer.

Points to remember

Although the rate of the S_N1 reaction of this compound with water is 0, a small rate is observed as a result of an S_N2 reaction.



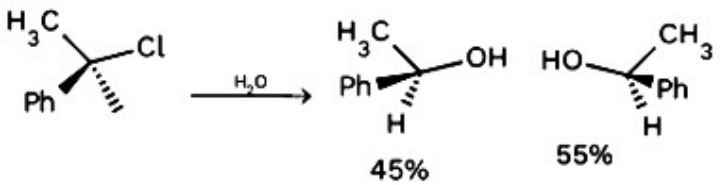
if the leaving group in an S_N1 reaction is attached to an asymmetric carbon, a pair of enantiomers will be formed as products



If the nucleophile attacks the side of the carbon from which the leaving group departed, the product will have the same relative configuration as that of the reacting alkyl halide. If, however, the nucleophile attacks the opposite side of the carbon, the product will have the inverted configuration relative to the configuration of the alkyl halide

If the leaving group in a S_N1 reaction is attached to an asymmetric carbon, a pair of enantiomers will be formed as products

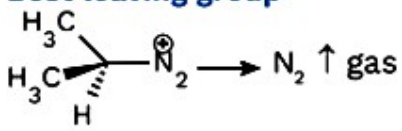
In S_N1 reaction partial racemization and slightly inversion takes place.



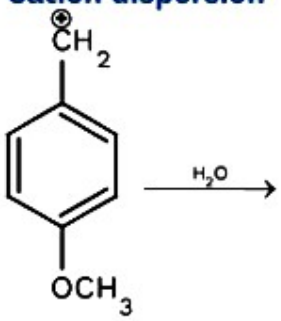
90% Racemization & 10% inversion

Perfect Racemisation

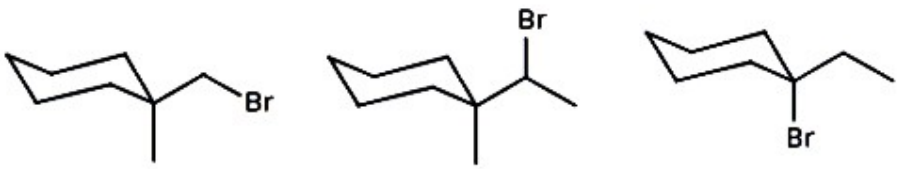
(A) Best leaving group



(B) Cation dispersion



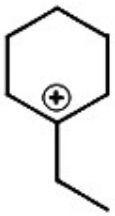
Q71 **Which of the following alkyl halides is most likely to undergo substitution by an S_N1 mechanism?**



(a) (b) (c)

A71 (c)

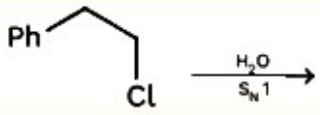
As it forms 3° carbocation



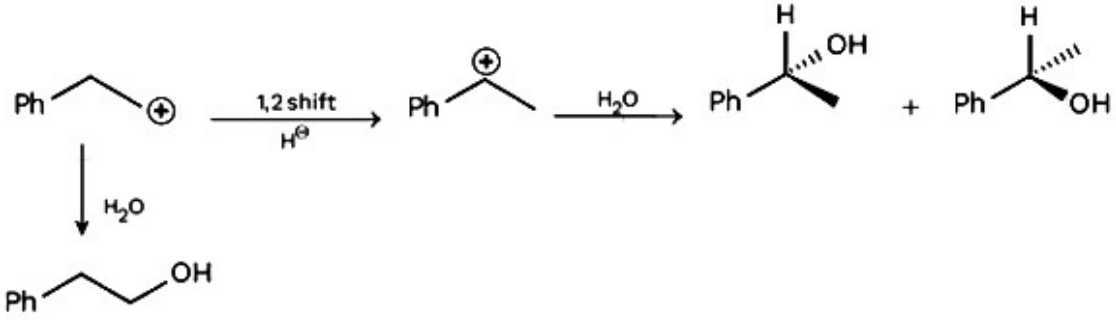
More stable carbocation than other 2.

Number of products formed in solvolysis reaction

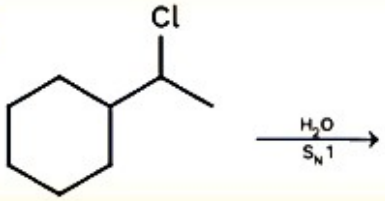
Q72



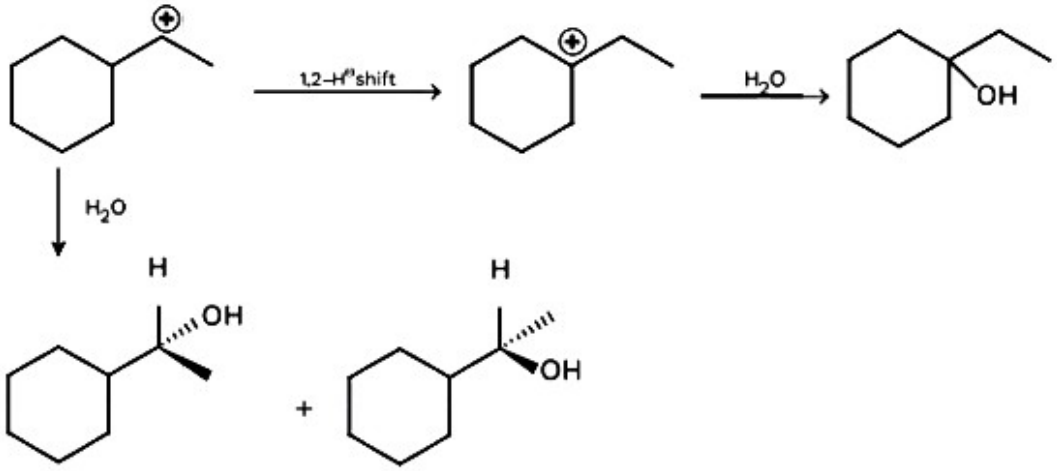
A72 (3)



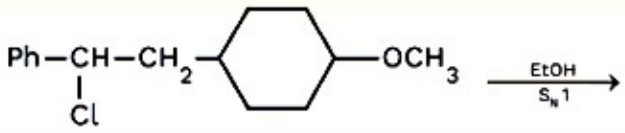
Q73



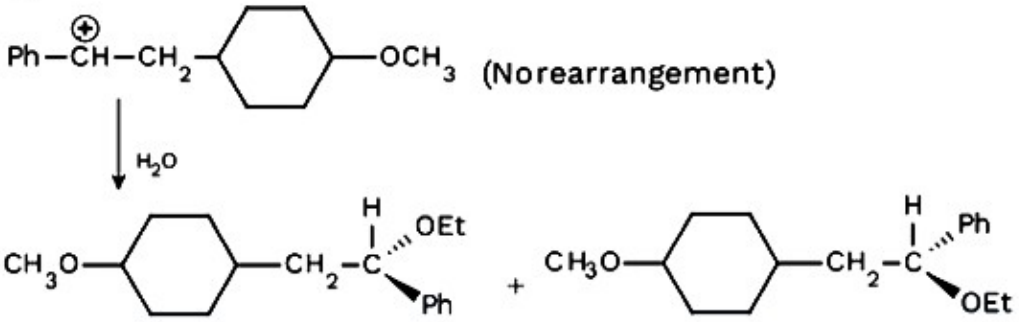
A73 (3)



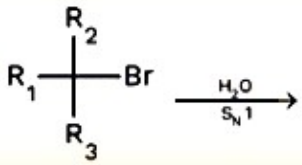
Q74



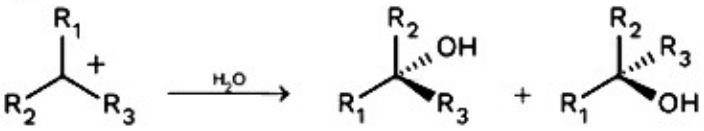
A74 (2)



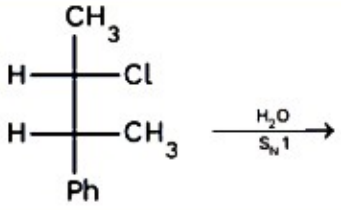
Q75



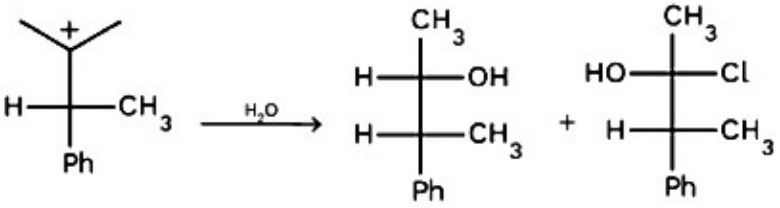
A75 (2)



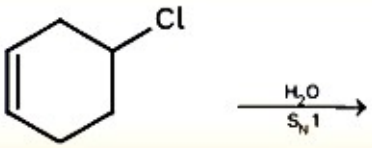
Q76



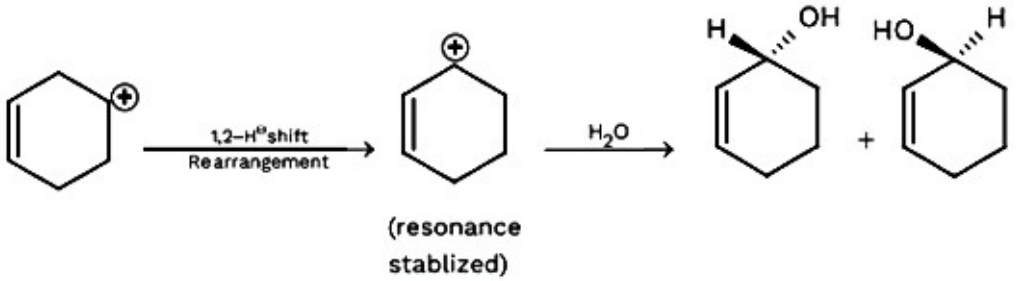
A76 (2)



Q77

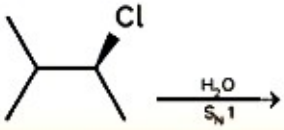


A77 (4)

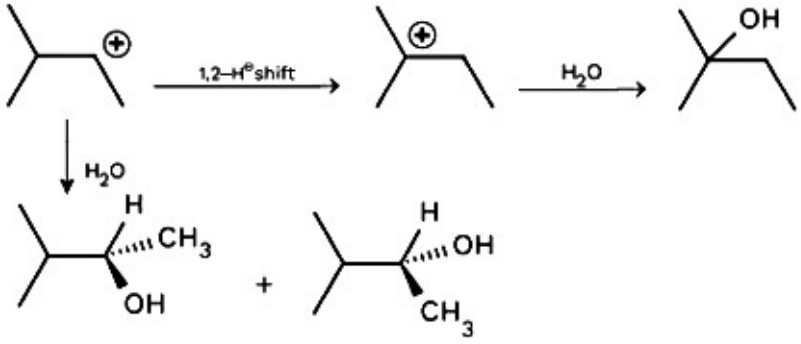


(resonance stablized)

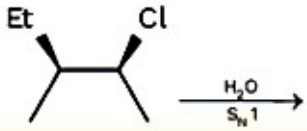
Q78



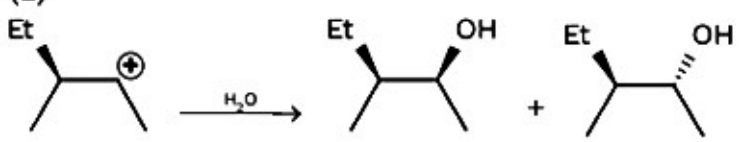
A78 (3)



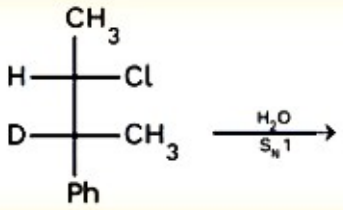
Q79



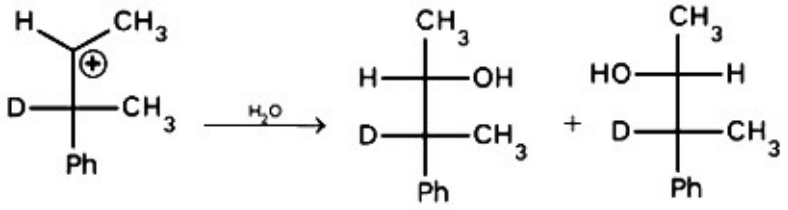
A79 (2)



Q80

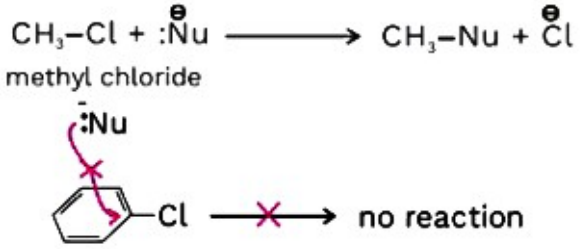


A80 (2)



Nucleophilic substitution of Aromatic halogen:

Halogens attached to an aromatic ring do not ordinarily undergo displacement (S_N1 or S_N2). Thus, unlike alkyl halides, aryl halides do not react with nucleophilic reagents under normal laboratory conditions.



The low reactivity of aromatic halogen to nucleophiles is attributed to the facts:

(1) The halogen has a positive charge on it.

(2) Aryl halides could be represented as resonance hybrid whereby C–X bond in them possesses a double bond character and is stronger than C–X bond in alkyl halides.



For these reasons, the substitution of halogen in aryl halides by S_N2 mechanism is significantly more tough than with alkyl halides.

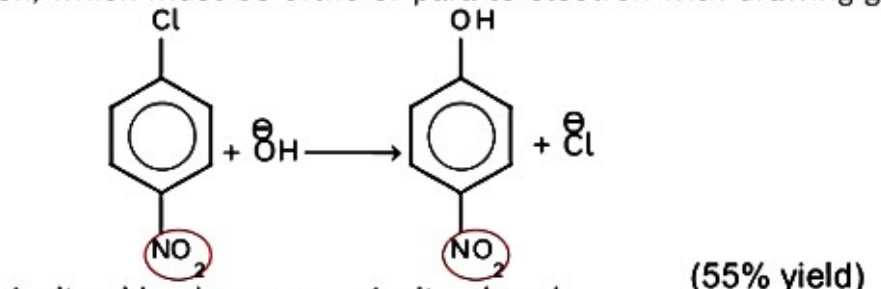
S_N2 reactions are not likely because the usual backside approach of the nucleophile required for such a reaction is blocked by the aromatic ring in aryl halides.

Plain aryl halides are ordinarily inert to most nucleophiles e.g., hydroxide, cyanide ion, or with ammonia

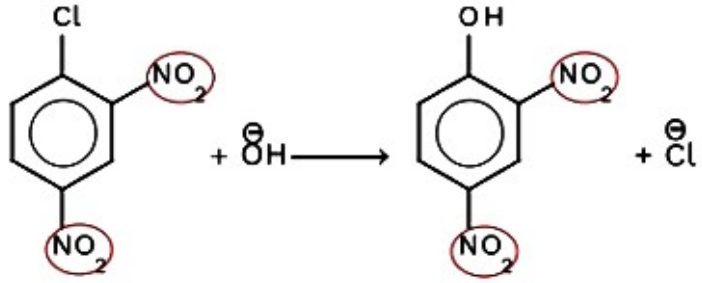
Aryl halides experience nucleophilic displacement with increasing readiness if a strongly electron attracting group (like NO_2) is present in the 2-position (orthro) or in 4-position (para). Hence the nucleophilic substitution with ortho or para nitro chlorobenzene is facilitated.

Condition for SN-Ar:

NO_2 and halogen, which must be ortho or para to electron with drawing group.

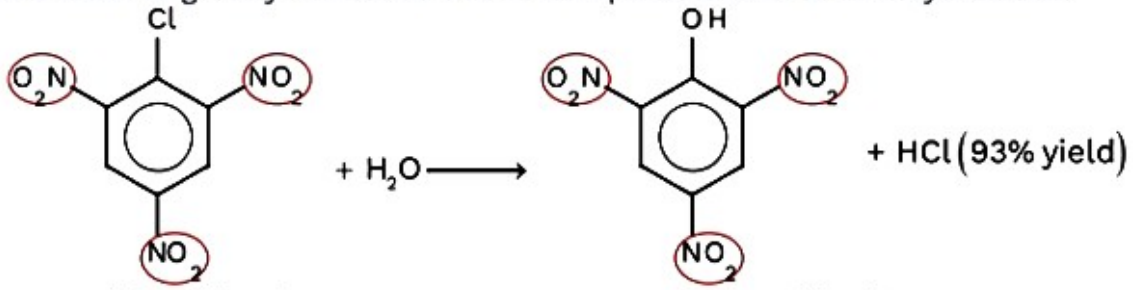


4-nitrochlorobenzene 4-nitrophenol (55% yield)



2,4-dinitrochlorobenzene 2,4-dinitrophenol (76% yield)

When three nitro groups are located in 2,4 and 6-positions to a halogen, the reactivity of the latter is greatly increased and is comparable to that of alkyl halides.

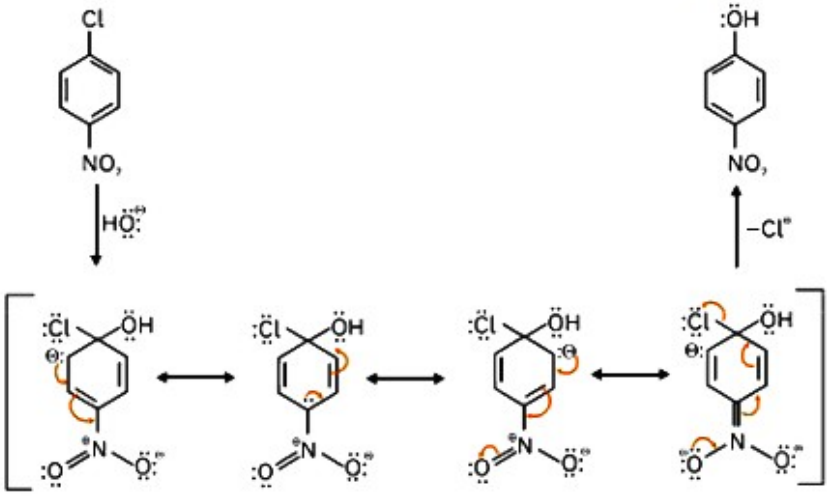


2,4,6-trinitrochlorobenzene (picry chloride) 2,4,6-trinitrochlorobenzene (picry acid)

TWO STEPS INVOLVED-

In the first step, aromatic ring is attacked by a nucleophile, forming the intermediate **Meisenheimer complex** which is an anion with highly delocalized charge.

In the second step, of SNAr mechanism involves loss of a leaving group to restore geometry.



Meisenheimer Complex

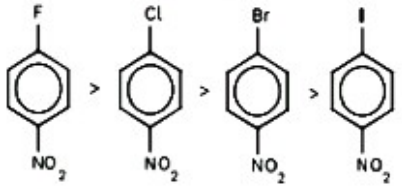
Characteristics:

(i) Nucleophilic substitution reaction by addition elimination step.

(ii) Formation of carbanion is rate determining step.

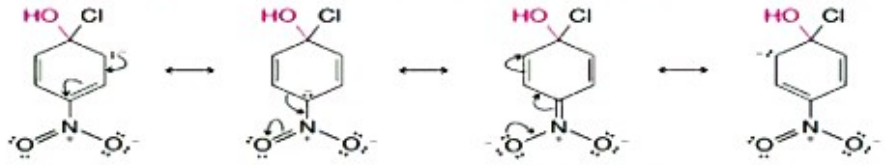
(iii) Rate is proportional to stability of carbanion, that is more the rate of SNAr if more stable carbanion (2nd order reaction)

(iv) Fluorine is most reactive towards SNAr because of its –I effect, stable carbanion will be formed in its rate determining step.



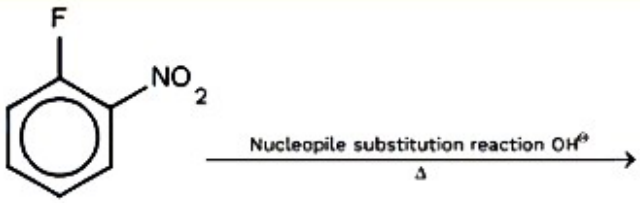
(v) More the NO_2 group at ortho, para position, more rate of reaction because carbanion is more stable by resonance.

- The carbanion is stabilized by electron-withdrawing groups in the positions ortho and para to the halogen atom. If we examine the following resonance contributor for a Meisenheimer intermediate, we can see how:

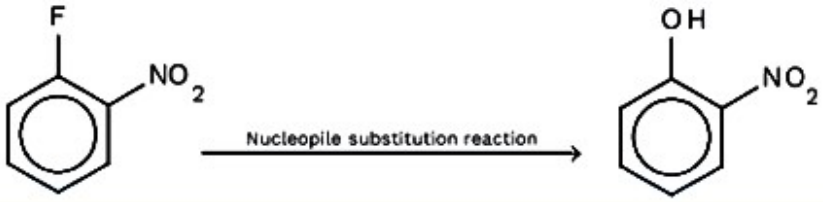


Especially stable (Negative charges are both on oxygen atoms.)

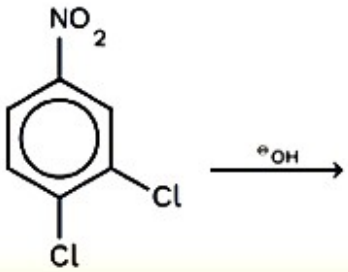
Q81



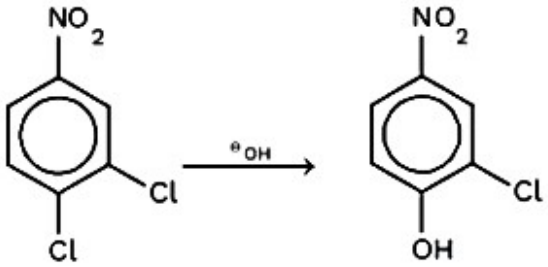
A81



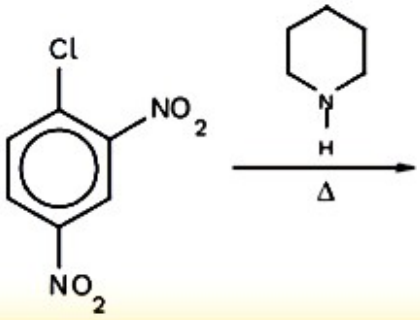
Q82



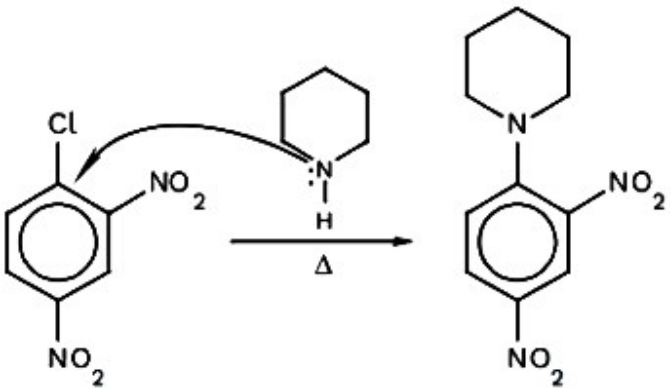
A82



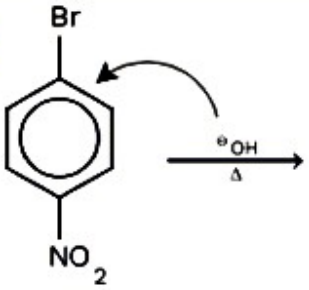
Q83



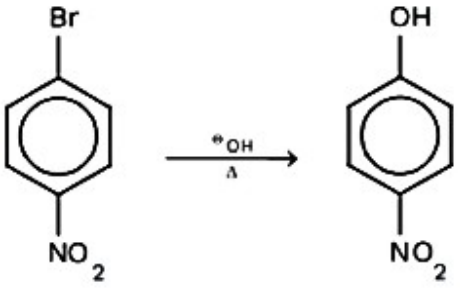
A83



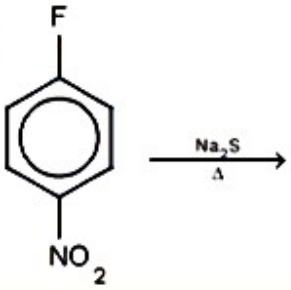
Q84



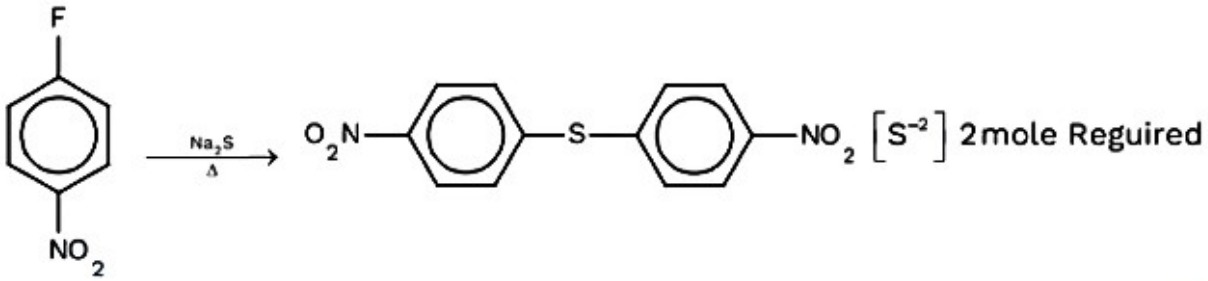
A84



Q85

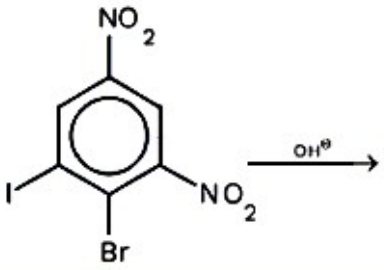


A85

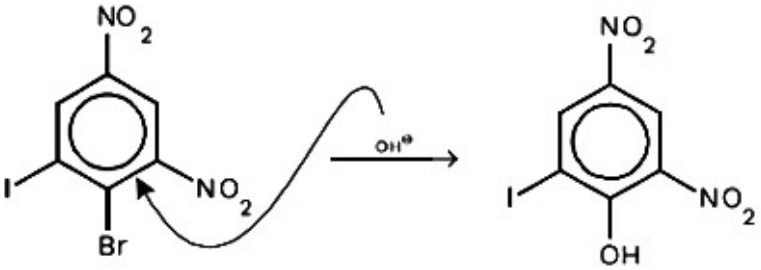


2mole Reguired

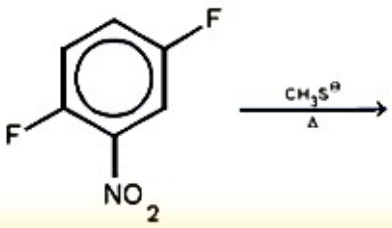
Q86



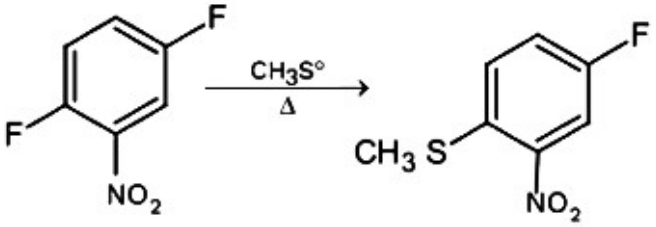
A86



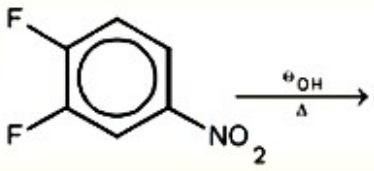
Q87



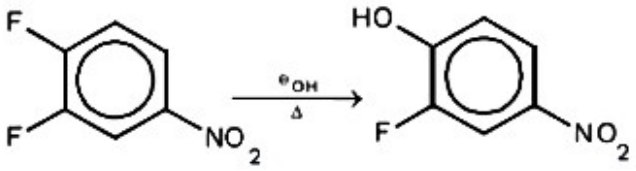
A87



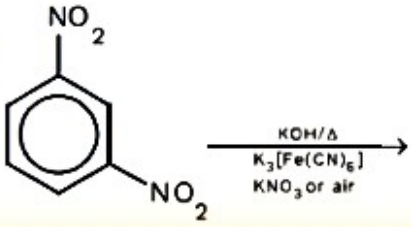
Q88



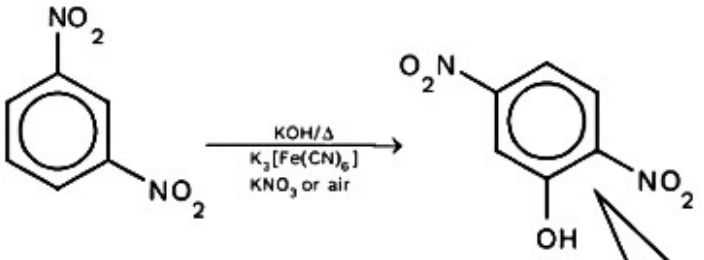
A88



Q89



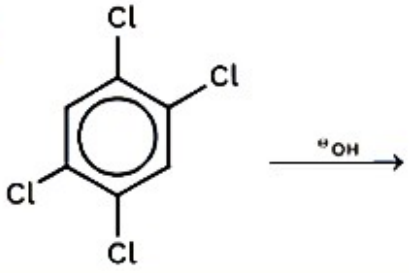
A89



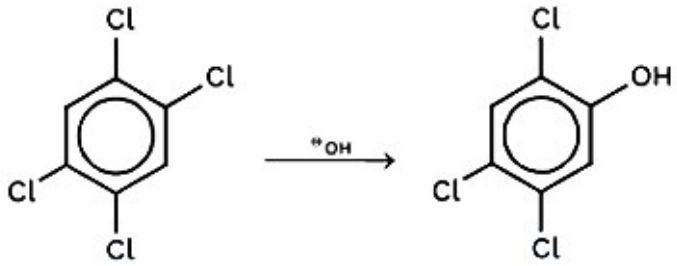
The role of oxidising agent is to oxidise the intermediate to form

$$H^{\oplus} + Fe^{+3} \longrightarrow Fe^{2+} + H \longrightarrow \frac{1}{2}H_2$$

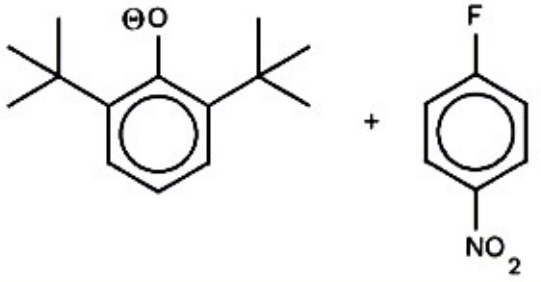
Q90



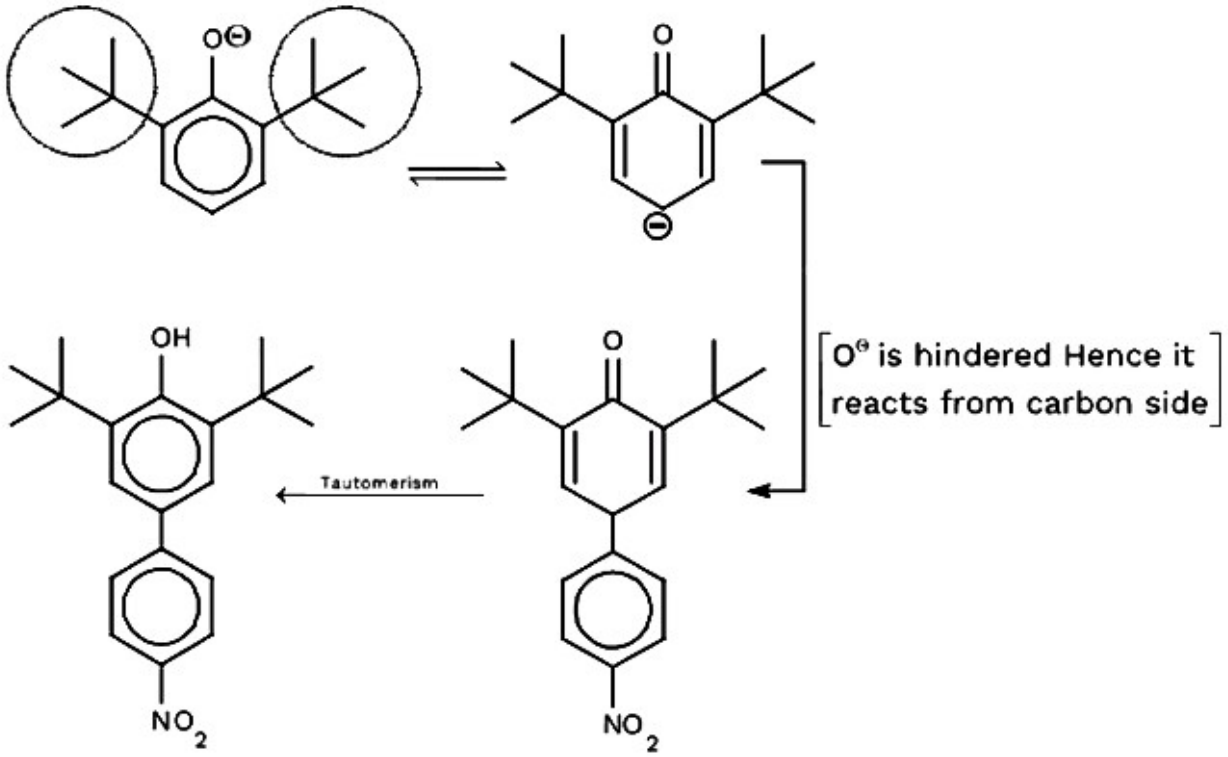
A90


Cl
Cl
Cl
Cl
⊖OH
Cl
OH
Cl
Cl

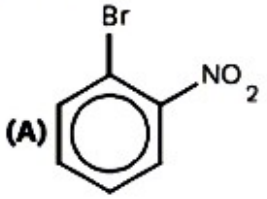
Q91

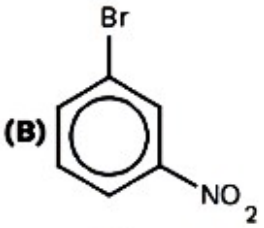

ΘO
+
F
NO2

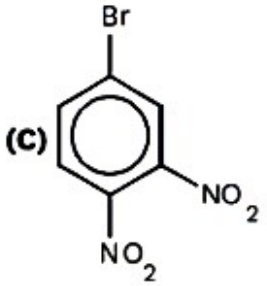
A91

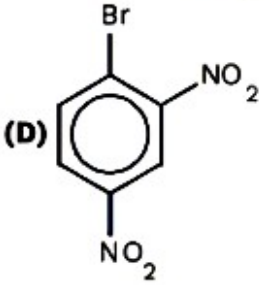

OΘ
O
OH
O
Tautomerism
NO2
NO2
[Oθ is hindered Hence it reacts from carbon side]

Q92 **Compare rate of SNAr :**









A92 (D > C > A > B)

NO_2 on ortho, para provides more rate.

D → Both O–NO_2 are at ortho/para w.r.t. Br

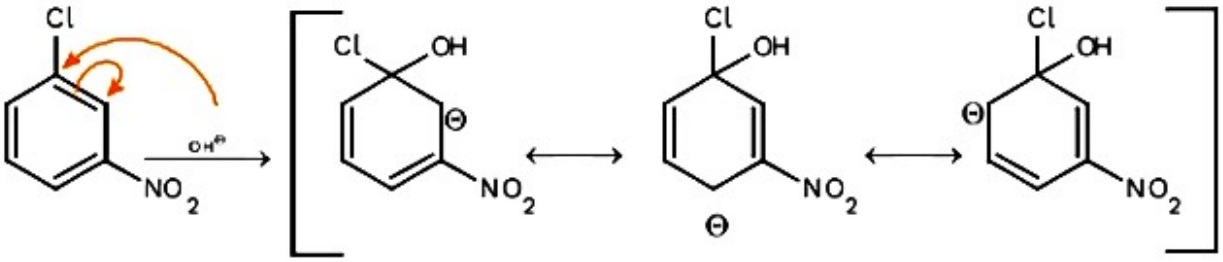
C → Only one NO_2 is at para and other is at meta w.r.t. Br

A → NO_2 at ortho (single NO_2)

B → NO_2 at meta

Q93 **Draw resonating structures for the carbanion that would be formed if meta-chloronitrobenzene could react with hydroxide ion. Why doesn't it react?**

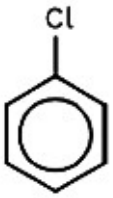
A93



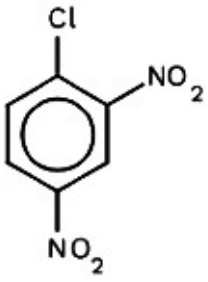
In any situation the carbanion formed doesn't get stabilized by -M of $-NO_2$. Which makes it an unstable carbanion and hence it doesn't react with hydronide.

Q94 **List the following compounds in order of decreasing reactivity toward nucleophilic aromatic substitution:**
(A) chlorobenzene **(B) 1-chloro-2,,4-dinitrobenzene**
(C) p-chloronitrobenzene

A94



(A)



(B)



(C)

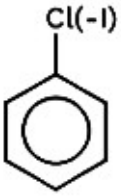
Reactivity : B > C > A
B is more reactive as it forms highly resonance stabilized carbanion intermediate due to presence of two $-NO_2$ groups at ortho & para positions.

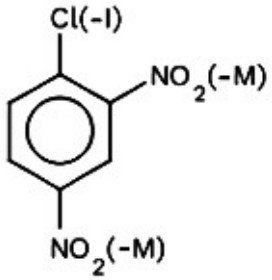
Q95 **List the same compounds in order of decreasing reactivity toward electrophilic aromatic substitution.**

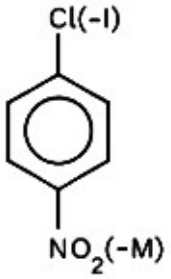
A95 For EAS reaction Reactivity $\propto$ '+' effect

$$\propto \frac{1}{\text{'}-\text{'effect}}$$

Hence,







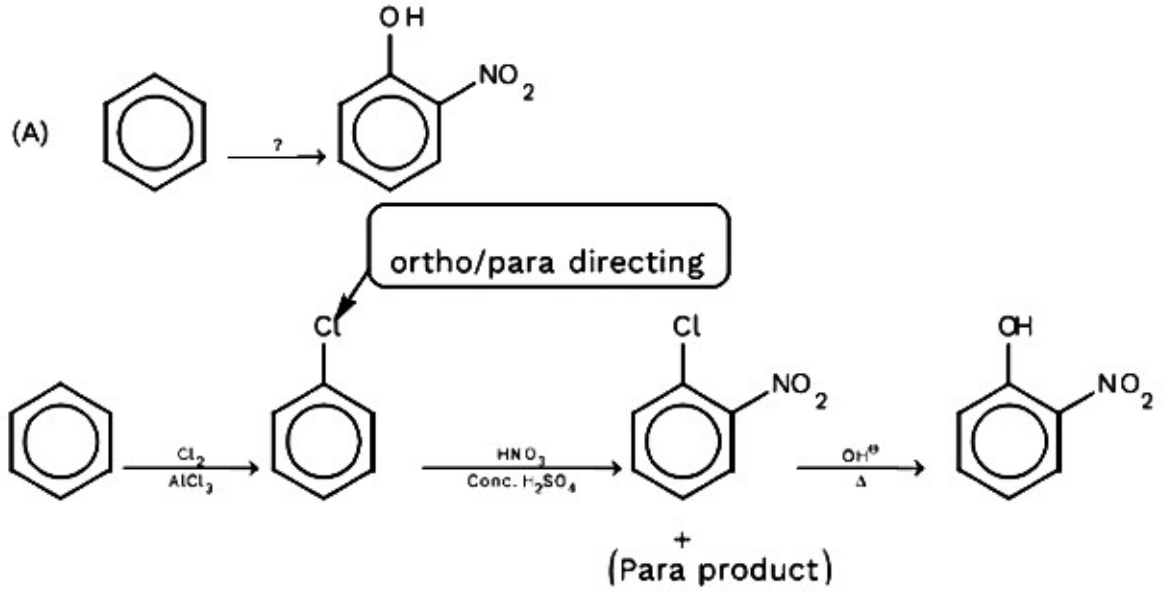
Rate for EAS : A > C > B

Q96 **Show how each of the following compounds could be synthesized from benzene :**

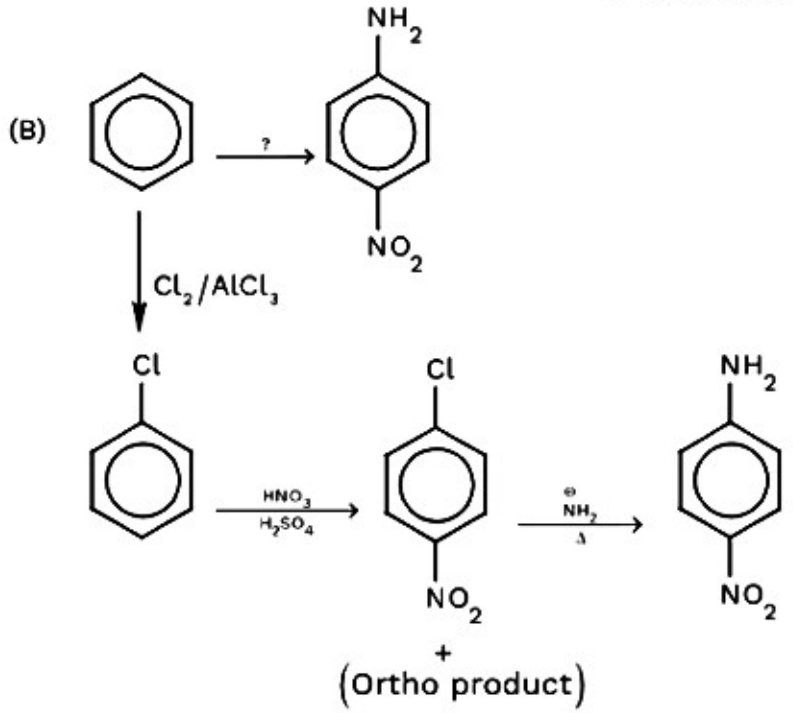
(A) *o*-nitrophenol **(B) *p*-nitroaniline** **(C) *p*-bromoanisole**

A96

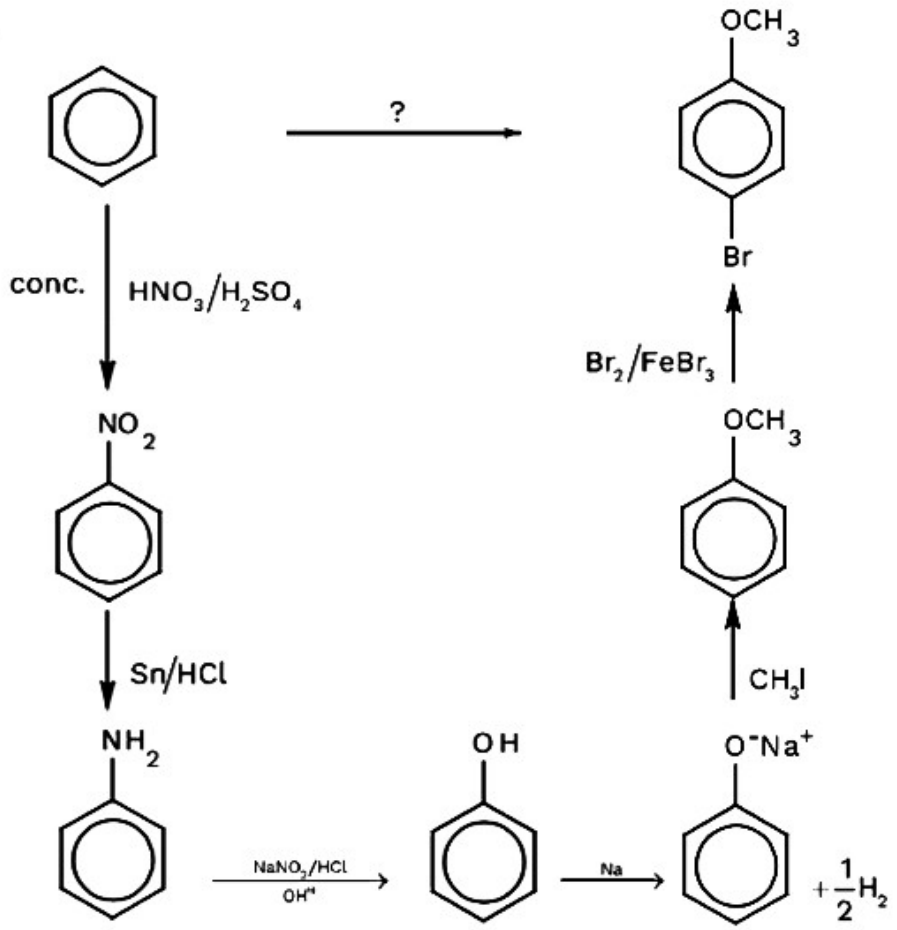
(A)



(B)

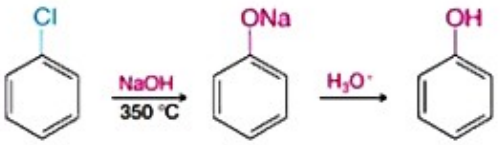


(C)



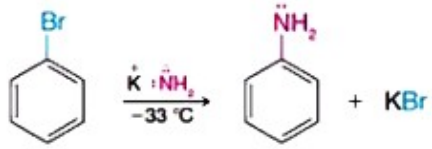
Benzyne: Nucleophilic Aromatic Substitution by Elimination-Addition

Although aryl halides such as chlorobenzene and bromobenzene do not react with most nucleophiles under ordinary circumstances, they do react under highly forcing conditions. Chlorobenzene can be converted to phenol by heating it with aqueous sodium hydroxide in a pressurized reactor at 350°C:



Phenol

Bromobenzene reacts with the very powerful base, $^-NH_2$, in liquid ammonia:

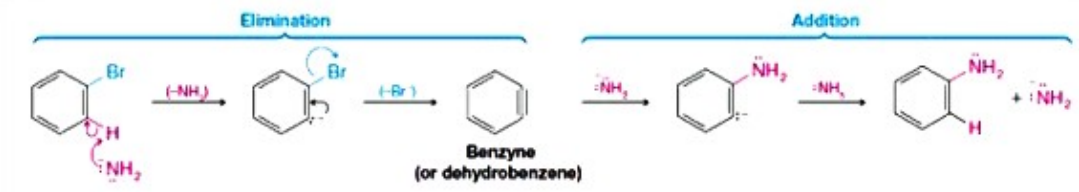


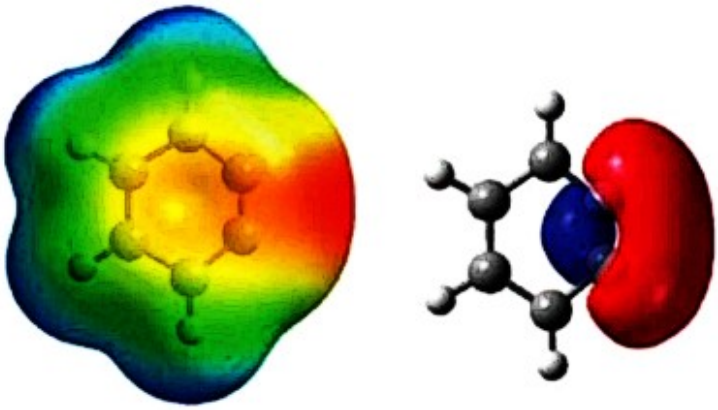
Aniline

These reactions take place through an elimination-addition mechanism that involves the formation of a highly unstable intermediate called benzyne (or dehydrobenzene).

In the first step (see the following mechanism), the amide ion initiates an elimination by abstracting one of the ortho protons because they are the most acidic. The negative charge that develops on the ortho carbon is stabilized by the inductive effect of the bromine. The anion then loses a bromide ion. This elimination produces the highly unstable, and thus highly reactive, **benzyne**. Benzyne then reacts with any available nucleophile (in this case, an amide ion) by a two-step addition reaction to produce aniline.

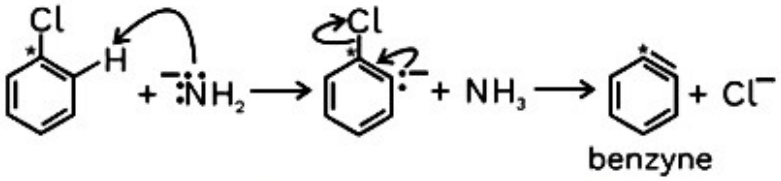
[A Mechanism for the Reaction: The Benzyne Elimination-Addition**]**



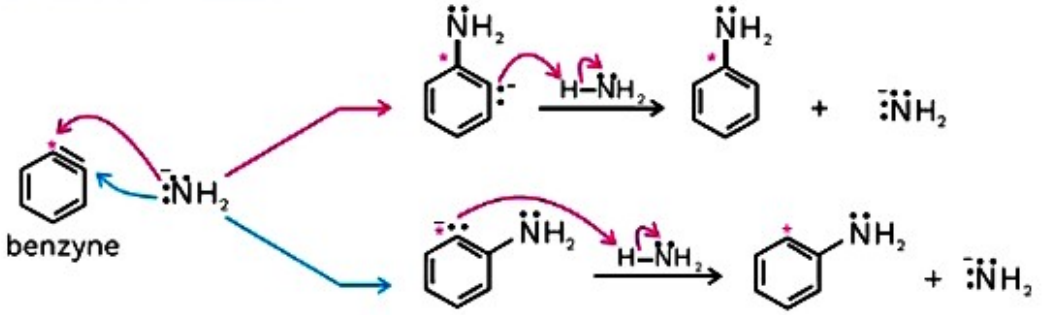


(A) (B)

Benzyne is an extremely reactive species. In a molecular structure with a triple bond, the two sp hybridized carbons and the atoms attached to these carbons $(C-C\equiv C-C)$ are linear because the bond angles are 180°. Four linear atoms cannot be introduced into a six-membered ring, so the $C-C\equiv C-C$ system in benzyne is distorted: The original π bond is unchanged, but the sp^2 orbitals that form the new bond are not parallel to one another. So, they cannot overlap as well as the p orbitals that form a normal π bond, resulting in a much weaker and much more reactive bond.

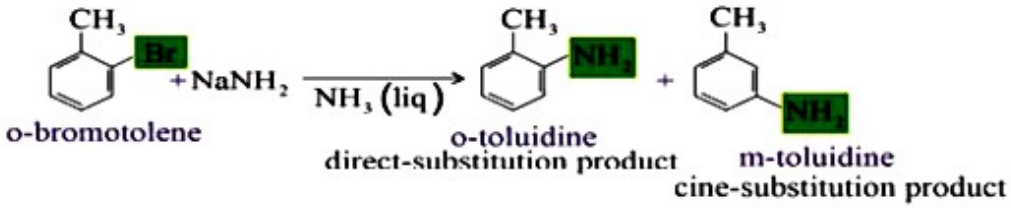


The attacking nucleophile can attack either of the carbons of the "triple bond" of benzyne. Protonation of the resulting anion forms the substitution product. The overall reaction is an elimination-addition reaction: Benzyne is formed in an elimination reaction and immediately undergoes an addition reaction.

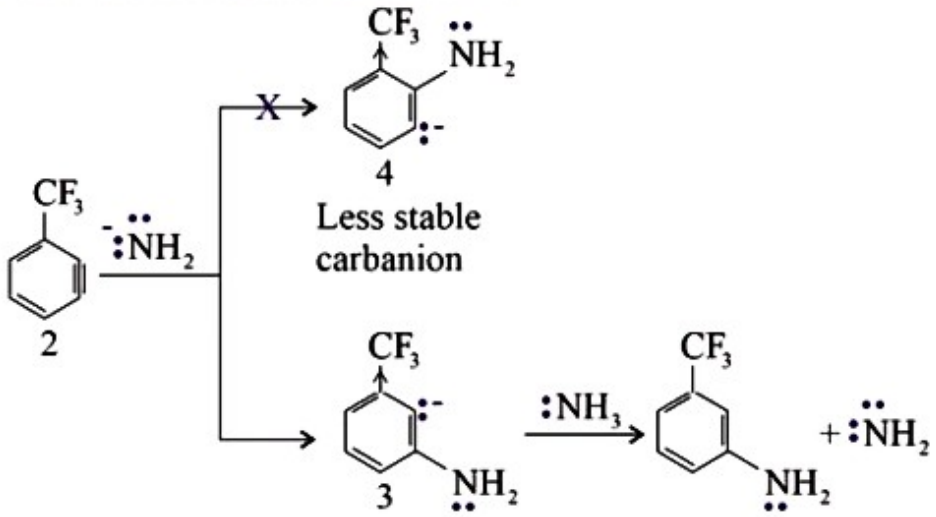


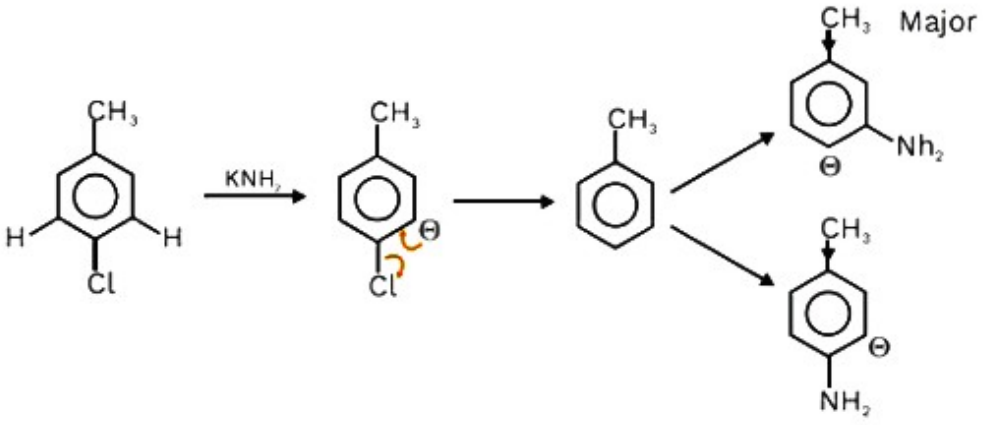
Substitution at the carbon that was close to the leaving group is called direct substitution. Substitution at the adjacent carbon is called cine substitution (cine comes from kinesis, which is Greek for "movement"). In the following reaction, o-toluidine is the direction-substitution product; m-toluidine is the cine-substitution product.

Substitution at the carbon that was attached to the leaving group is called direct substitution. Substitution at the adjacent carbon is called cine substitution (cine comes from kinesis, which is Greek for: movement"). In the following reaction, o-toluidine is the direct-substitution product; m-toluidine is the cine-substitution product.



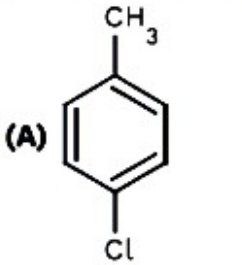
This benzyne added an amide ion in the way that produces the more stable carbanion 3 rather than the less stable carbanion 4:

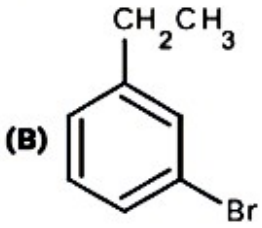


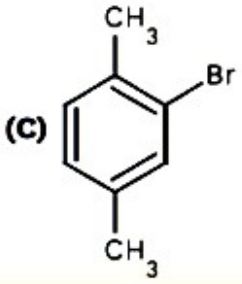


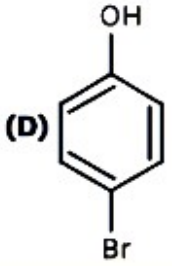
Illustration

Q97 **Give the product that will be obtained from the reaction of the following compounds with sodium amide in liquid ammonia:**

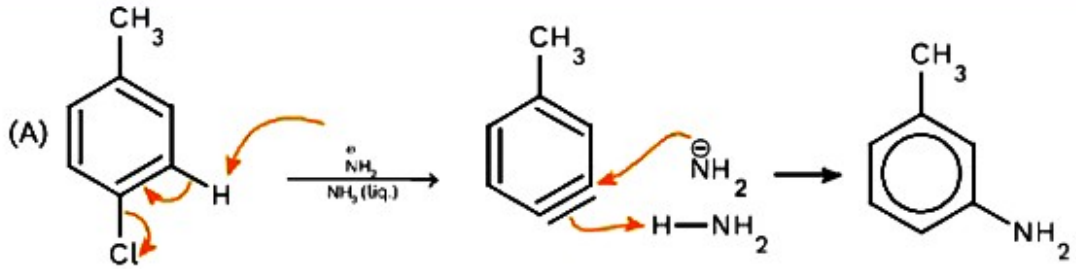


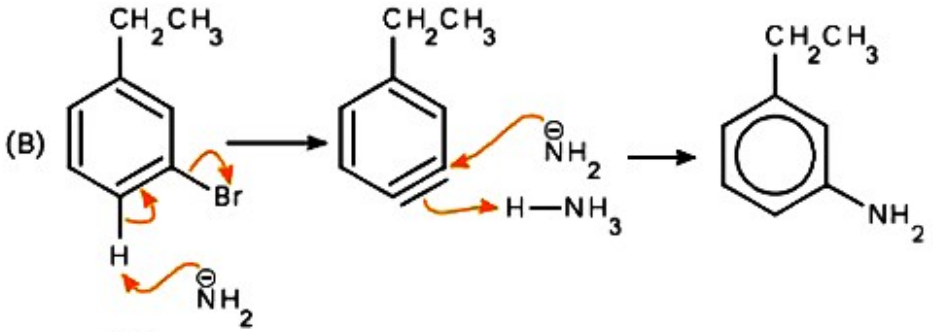


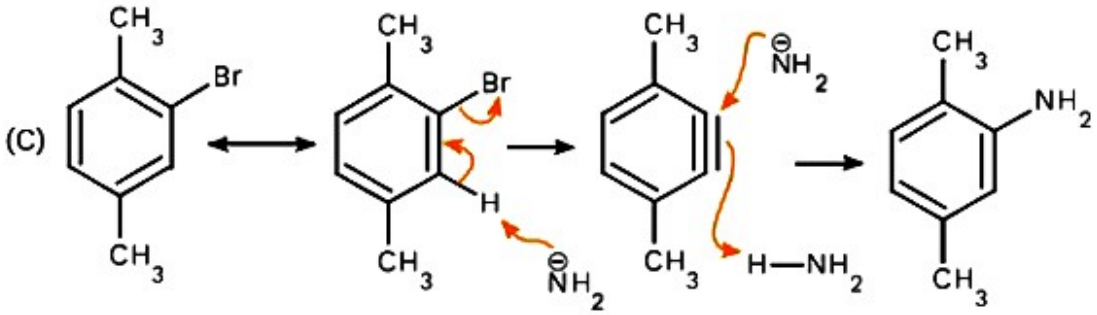


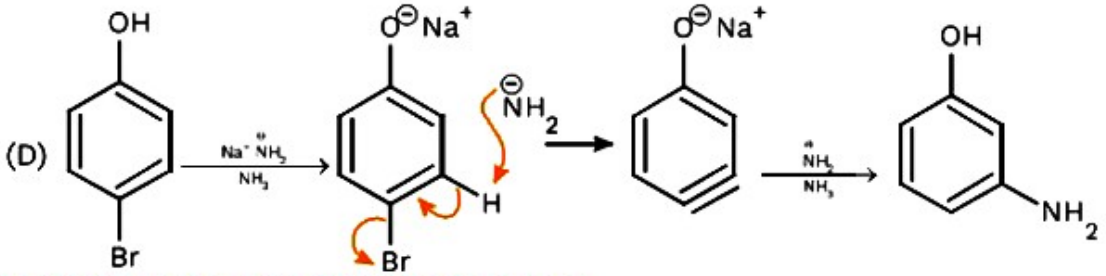


A97



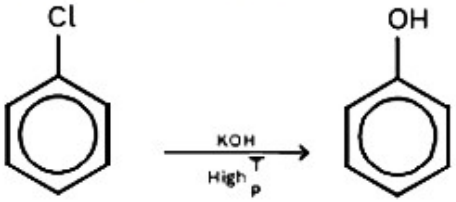






(SNEA) Dow's Process condition for benzyne:

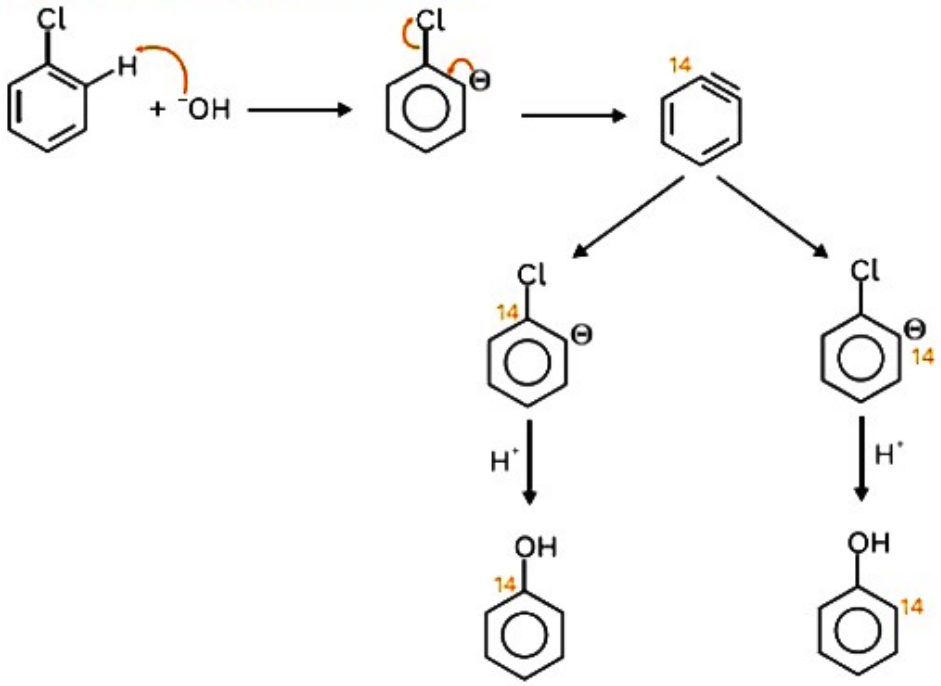
Arylhalide + strong base + b-hydrogen in arylhalide



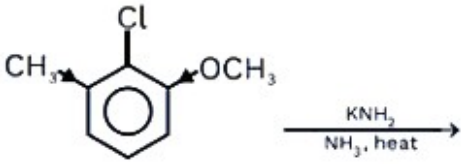
Benzyne mechanism involves reaction of unactivated aryl halide having at lest one hydrogen at ortho position with strong base like KNH_2 or $NaNH_2$ in liquid ammonia. This reaction proceed via-benzyne intermediate.

This mechanism involves elimination followed by addition, hence, it is also called elimination – addition mechanism of aromatic nucleophilic substitution or SNEA.

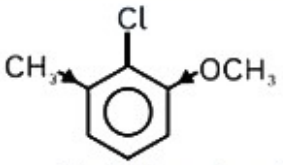
[A Mechanism for the Reaction:]



Q98

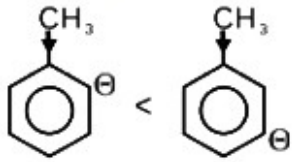


A98

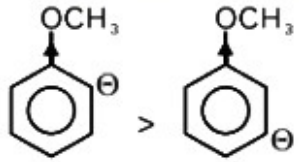


No reaction (No β-hydrogen)

Stability of carbanion on the ring is decided by inductive effect only.

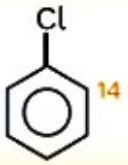


Case I



Case II

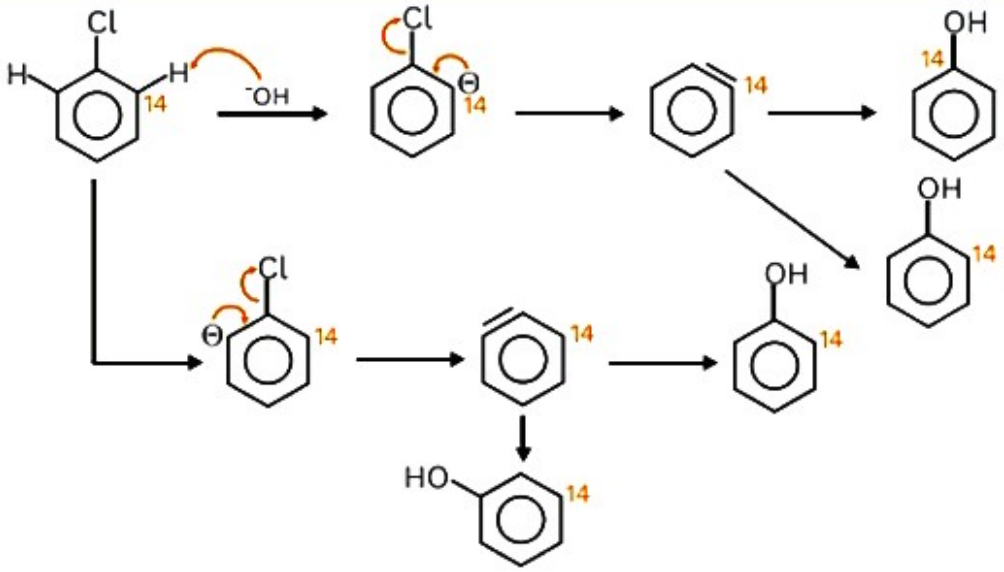
Q99

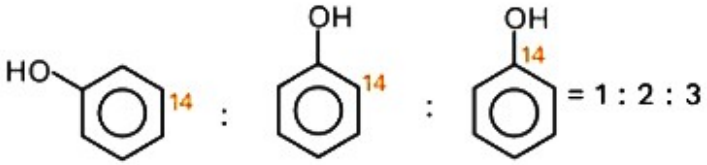


KOH → A + B + C

1 : 2 : 1

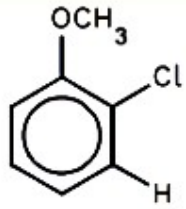
A99



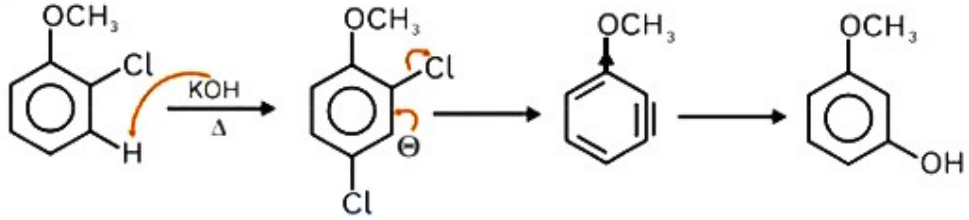


= 1 : 2 : 3

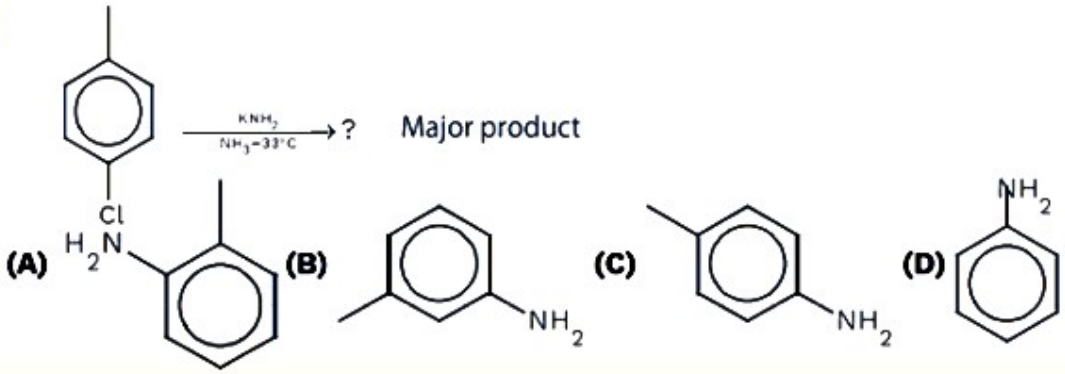
Q100



A100

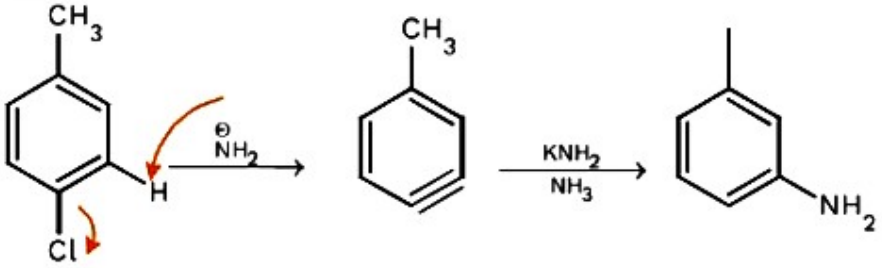


Q101



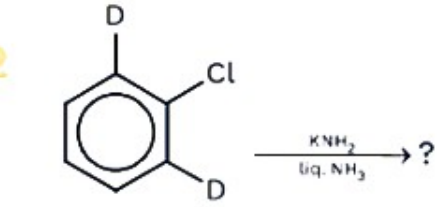
A101

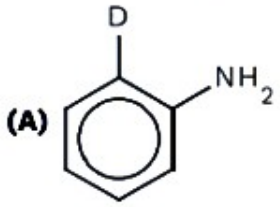
(B)

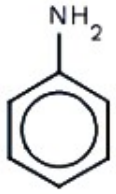


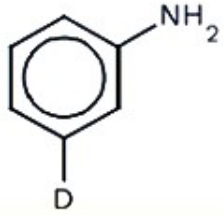
Meta Amino product (Benzyne mechanism)

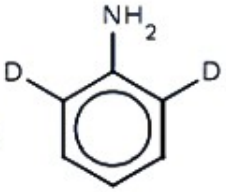
Q102



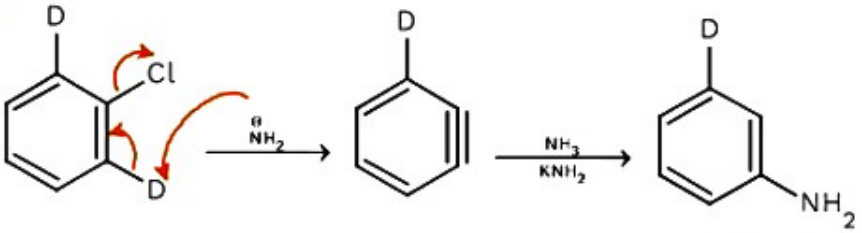
(A) 

(B) 

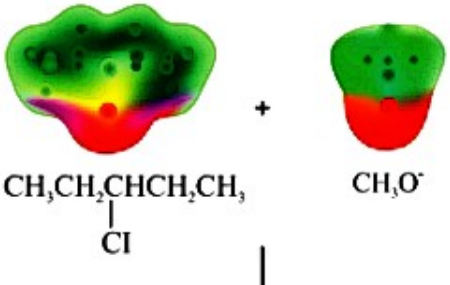
(C) 

(D) 

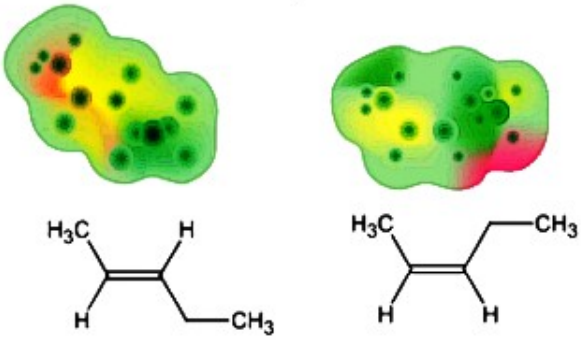
A102 (C)

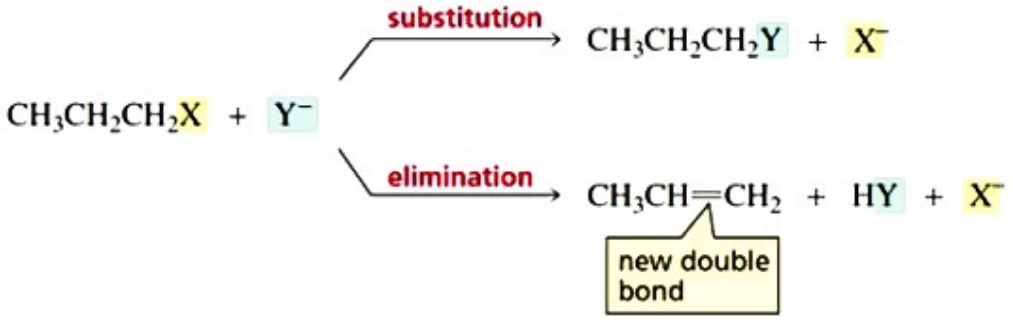


In addition to undergoing the nucleophilic substitution reactions alkyl halides undergo *elimination* reactions. In an **elimination reaction**, groups are eliminated from a reactant molecule .



For example, when an alkyl halide experiences an elimination reaction, the halogen (X) is removed from one carbon and a proton is removed from an adjacent carbon. A double bond is formed between the two carbons from which the atoms are eliminated. Therefore, *the product of an elimination reaction will be alkene..*

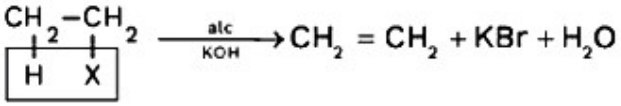




There are two important elimination reactions: E1 and E2.

The E2 Reaction :

(A) Dehydrohalogenation (–HX)

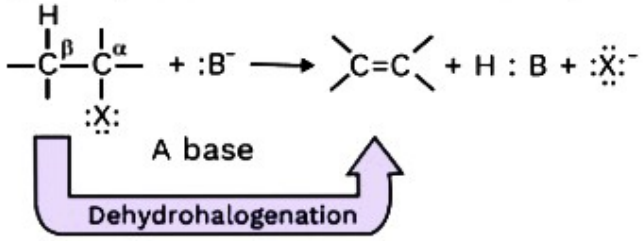


Introduction of Dehydrohalogenation:

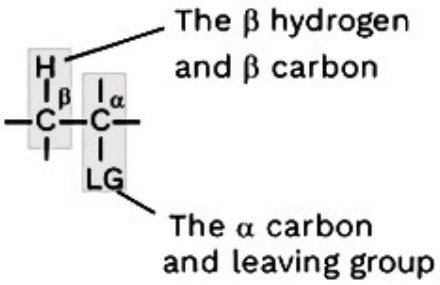
Dehydrohalogenation is a widely used method for synthesizing alkenes via the elimination of HX from adjacent atoms of an alkyl halide. Heating the alkyl halide with a strong base causes the reaction to take place.

The following are two examples :

Any type of halogen atom can participate in elimination of HX to form an alkene. In general, we call these reactions dehydrohalgenation:

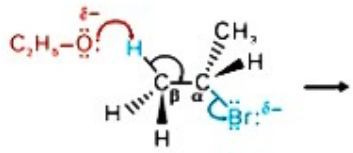


These reactions are often named β eliminations., also often referred to as 1,2 eliminations.

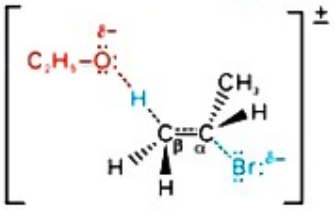


Reaction

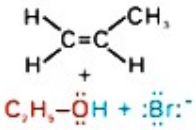
$C_2H_5O^- + CH_3CHBrCH_3 \rightarrow CH_2{=}CHCH_3 + C_2H_5OH + Br^-$



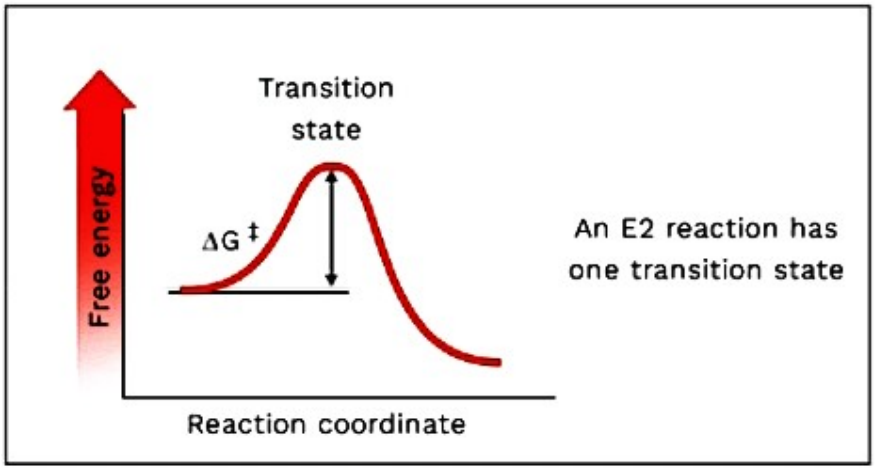
The basic ethoxide ion begins to remove a proton from the b carbon using its electron pair to form a bond to it. At the same time, the electron pair of the β C–H bond begins to move in to become the π bond of a double bond, and the bromine begins to depart with the electrons that bonded it to the α carbon.



Transition state
Partial bonds in the transition state extend from the oxygen atom a that is removing the β hydrogen, through the carbon skeleton of the developing double bond. to the departing leaving group. the flow of electron density is from the base toward the leaving group as an electron pair fills the π bonding orbital of the alkene.

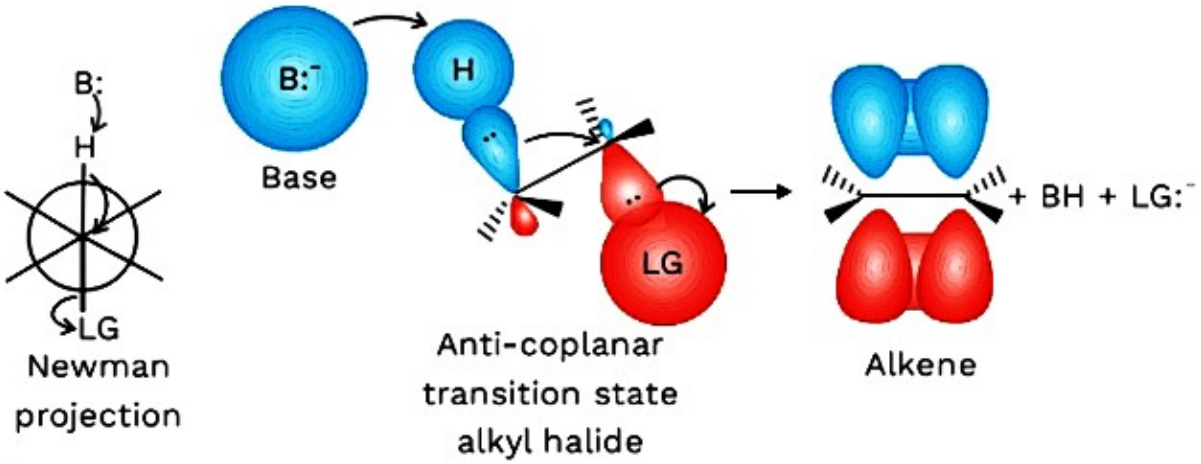


At completion of the reaction, the double bond is fully formed and the alkene has a triogonal planar geometry.

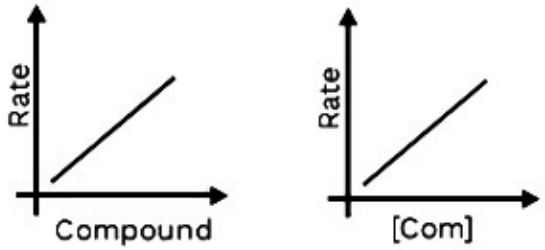


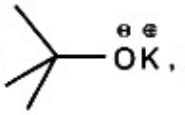
The alignments of the β-hydrogen and the leaving group are not random.
A conformation is required where they all are in the same plane, as shown above and in the

example that follows :



* E_2 is a bimolecular reaction because Rate depends on two factors Rate a [Compound]^1 [Base]^1
* Second order, bimolecular



* In E_2 reaction anti periplannar transition state form, it means b-H and leaving group are anti to each other.
* For dehgdrohalogenation alc. KOH (KOH + C_2H_5OH), alc 

, alc $NaNH_2$ following reagents are used.
* In alc. KOH, E_2 preferred while in aq. KOH S_N2 preferred for 1°.

Bases support in Dehydrohalogenation Reaction

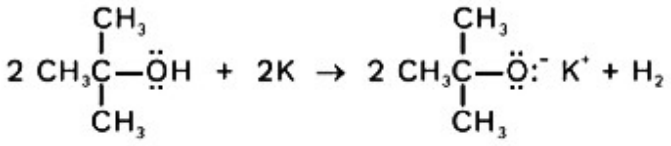
Numerous bases can be utilized for dehydrohalogenations. KOH dissolved in ethanol (KOH/EtOH) is sometimes used, but the conjugate bases of alcohols, such as sodium ethoxide (EtONa), often offer distinct advantages.

The conjugate base of an alcohol (an alkoxide) can be prepared by treating an alcohol with an alkali metal. For example:

$$2\ R{-}\ddot{O}H + 2\ Na \rightarrow 2\ R{-}\ddot{O}{:}^{-}Na^{+} + H_2$$

Alcohol → Sodium alkoxide

Potassium *tert*-butoxide (*t*-BuOK) is another extremely effective base for dehydrohalogenation. It can be completed by the reaction below or bought as a solid.

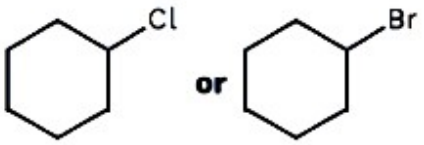


tert-Butaol (excess)

Potassium tert-butoxide

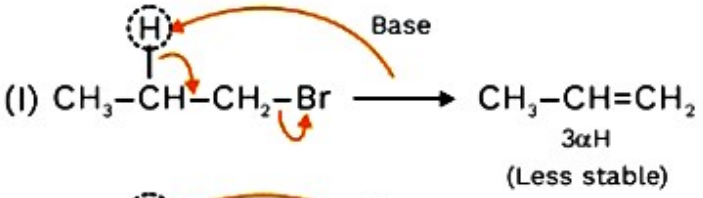
Q103 **Which of the alkyl halide is more reactive in an E2 reaction ?**

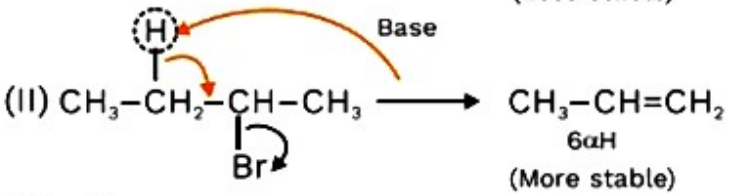
(A) $CH_3CH_2CH_2Br$ or $CH_3CH_2\underset{\underset{Br}{|}}{C}HCH_3$

(B)  **or**

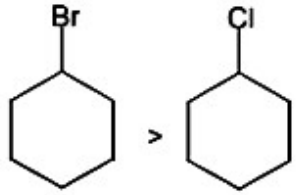
(C) $CH_3\underset{\underset{Br}{|}}{\overset{\overset{CH_3}{|}}{C}}HCH_2CHCH_3$ or $CH_3CH_2CH_2\underset{\underset{Br}{|}}{\overset{\overset{CH_3}{|}}{C}}CH_3$

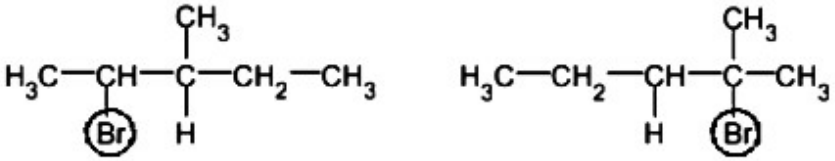
(D) $CH_3\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}CH_2Cl$ or $CH_3C\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}H_2CH_2Cl$

A103 (A) (I) 



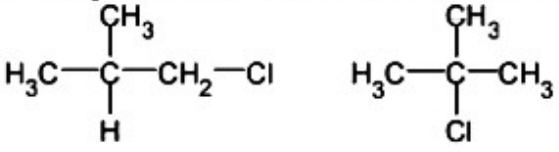
(II > I)

(B)  (Leaving group ability)

(C) 

2 3°

3° alkyl halide will be more reactive

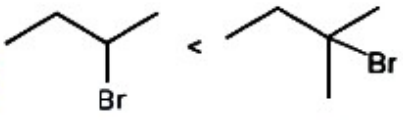
(D) 

1 3°

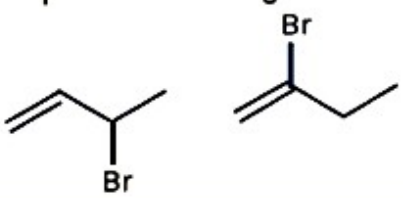
3° Alkyl halide will be more reactive.

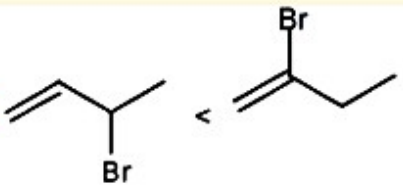
Identify rate of E2 elimination :

Q104 

A104 

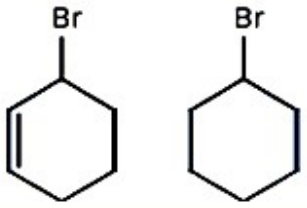
1 3°

Q105 

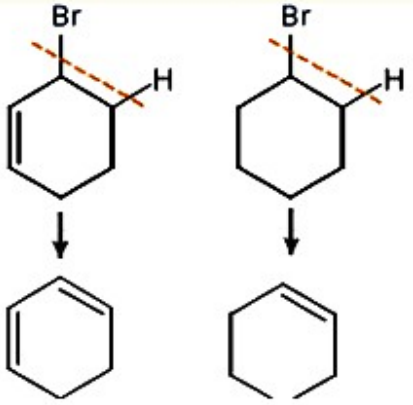
A105 

No elimination possible

Q106

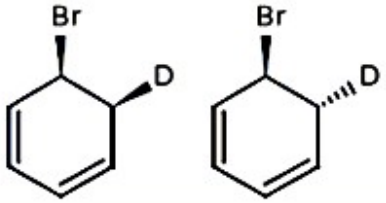


A106

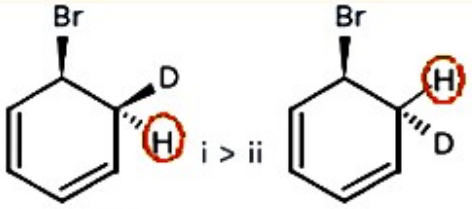


Conjugated alkene more stable

Q107

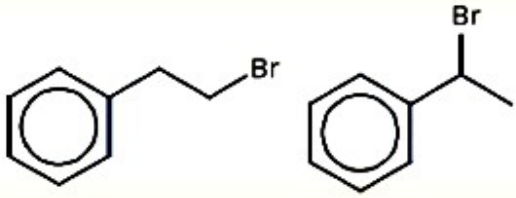


A107

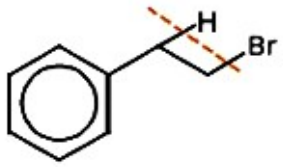


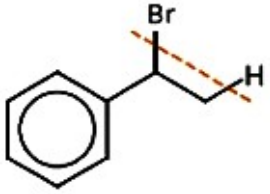
* Br and H are anti
* Br and D are anti
* Removing D is more difficult.

Q108



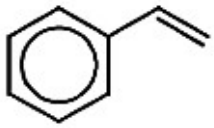
A108

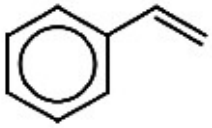




1° alkyl halide

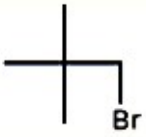
2° alkyl halide

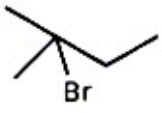




2° > 1°

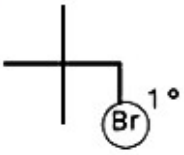
Q109

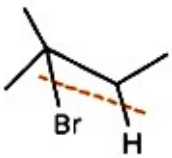


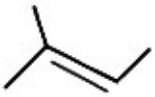


A109

(b > a)

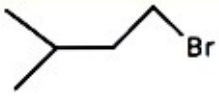


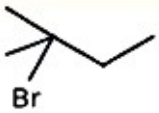




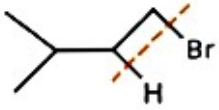
* No αH, No elimination

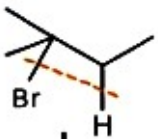
Q110

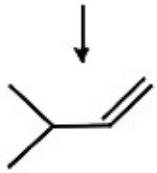


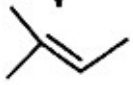


A110







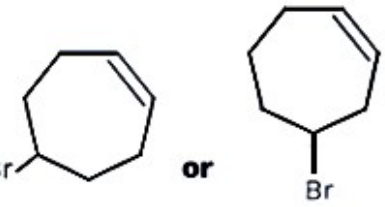


(i) (ii)

(ii > i)

Q111 **Which alkyl halide would you suppose to be more reactive in an E2 reaction?**

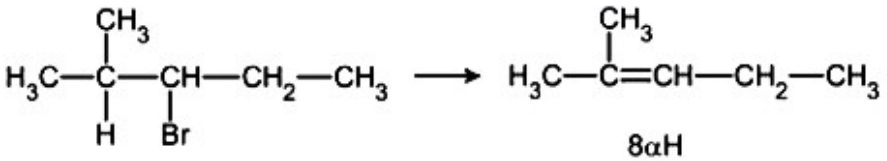
a) $CH_3CHCHCH_2CH_3$ (Br on 2nd C) or $CH_3CHCH_2CHCH_3$ (Br on 4th C)

b)  **or**

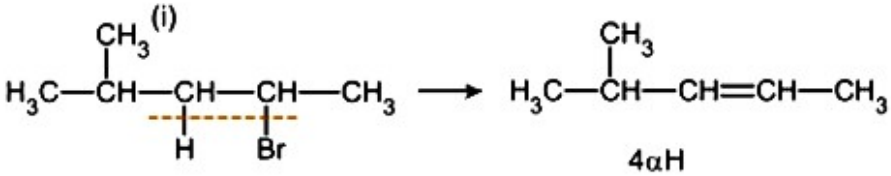
c) $CH_3CH_2CH_2CHCH_3$ (Br on 4th C) or $CH_3CH_2CHCH_2CH_3$ (Br on 3rd C)

d) $Ph{-}CH{=}CH{-}CH_2{-}CH_3$ or $Ph{-}CH_2{-}CH{=}CH{-}CH_3$

A111

a) 

8αH

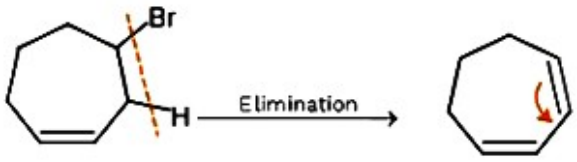
(i) 

4αH

(ii)

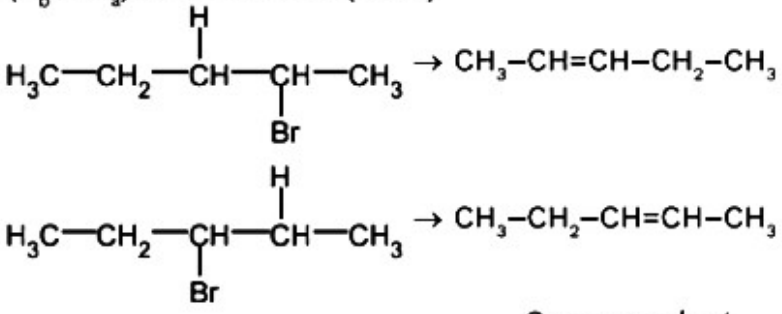
(i > ii)

b) 



Conjugated, more stable

c) (H_b > H_a) acidic nature (a > b)

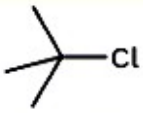
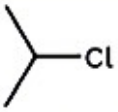
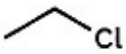


Same product

Now acidic nature of β-H is only deciding factor

d) $Ph-CH=CH-CH_2-CH_3$ * Resonance

$Ph-CH_2-CH=CH-CH_3$ * Not conjugated

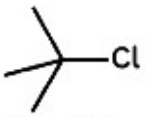
Q112 **(A)**  **(B)**  **(C)**  **(D) CH_3-Cl**

Most reactive towards SN_1 reaction?

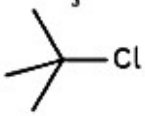
Most reactive towards SN_2 reaction?

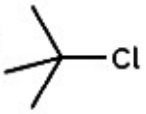
Most reactive towards E_2 reaction?

Most reactive towards E_1 reaction?

A112 Most reactive towards SN_1 reaction : 

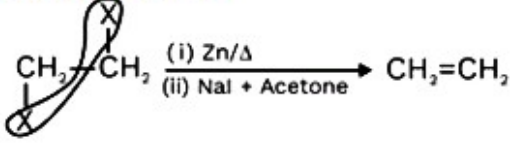
Most reactive towards SN_2 reaction : CH_3-Cl

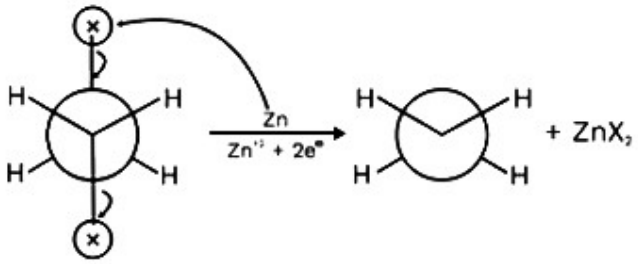
Most reactive towards E_2 reaction : 

Most reactive towards E_1 reaction : 

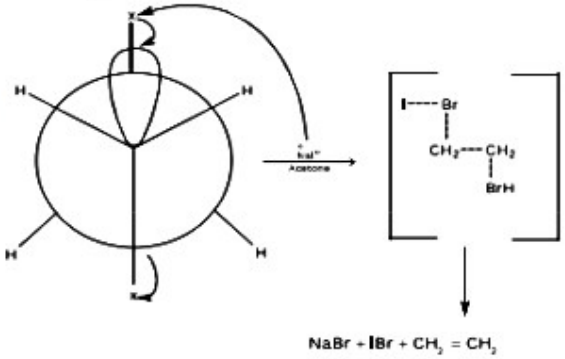
E2 Bimolecular Elimination

(B) Dehalogenation

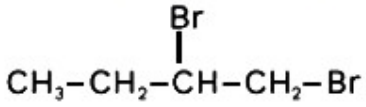


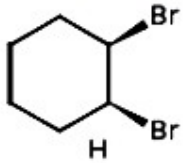


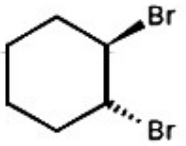
(C)

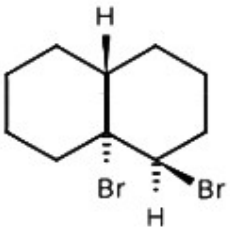


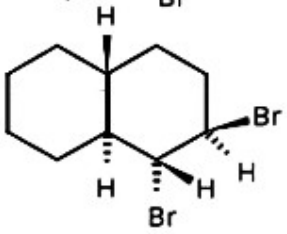
Forecast the products of the following eliminations of vicinal dibromides with potassium iodide. Remember to consider the perometric constrains of the E2 reaction..

(A) 

(B) 

(C) 

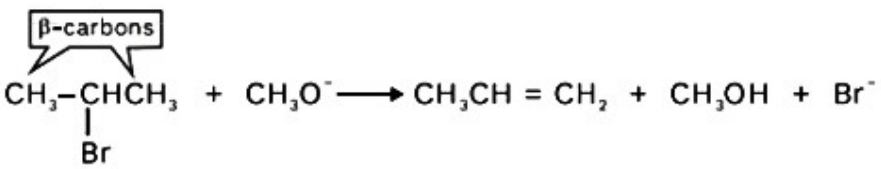
(D) 

(E) 

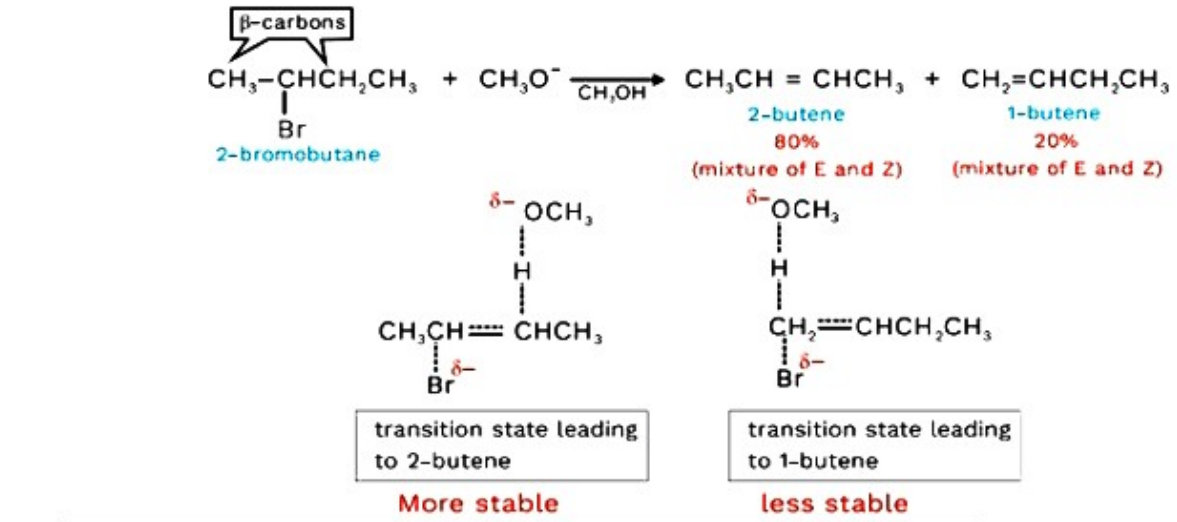
The Regioselectivity of the E2 Reaction

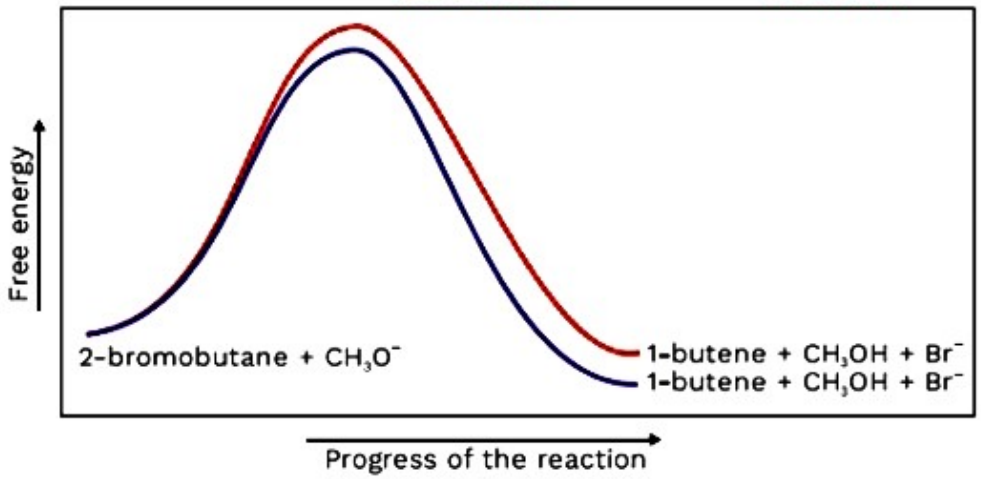

Alexander M. Zaitsev (1841–1910)

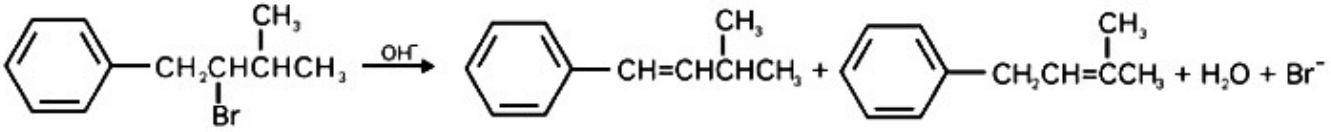
An alkyl halide such as 2-bromopropane has two β-carbons from which a proton can be removed in an E_2 reaction. Because the two β-carbons are identical, the proton can be removed with equal yield from both side. The output of this elimination reaction is propene.



According to **Zaitsev's rule**, the more substituted alkene will be the one formed by removing a proton from the -carbon that is bonded to two hydrogens, rather than from the -carbon that is bonded to three hydrogens.







2-bromo-3-methyl-1-phenylbutane

3-methyl-1-phenyl-1-butane
the double bond is conjugated with the benzene ring
Major Product

3-methyl-1-phenyl-1-butane
the double bond is not conjugated with the benzene ring
Minor Product

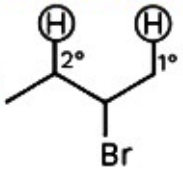
HOFFMANN PRODUCTS

When a sterically hindered alkyl halide undergoes an E_2 reaction with a variety of alkoxide ions, the percentage of the less substituted alkene increases as the size of the base increases .

If the alkyl halide is not sterically hindered and the base is only moderately hindered , the major product will still be the more stable product.

In which condition Hofmann alkene formed as a major product.

(A) Bulky Base :



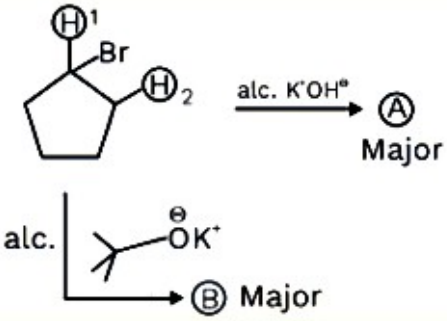
$CH_3-CH=CH-CH_3$

Satyzeff

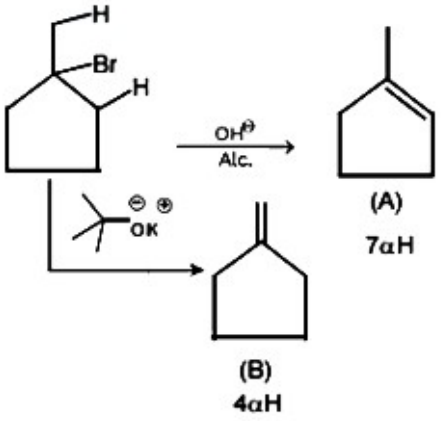
$CH_3CH_2-CH=CH_2$

Hofmann

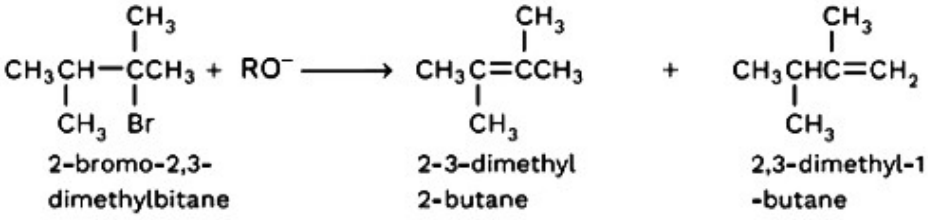
Q113

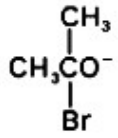
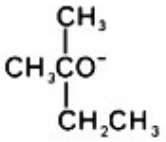
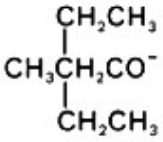


A113



Products in an E_2 Reaction

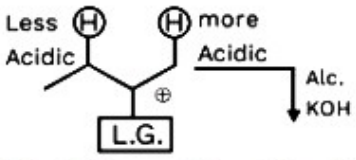


Base	More substituted product	Less substituted product
$CH_3CH_2O^-$	79%	21%
	27%	73%
	19%	81%
	8%	92%

In which condition Hofmann alkene formed as a major product.

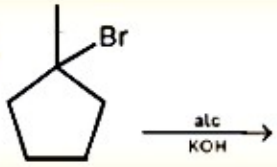
(B) Bad leaving group : $-\overset{\oplus}{N}R_3 \mid -\overset{\oplus}{S}R_2 \mid -F$

sp^3

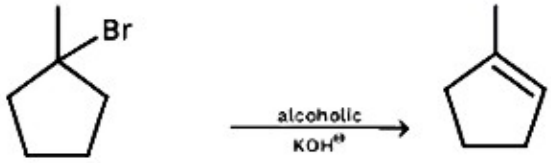


		$CH_3-CH=CH-CH_3$	$CH_3-CH_2-CH=CH_2$
(a)	$-\overset{\oplus}{N}R_3$	5%	95%
(b)	$-\overset{\oplus}{S}R_2$	22%	78%
(c)	$-F$	30%	70%

Q114



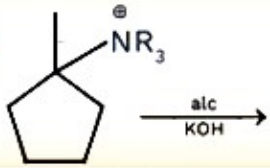
A114



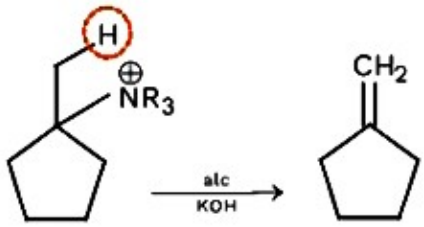
Zaitsev's rule

alkene

Q115



A115



Hoffmann product

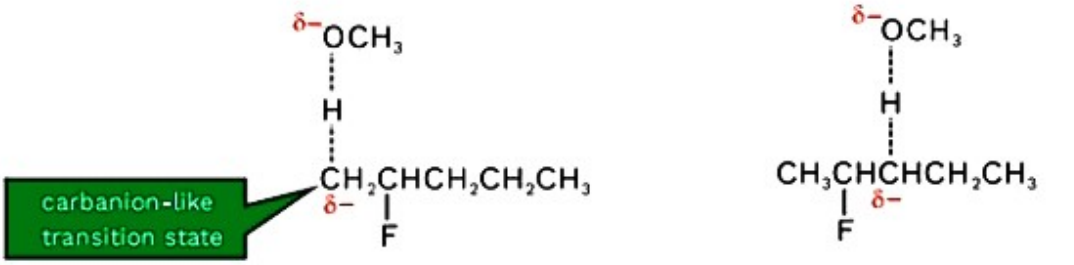
alkene less substituted

Reason behind formation of less substituted alkene in case of bad leaving group :

When a hydrogen atom and a chlorine, bromine, or iodine are eliminated from an alkyl halide, the halogen starts to leave as soon as the base starts to remove the proton. Consequently, a negative charge does not shape up on the carbon that is losing the proton. Thus, the change in state resembles an alkene rather than a carbanion .

The fluoride ion, however, is the toughest base of the halide ions and, therefore, the weakest eliminating group. So, when a base starts to eliminate a proton from an alkyl fluoride, there is fewer inclination for the fluoride ion to be removed than for the other halide ion.

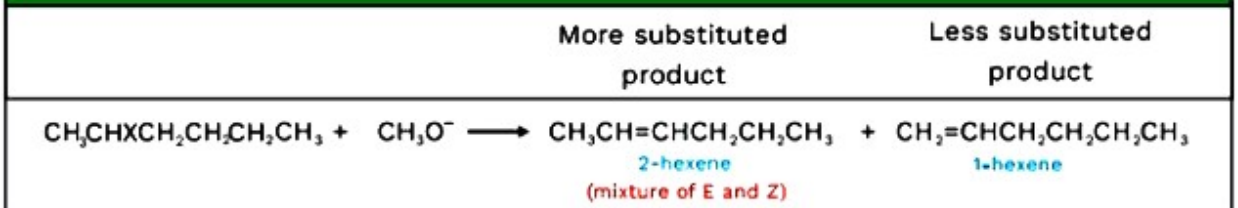
As a consequence, negative charge develops on the carbon that is losing the proton, causing the transition state to resemble a carbanion rather than an alkene



More stable less stable

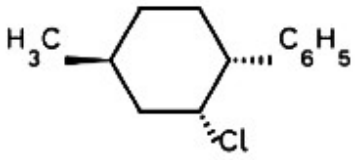
Main product of an E_2 elimination reaction is the more stable alkene except if the reactants are sterically hindered or the leaving group is poor (e.g., a fluoride ion), in which case the major product will be the less stable alkene.

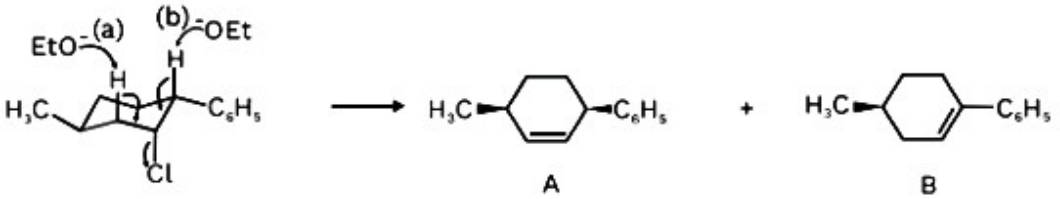
Products Obtained from the E_2 Reaction of CH_3O^- and 2-Halohexanes



Leaving group	Conjugate acid	pK_a	More substituted product	Less substituted product
X = I	HI	−10	81%	19%
X = Br	HBr	−9	72%	28%
X = Cl	HCl	−7	67%	33%
X = F	HF	3.2	30%	70%

Predict the major product designed when the following compound is subjected to dehydrochlorination with sodium ethoxide in ethanol.

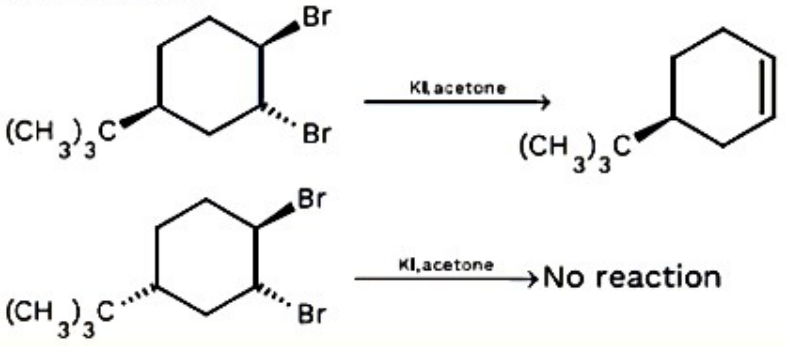




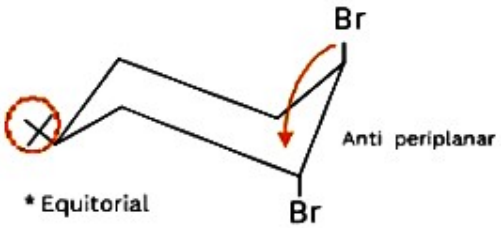
Disubstituted, less stable (minor product)

Trisubstituted, more stable (major product)

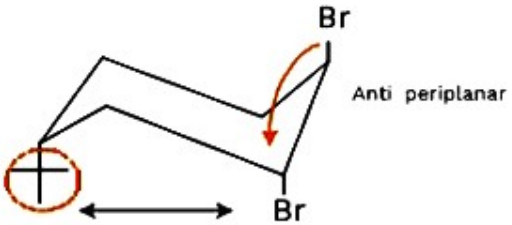
Q116 **The following compounds show different rate of debromination. One reacts quite fast, and the other seems not to react at all explain this surprising difference in rates.**



A116



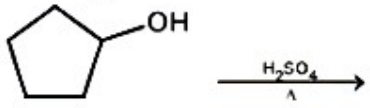
* Equitorial
* Stable



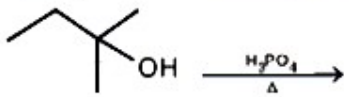
Unstable

Q117 **Predict the products of the following reactions. When more than one product is expected, predict which will be the major product.**

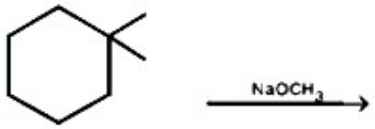
(a)



(b)

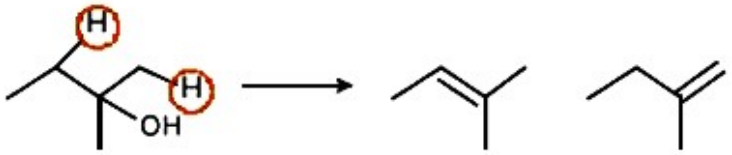


(c)



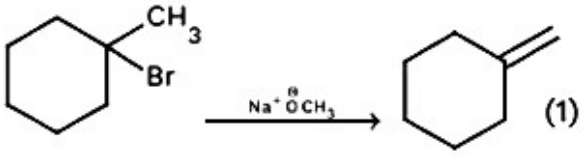
A117 (a) Single product

(b)



(1) Major (2)

(c)



(1) Minor

(2) Major

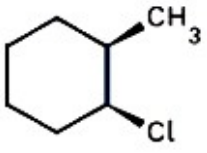
Q118 **Predict the dehydrohalogenation product(s) that result when the following alkyl-halides are heated in alcoholic KOH. When more than one product is formed, predict the major and minor products.**

(A) $(CH_3)_2CH-C(CH_3)_2$ with Br

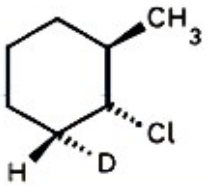
(B) $(CH_3)_2CH-C(Br)-CH_3$

(C) $(CH_3)_2C(Br)-CH_2-CH_3$

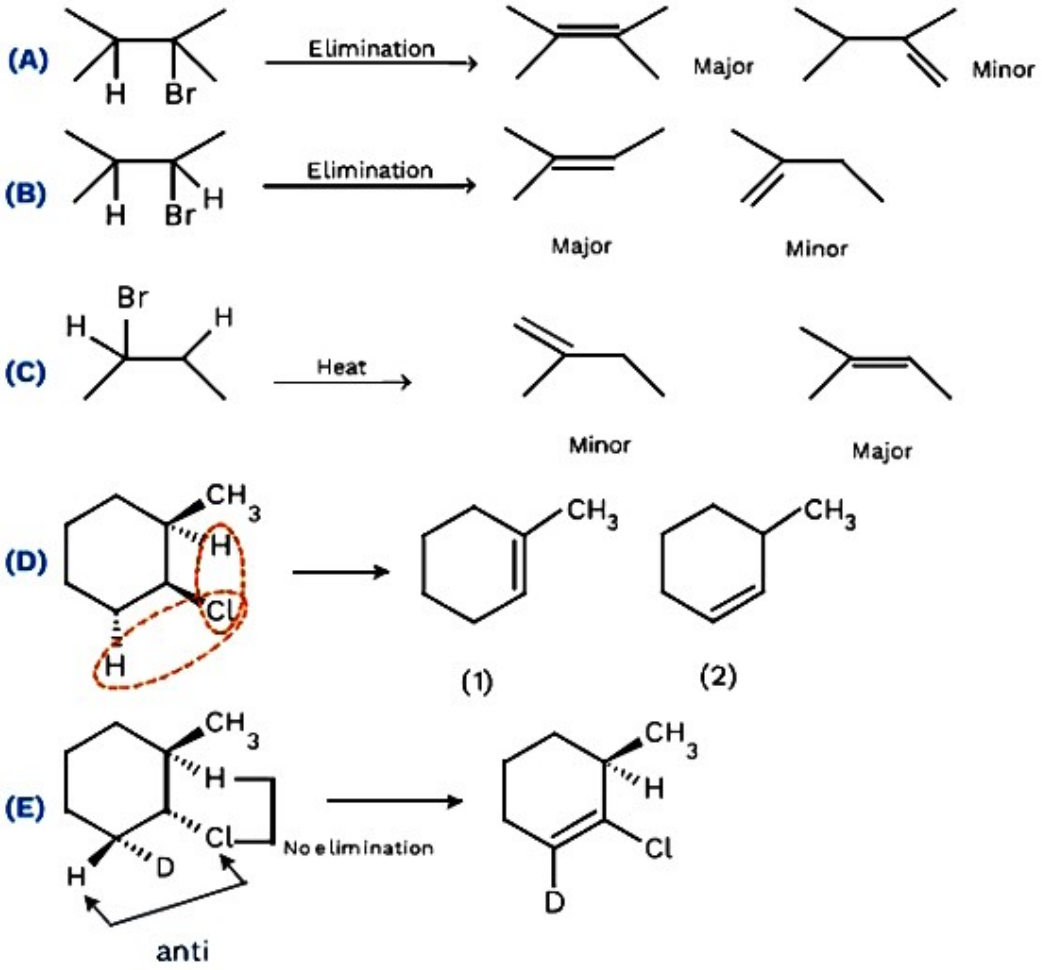
(D)



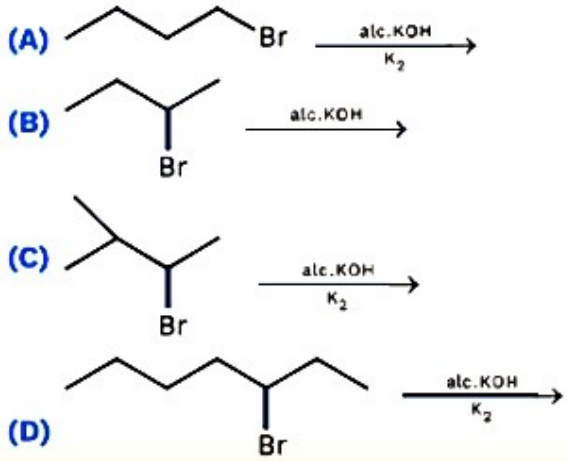
(E)



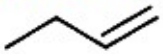
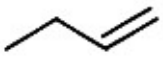
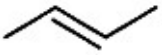
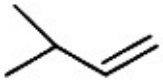
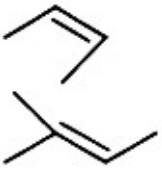
A118



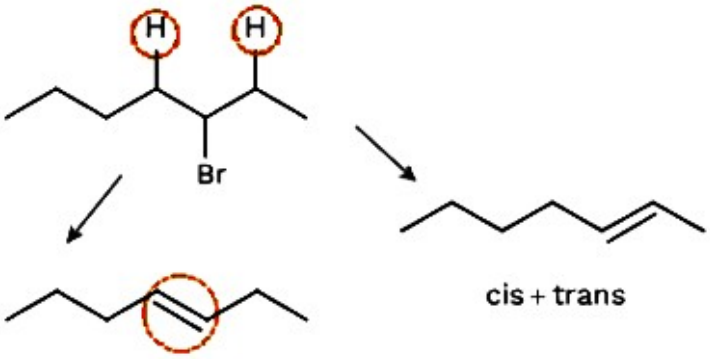
Q119 **Dehydrohalogenation**



A119

	Excluding Stereo isomer	Including Stereo isomer
(A)		
(B)	 	 
(C)		

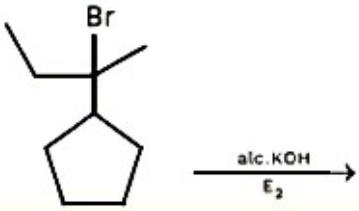
(D) Excluding and including both are same



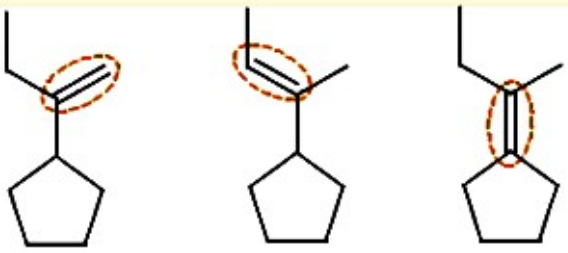
cis + trans

cis + trans

Q120



A120



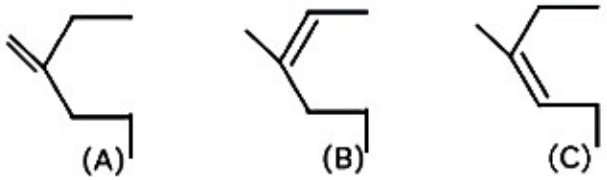
(1) cis + trans (1)

Total = 4 (including stereo)

Q121



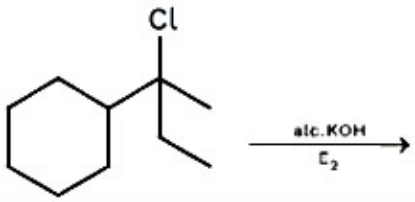
A121



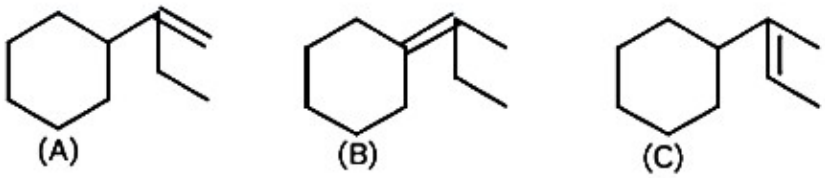
cis + trans

Total = 4 (including stereo)

Q122

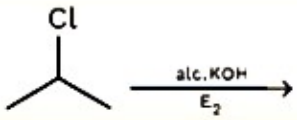


A122

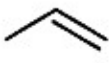


Total = 4 (including stereo)

Q123

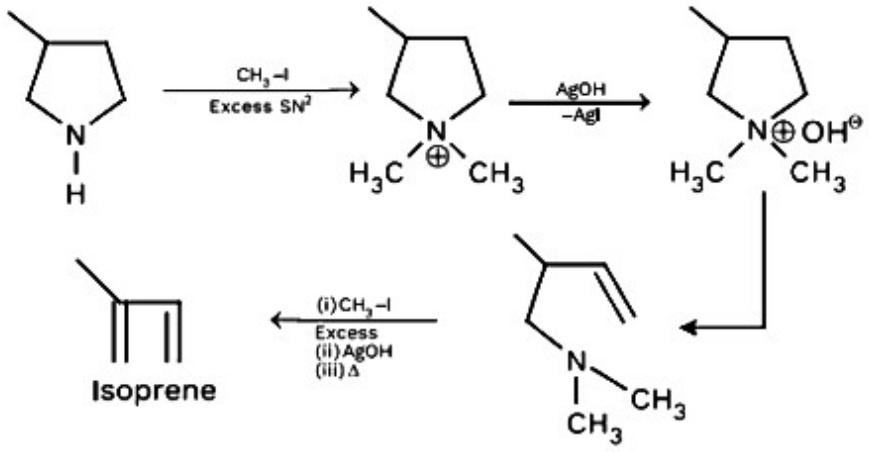


A123

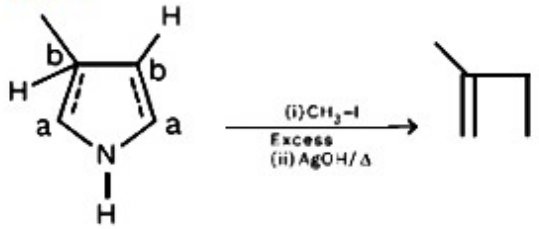


single compound

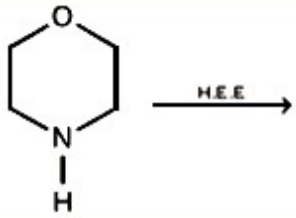
Hofmann Exhaustive Elimination :



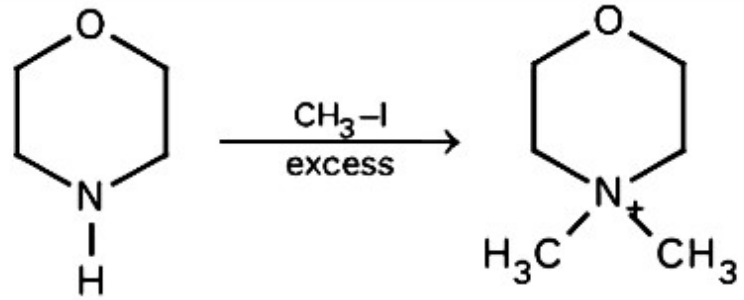
Hint :

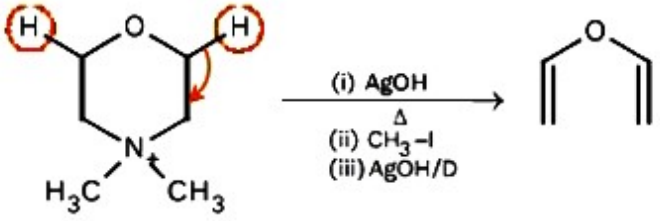


Q124

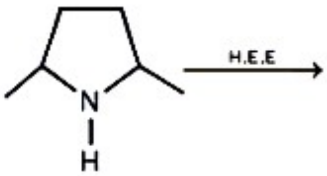


A124

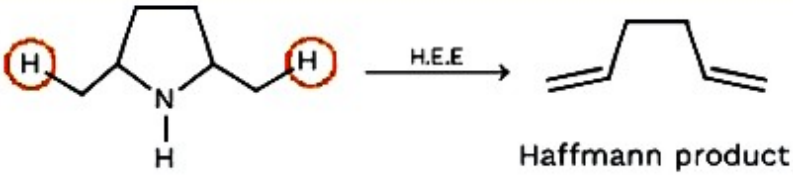




Q125

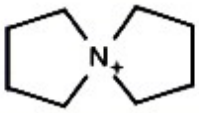


A125

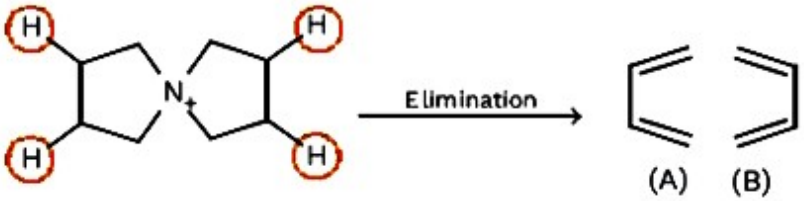


Haffmann product

Q126

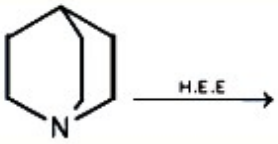


A126

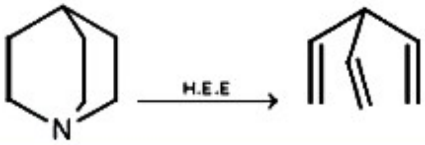


(A) (B)

Q127



A127



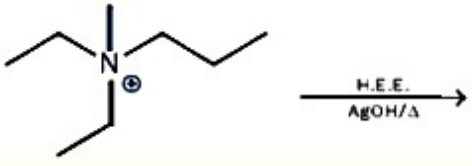
Q128



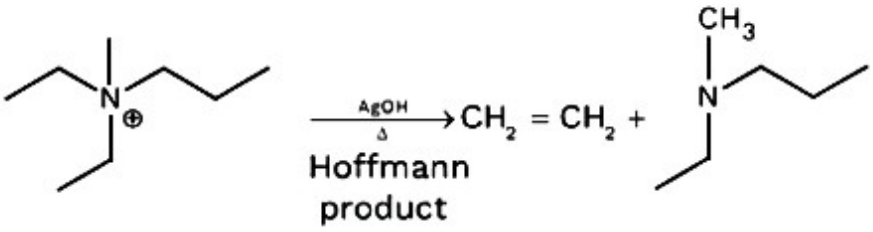
A128



Q129



A129

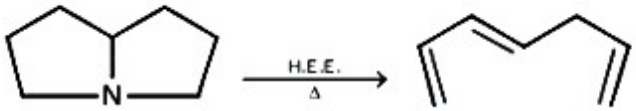


Hoffmann product

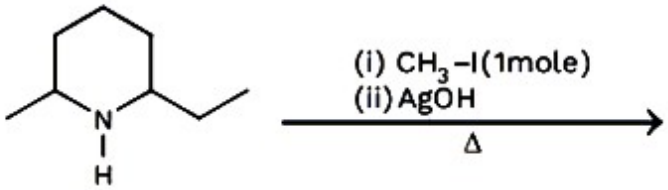
Q130



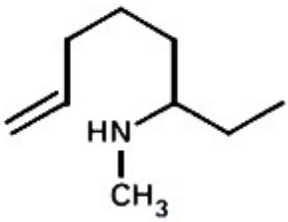
A130



Q131

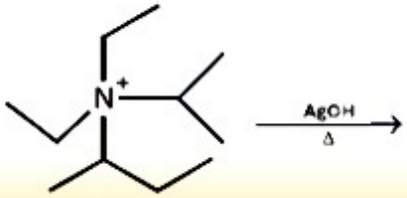


A131

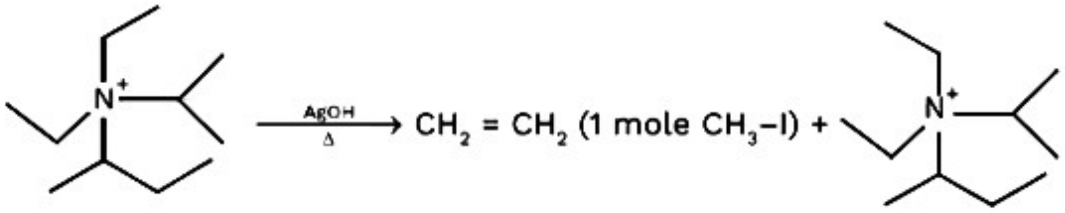


(Single mole $CH_3 - I$)

Q132



A132

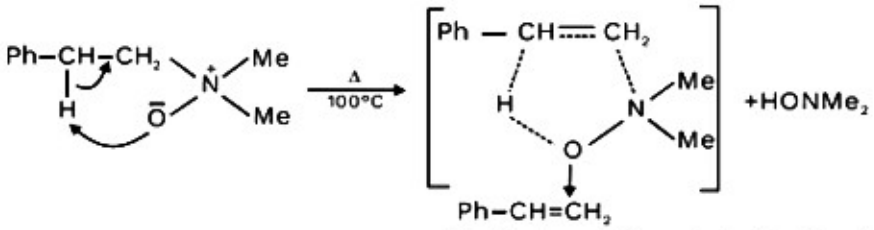


Pyrolytic Syn Elimination Ei (Elimination Intramolecular)

A number of organic compounds such as an amine oxide, ester or xanthate undergo pyrolytic elimination of HX (in absence of any reagent), either in an inert solvent or, in some cases, in the gas-phase. In general these eliminations follows the rate law : Rate = K [substrate]

Cope Reaction

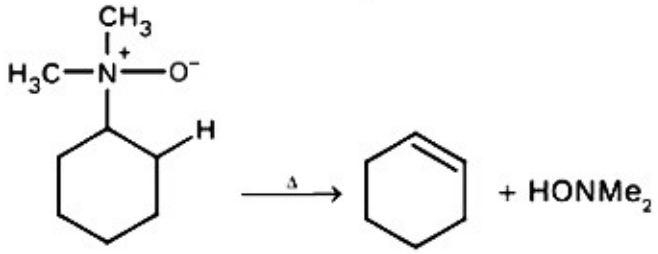
When a tertiary amine oxide is heated at approximately 100°C, elimination reaction takes place as follows:



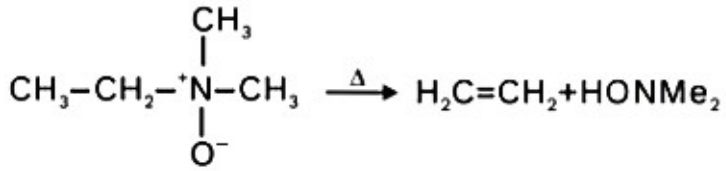
Cyclic transition state (Pericyclic reaction)

Elimination of the above type is named as Cope reaction. H and NMe_2O must lie on the same side for elimination as Me_2NOH. The reaction proceeds via a five-membered cyclic transition state. Some of Cope reaction are:

1.

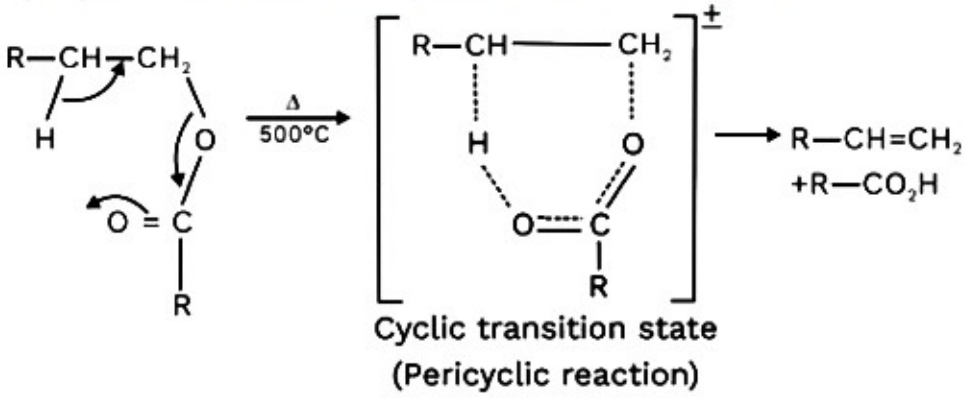


2.



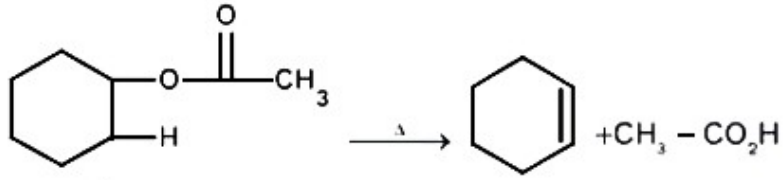
Pyrolysis of Esters

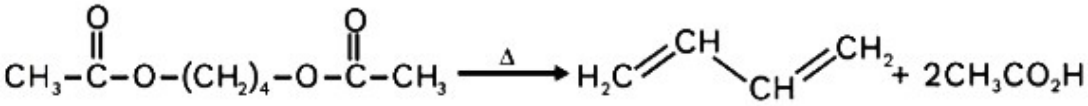
Pyrolysis of an ester takes place when it is heated at 500°C.

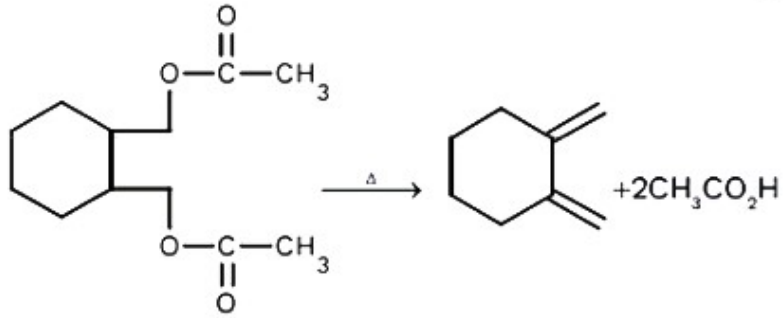


Cyclic transition state
(Pericyclic reaction)

Some examples of pyrolysis of esters are:

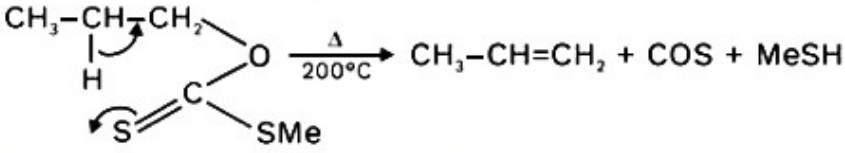
1. 

2. 

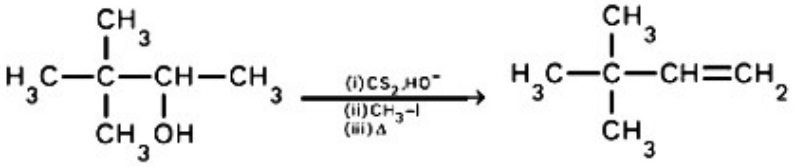
3. 

Pyrolysis of Xanthates or Chugaeve reaction

o-Alkyl-S-methyl xanthate is prolyzed at 200°C to give an alkene, carbon oxysulphide and methanethiol as the products.

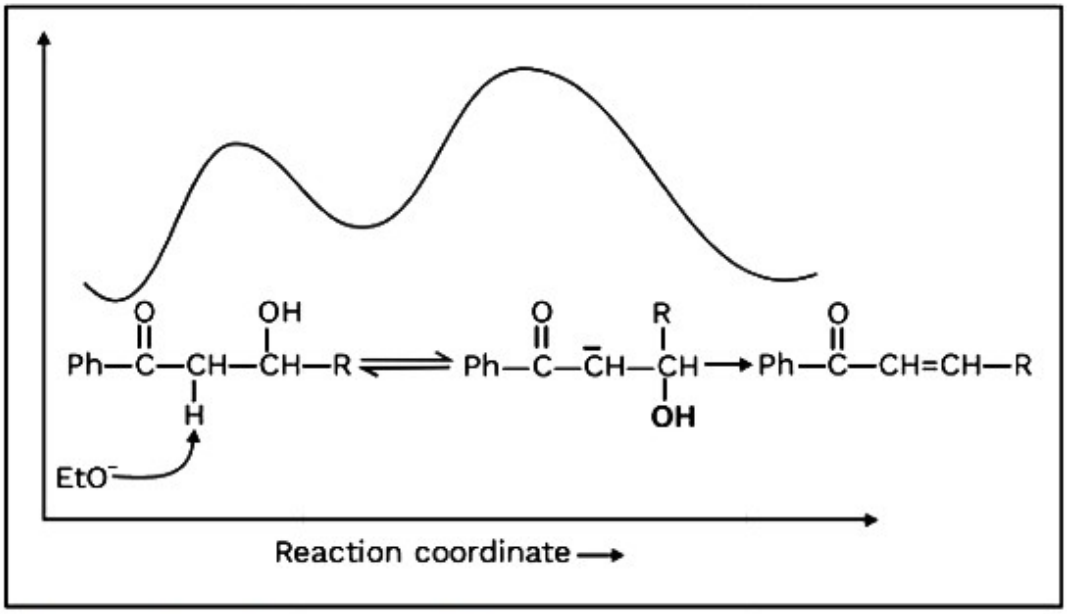


This type of thermal decoposition is called normal Chugaev reaction.

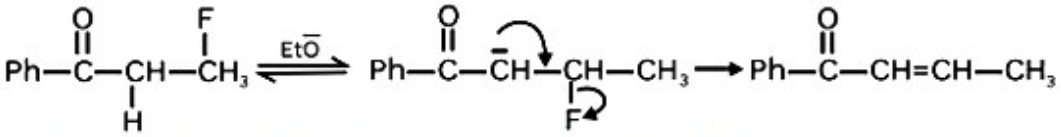


E1CB (Elimination from Conjugated Base)

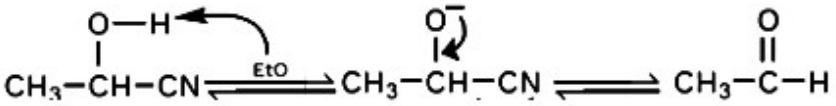
In E_{1CB}, the formation of the carbonation intermediate is fast and reversible, and the loss of the leaving group is slow and rate limiting. This elimination is bimolecular.



- **The examples of E1CB are as follows :**



- **The base induced elimination of cyanohydrin is by E1CB.**



Q133 **Which one of the following compounds will derive Saytzeff product in E_2 reaction?**

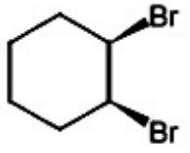
(A) $H_3C-CH_2-\underset{Br}{CH}-CH_3$

(B) $H_3C-CH_2-\underset{F}{CH}-CH_3$

(C) $H_3C-CH_2-CH_2-\overset{\oplus}{N}(CH_3)_3\,\overset{\ominus}{O}H$

(D) $H_3C-CH_2-\underset{OCOCH_3}{CH}-CH_3$

A133 (A) $CH_3-CH_2-CH=CH_2$

(B) 

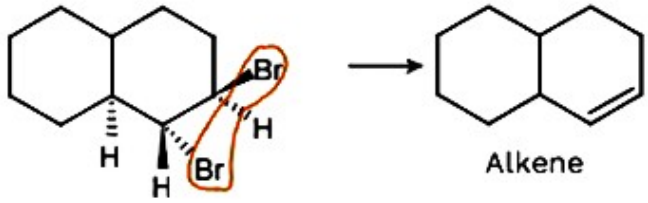
No reaction

(C)

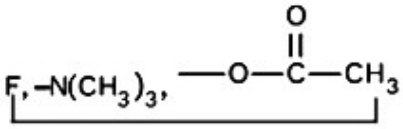


Anti Periplanar

(D)



Alkene



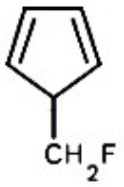
Weak leaving group

* Br is better leaving group support elimination

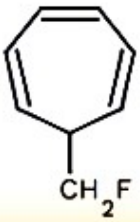
Q134 **Which compound is most reactive for E_{1CB} reaction?**

(A) $CH_3—CH_2F$

(B)



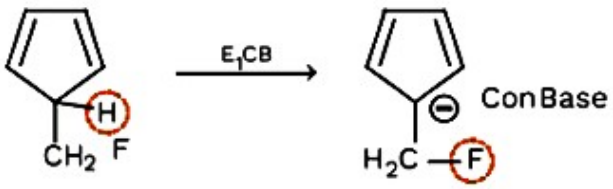
(C)



(D) $C_6H_5—CH_2—CH_2F$

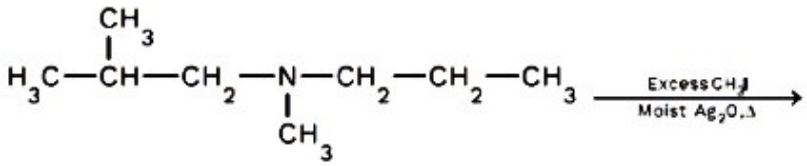
A134 (b)

Bad leaving group and stable conjugate base is regeared for E_1 CB.



Bad L.G.

Q135



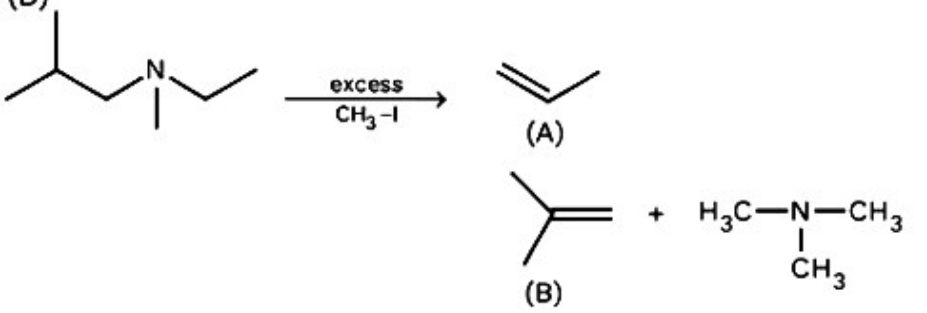
Product mixture $\xrightarrow{O_3/Zn-H_2O}$ **Final product mixture. The final product mixture contains :**

(A) $CH_3CHO + CH_3COCH_3$ **(B)** $CH_3CHO + CH_3CH_2CH_3 + HCHO$

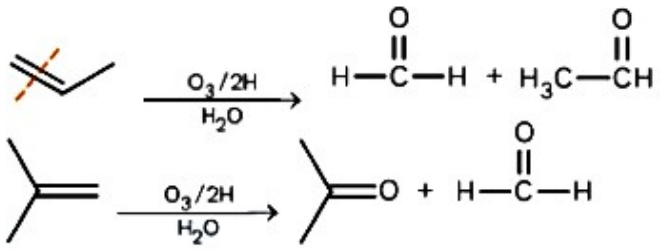
(C) $CH_3CHO + HCHO$ **(D)** $CH_3CHO + CH_3COCH_3 + HCHO$

A135

(D)



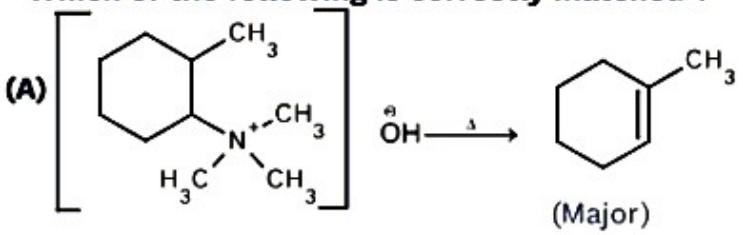
Ozonolysis of A :



Q136

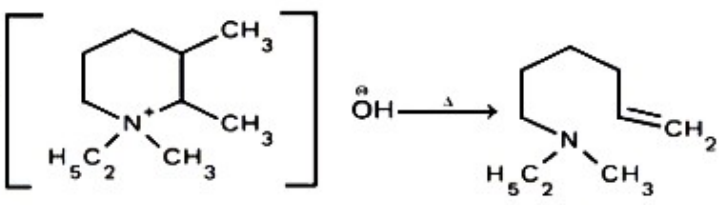
Which of the following is correctly matched ?

(A)



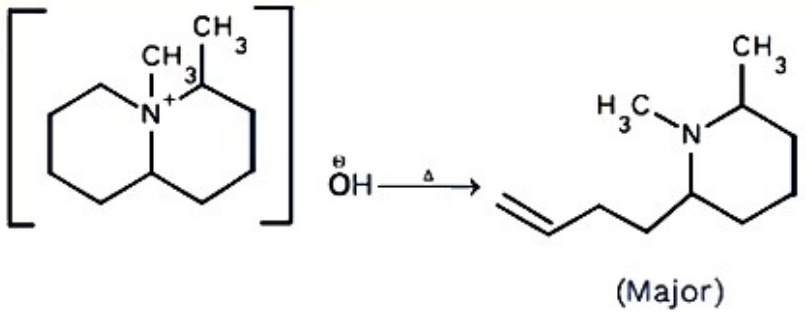
(Major)

(B)



(Major)

(C)

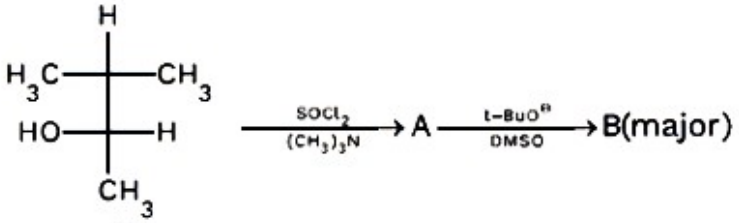


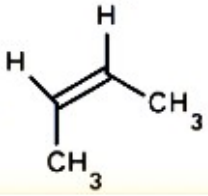
(Major)

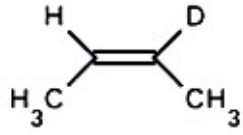
(D) All of these

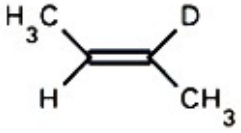
A136 (C)
Least substituted alkene will be final product.

Q137 **The final product of the given reaction sequence is :**



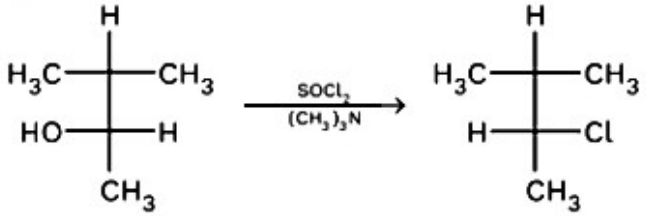
(A) 

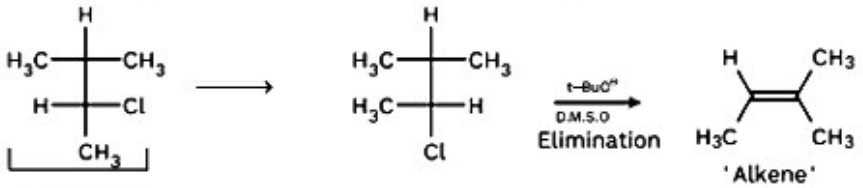
(B) 

(C) 

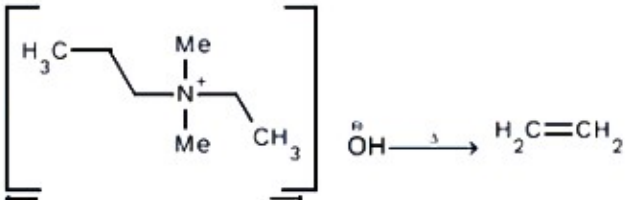
(D) none of these

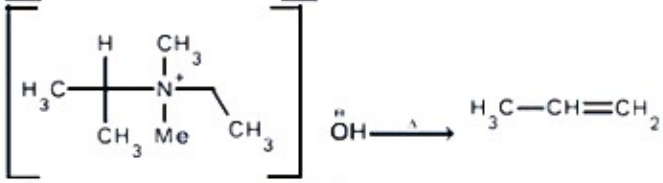
A137 (D)

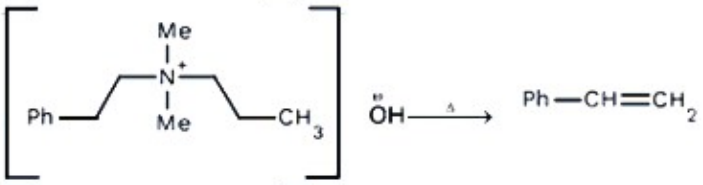


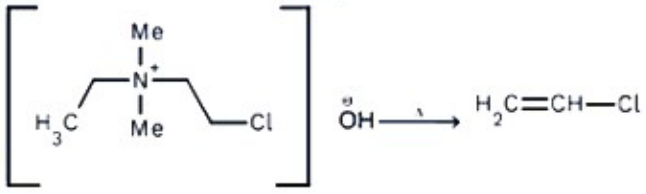


Q138 Which of the following reaction is correct regarding the formation of major product (alkene)?

(A) 

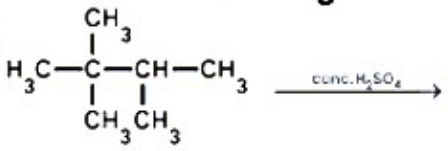
(B) 

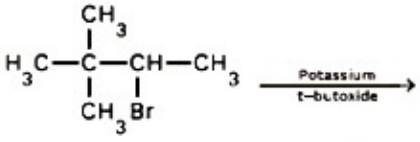
(C) 

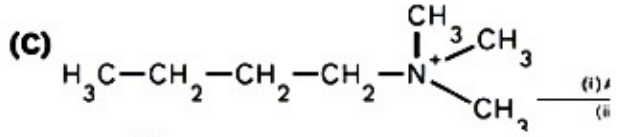
(D) 

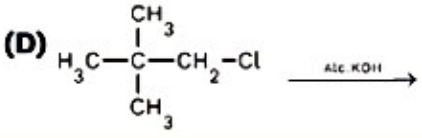
A138 (ABCD)

Q139 In which of the following Hoffman's elimination product is more?

(A) 

(B) 

(C) 

(D) 

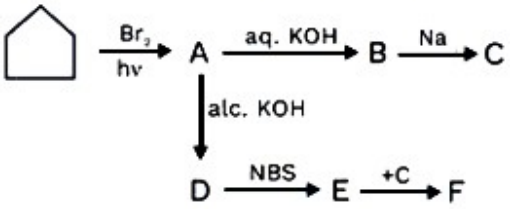
A139 (BC)

Q140 **Match the following :**

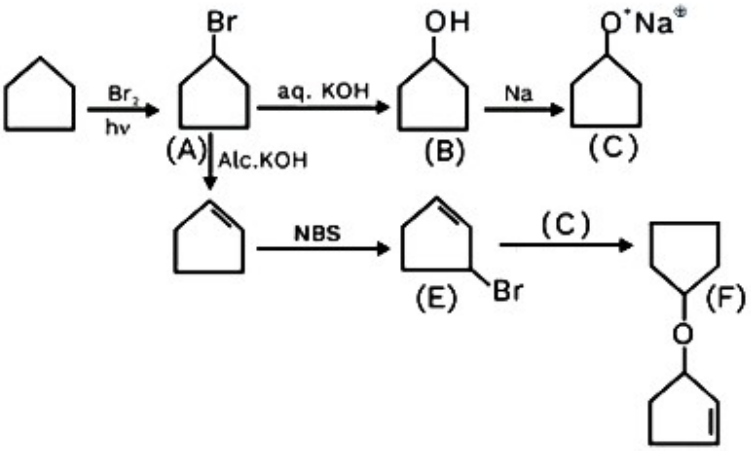
Column I	Column II
(A) $CH_3-CHBr-CD_3$ on treatment with alc. KOH gives $CH_2=CH-CD_3$ as a major product.	(P) E1 reaction
(B) $Ph-CHBr-CH_3$ reacts faster than $Ph-CHBr-CD_3$.	(Q) E2 reaction
(C) $Ph-CD_2-CH_2Br$ on treatment with $C_2H_5OD/C_2H_5O^-$ Gives $Ph-CD=CH_2$ as the major product.	(R) E1 cb reaction
(D) $PhCH_2CH_2Br$ and $PhCD_2Br$ react with same rate.	(S) Ist order reaction

A140 (A-Q; B-Q; C-RS; D-PS)

Q141 **Obtain A,B,C,D,E and F in the following series of reaction.**



A141



Q142 **Match the List I with List II and select the correct answer using the codes given below the Lists.**

List I	List II
(A) E1CB	**(1) 3° Amine oxide**
(B) Saytzeff alkene as major product	**(2) Xanthate**
(C) E2	**(3)** $CH_3-CH_2-\underset{Cl}{CH}-CH_2$
(D) Ei	**(4)** $C_6H_5-CH_2-\underset{F}{CH}-CH_3$

A142 (A-4; B-3,4; C-3; D-1,2)

Q143 **Which of the following will give syn-elimination**

(A) $CH_3CH_2CH_2-OCOCH_3$

(B) $C_6H_5-CH_2-\underset{Cl}{CH}-CH_3$

(C) $CH_3CH_2CH_2-O-\overset{\overset{S}{\|}}{C}-S-CH_3$

(D) $CH_3CH_2CH_2\overset{\oplus}{N}(CH_3)_2-O^{\ominus}$

A143 (ACD)

Q144 **Match the List-I (reaction with List II) (reaction intermediate) and select the correct answer using the codes given below the Lists.**

List-I	List-II
(A) $CF_3-CHCl_2 \xrightarrow[\Delta]{\text{alc.KOH}} CF_2=CCl_2$	**(1) Transition state only**
(B) $CH_3-\underset{CH_3}{\overset{CH_3}{C}}-OH \xrightarrow{H^{\oplus}} CH_3-\underset{CH_3}{C}=CH_2$	**(2) Carbocation**
(C) $CH_3-CH_2-Br \xrightarrow[\Delta]{\text{alc.KOH}} CH_2-CH_2$	**(3) Carbanion**
(D) $CH_3-\underset{CH_3}{\overset{Br}{C}}-CH_3 \xrightarrow[\Delta]{\text{alc.KOH}} CH_3-\underset{CH_3}{C}=CH_2$	**(4) Free radical**

A144 (A-3; B-2; C-1; D-2)

Q145 **Which alkyl halide undergo E_2 elimination**

(A) 

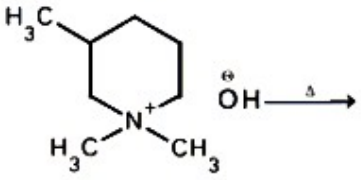
(B) 

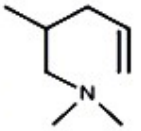
(C) 

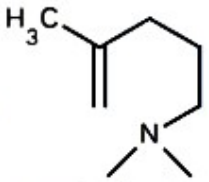
(D) 

A145 (AB)

Q146 **Which of the following are correct for given reaction :**



(A) Major product of reaction is 

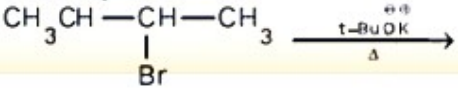
(B) Major product is 

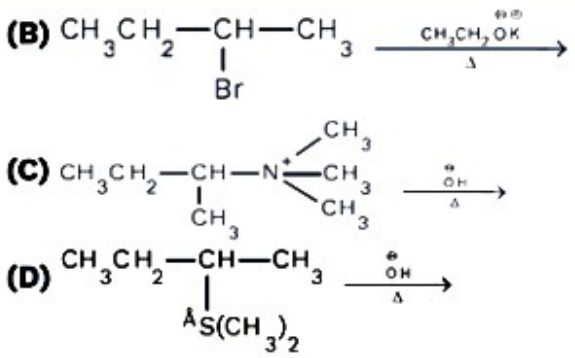
(C) The reaction is thermal elimination reaction

(D) The reaction is E_2 reaction

A146 (ACD)

Q147 **In which product formation takes place according to Hoffmann's rule**

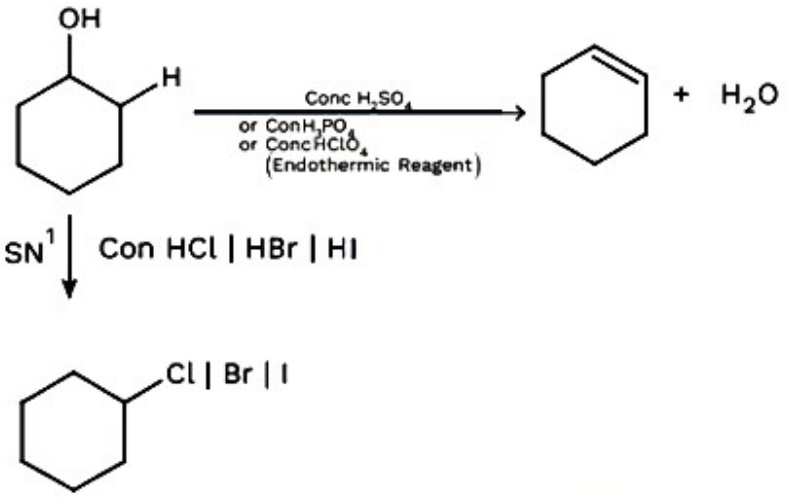
(A) 



A147 (ACD)

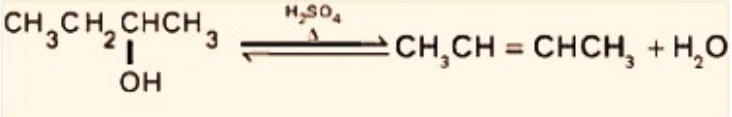
E1 (Unimolecular Elimination)
Dehydration of alcohol

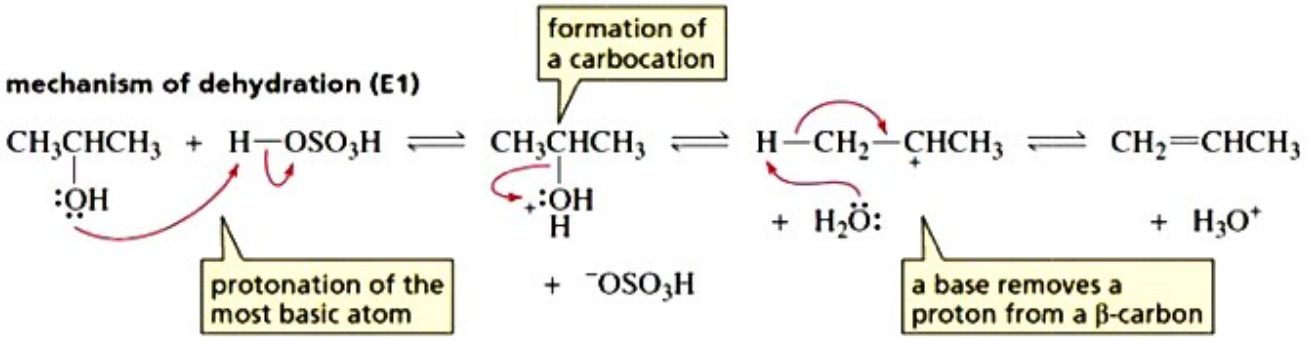
Basic difference in SN1 & E1



Elimination Reactions of alcohols: Dehydration

An alcohol can undergo an elimination reaction, forming an alkene by losing an OH from one carbon and an H from an next to carbon. Overall, this quantities to the elimination of a molecule of water. Elimination of water from a molecule is called **dehydration**. Dehydration of an alcohol requires an acid catalyst and heat. Sulfuric acid and phosphoric acid are the most commonly used acid catalysts.





Step 1. An acid always reacts with an organic molecule in the same way: It protonates the most basic (electron-rich) atom in the molecule. Thus, in the first step of the dehydration reaction, the acid protonates the oxygen atom of the alcohol

Step 2. Protonation converts the very poor leaving group OH^- into a good leaving group H_2O.

Step 3. In the next step, water departs, leaving behind a carbocation. A base in the reaction mixture (water is the base in highest concentration) removes a proton from α -carbon (α-carbon adjacent to the positively charged carbon), forming an alkene and regenerating the acid catalyst. Notice that the dehydration reaction is an E1 reaction of a β protonated alcohol.

Golden Points

A. In order to undergo dehydration, tertiary alcohols must be heated to about 50°C in 5% H_2SO_4

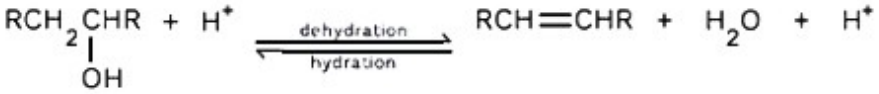
B. Secondary alcohols must be heated to about 100 °C in 75% H_2SO_4

C. Primary alcohols can be dehydrated only under extreme conditions (170°C in 95% H_2SO_4) and by a different mechanism because primary carbocations are too unstable to be formed.

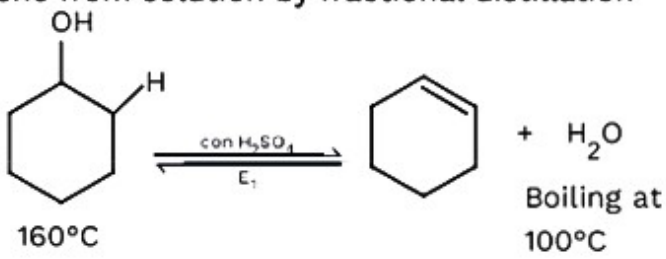
Q148 **Explain why the acid-catalyzed dehydration of an alcohol is a reversible re action, whereas the base-promoted dehydrohalogenation of an alkyl halide is an irreversible reaction**

A148 To stop the alkene formed in the dehydration reaction from adding water and reforming the alcohol, the alkene can be detached by distillation as it is formed, since it has a much lower boiling point than the alcohol. Eliminating a product dis-

places the reaction to the right. An alkene is hydrated (adds water) in the presence of an acid catalyst, thereby forming an alcohol. The hydration of an alkene is the opposite of the acid-catalyzed dehydration of an alcohol.

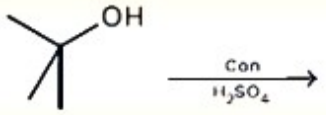


E_1 is reversible reaction to move in forward direction high temperature is required which separate alkene from solution by fractional distillation

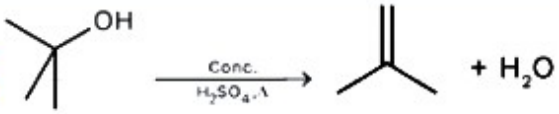


Draw products of unimolecular elimination (dehydration).
Example :

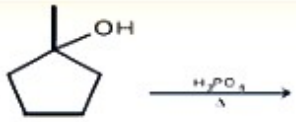
Q149



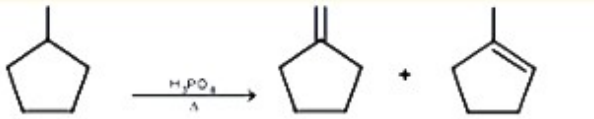
A149



Q150

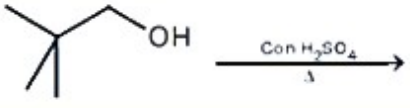


A150



(Hoffmann) (Saytzeff)
Major product

Q151



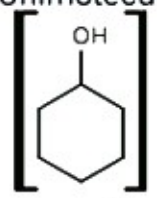
A151



Hoff mann Saytzeff

Some Facts:

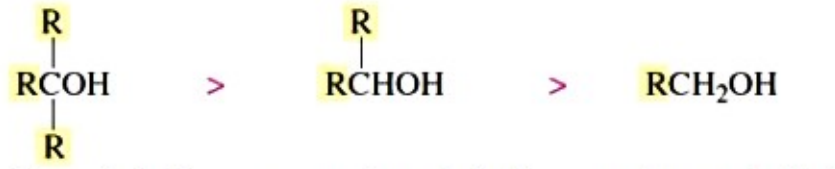
1. E_1 is a Unimolecular reaction because rate depends on con. of compound.

Rate $\propto$  [Con H_2SO_4] (Excess)

Pseudo 1st order

2. Multistep reaction process in which 1st step oxonium ion is formed.
3. 2nd step, cation formation is R.D.S. It means more the stable cation, more the rate of reaction.
4. Dehydrating agent present absorb water and form H-bonding.
   (a) $ConH_3PO_4$ **(b)** $ConH_2SO_4$ **(c)** $KHSO_4$ **(d)** Al_2O_3 **(e)** P_2O_5/P_4O_{10} **(f)** $POCl_3$ **(g)** ThO_2
5. Rate-determining step in the dehydration of a secondary or a tertiary alcohol is formation of a carbocation intermediate, the rate of dehydration parallels the ease with which the carbocation is formed.
6. Tertiary alcohols are the easiest to dehydrate because tertiary carbocations are more stable and therefore are easier to form than secondary and primary carbocations

relative ease of dehydration

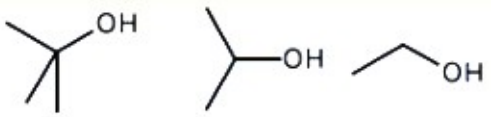


a tertiary alcohol **a secondary alcohol** **a primary alcohol**



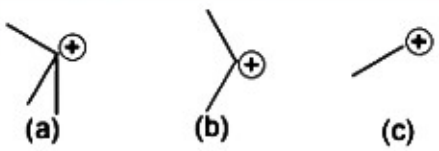
Compare rate of reaction of dehydration reaction :

Q152



(a) (b) (c)

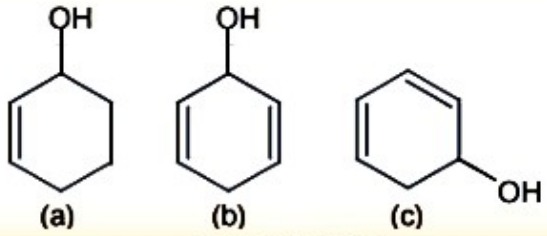
A152



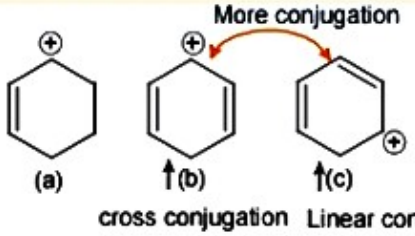
(a) (b) (c)

Rate of E_1 is proportional to stability of carbocation : a > b > C

Q153

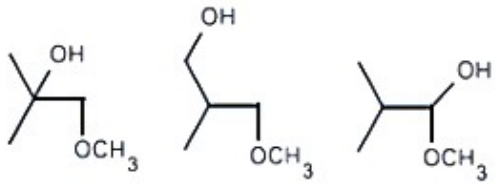


A153

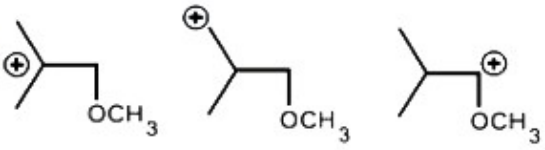


Linear conjugation is more stable than cross conjugation.
Hence, a < b < c order of E_1 rate

Q154

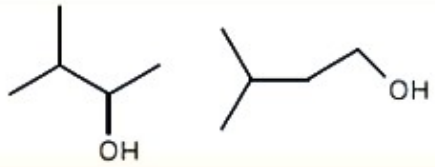


A154

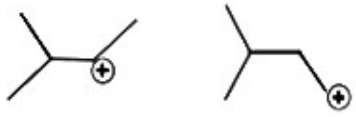


(a) (2° carbocation) (b) (1° carbocation) (c) (Resonance stabilized)
[8α–H]
Stability order : c > a > b ; Rate : c > a > b

Q155



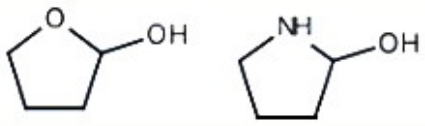
A155



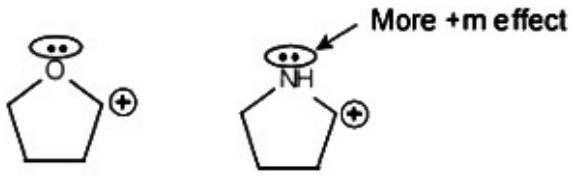
(4α–H) 2° (2αH) 1°

a > b

Q156

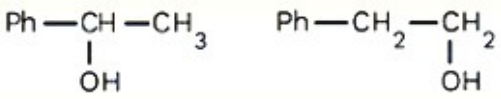


A156



b > a

Q157



A157

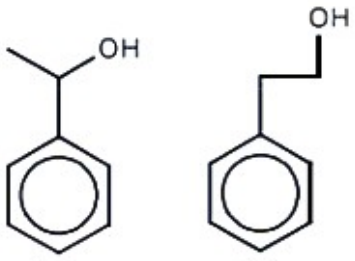


(Resonance stabilized) (1° carbocation)

(More stable)

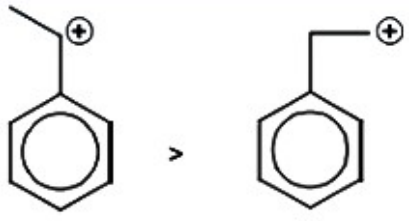
a > b

Q158



(a) (b)

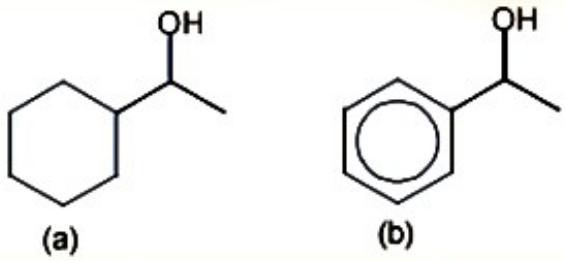
A158



(a) (b)

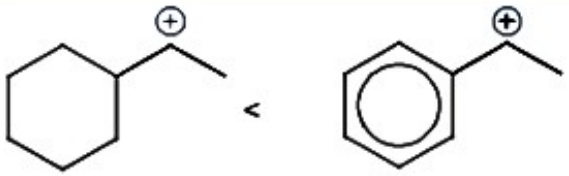
(Resonance stablized)

Q159



(a) (b)

A159

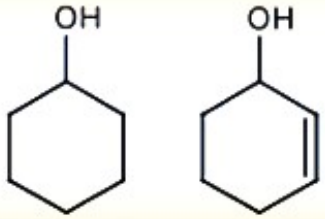


(a) < (b)

(Resonance stablized)

b > a

Q160



A160



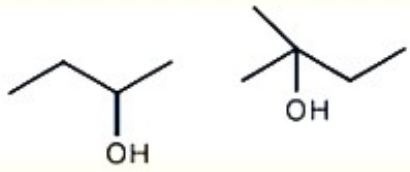
(a) (b)

2° 2° with

(Resonance stablized)

b > a

Q161



A161



2° 3°

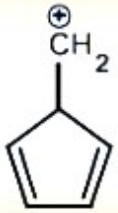
5α-H 8α- H

b > a

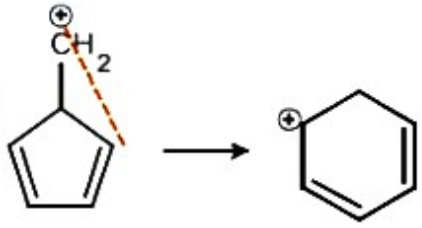
7. Rearrangement is possible due to formation of (positive charge) Carbocation in reaction.
8. Cation can be rearrange into more stable cation by
   1, 2 shit
   Ex. 1, 2 H – Shift
   Ex. 1, 2 CH_3 – Shift
   Ex. 1, 2 Ph – Shift

Which of the following carbocation will undergo rearrangement?

Q162

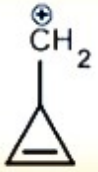


A162

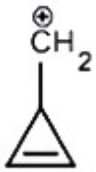


(Rearrangement via ring expansion)

Q163

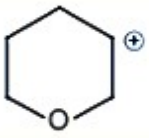


A163

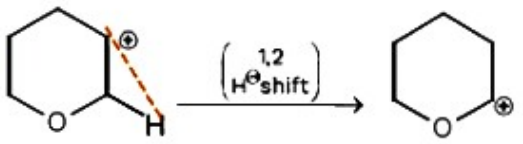


Highly stable carbocation due to dancing resonance.
Will not rearrange.

Q164

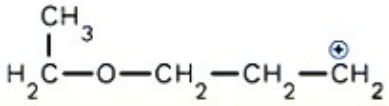


A164



(Resonance stabilized)

Q165



A165

$CH_3-CH_2-O-CH_2-CH_2-\overset{\oplus}{C}H_2$ [2α–H]

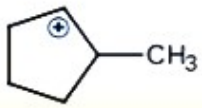
(1, 2-$H^{\ominus}$ shift.)

$CH_3-CH_2-O-CH_2-\overset{\oplus}{C}H-CH_3$ [5α–H]

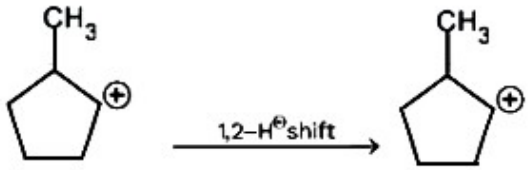
(1, 2-$H^{\ominus}$ shift.)

$CH_3-CH_2-O-\overset{\oplus}{C}H-CH_2-CH_3$ [Resonance stabilized]

Q166



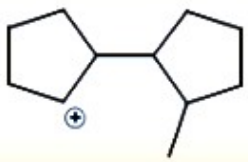
A166



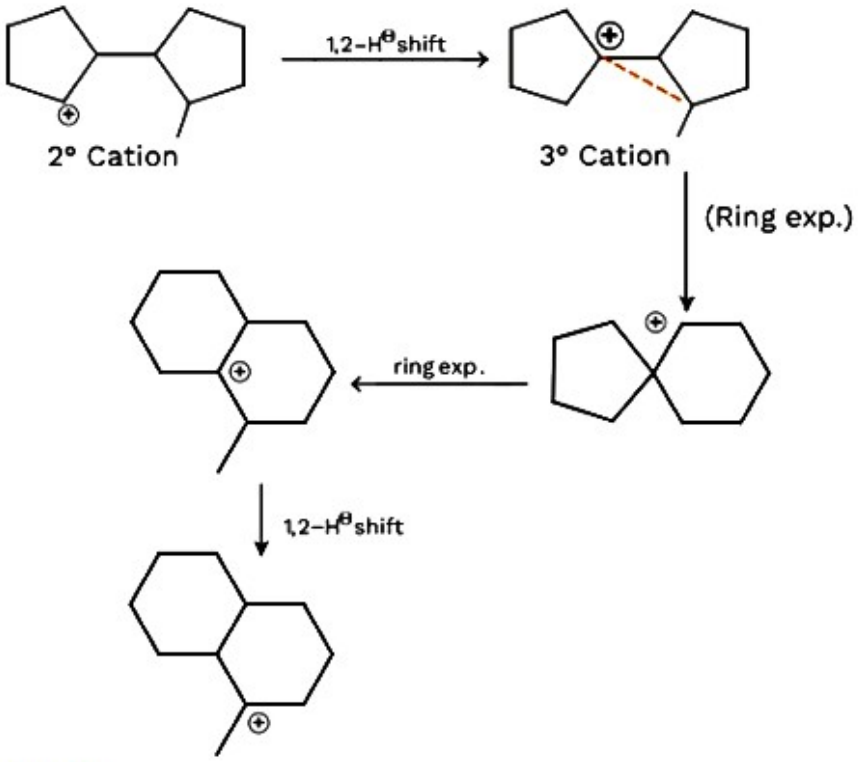
(3α-H) (7α-H)

3° cation

Q167

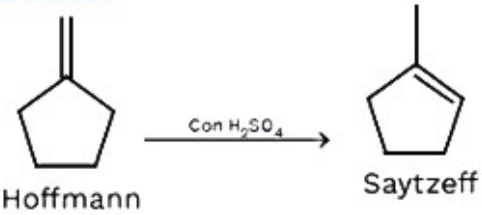


A167

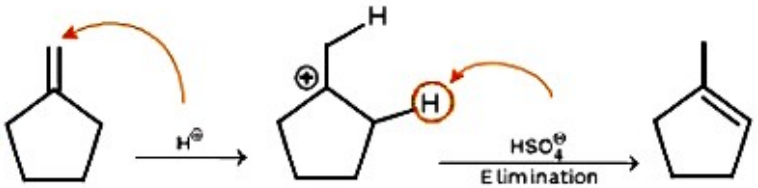


Isomerization

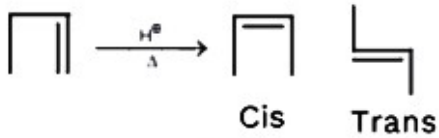
(a)



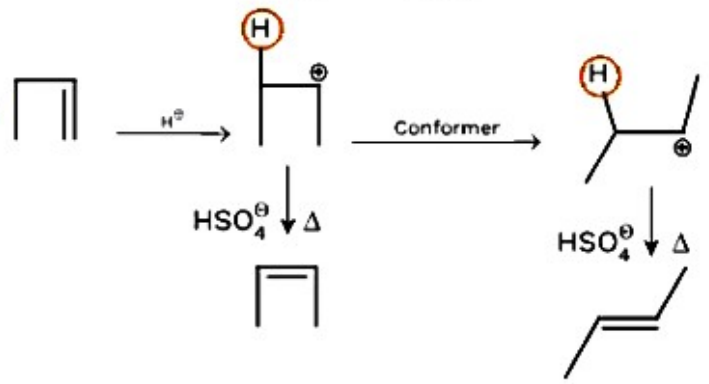
Ans.



(b)

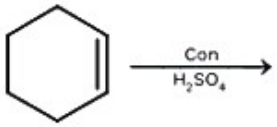


Ans.

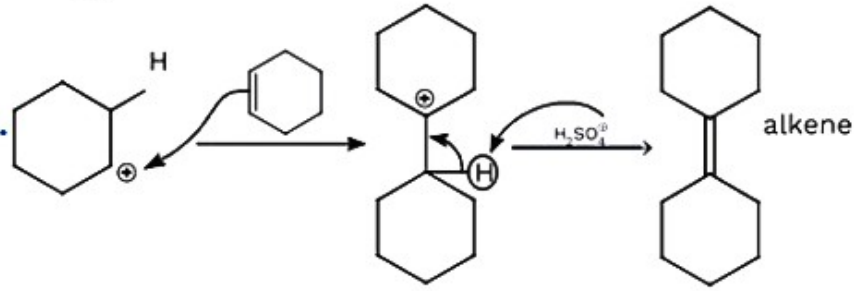


Polymerization

(a)

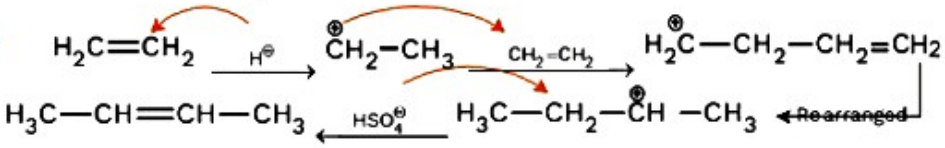


Ans.

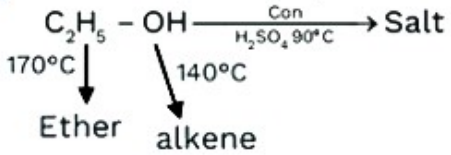


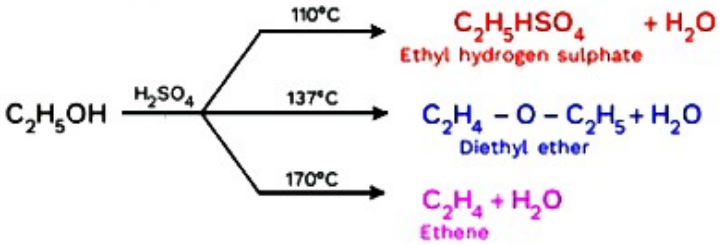
(b) $CH_2 = CH_2 \xrightarrow[H_2SO_4]{Con} CH_2 - CH = CH - CH_3$ Explain?

Ans.

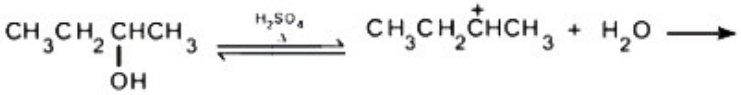


Special case (N.C.E.R.T)

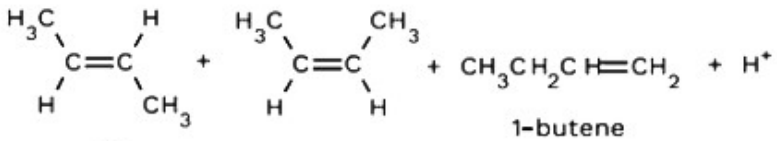




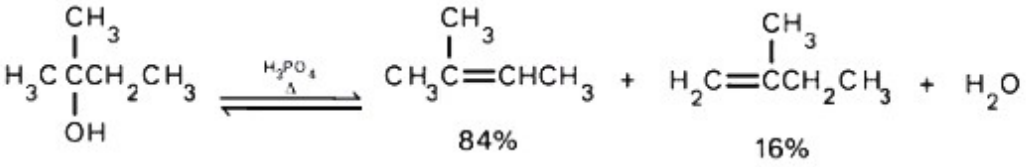
Stereochemical outcome of the E1 dehydration of an alcohol

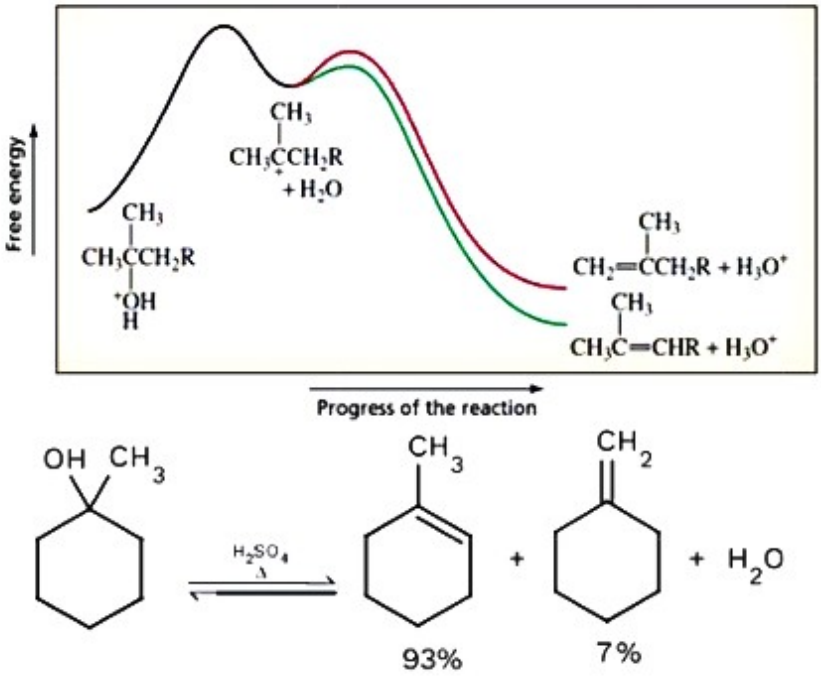


2-butanol



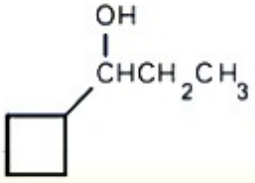
- When more than one elimination produce can be formed, the major product is the more substituted alkene, the one obtained by removing a proton from the carbon that is bonded to the fewest hydrogens
- The high substituted alkene is the major product because it is the more stable alkene, so it has the more stable transition state leading to its formation



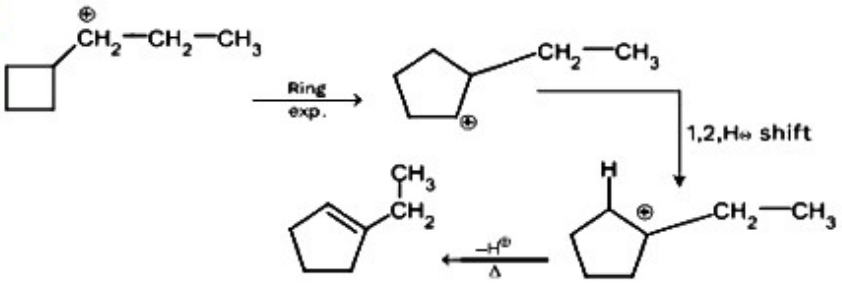


Give the major product formed when each of the following alcohols is heated in the presence of H_2SO_4.

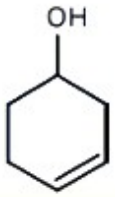
Q168



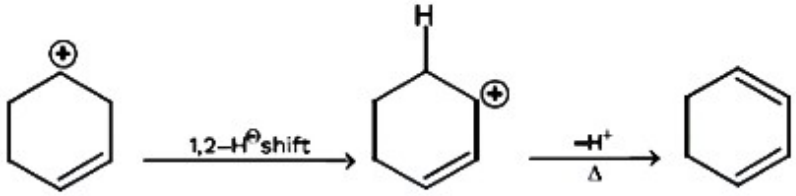
A168



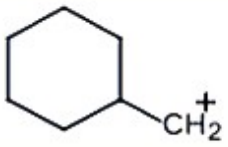
Q169



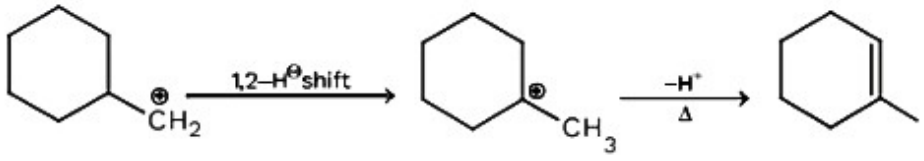
A169



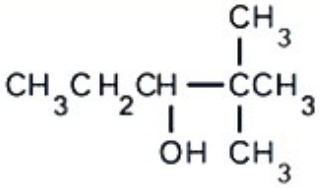
Q170



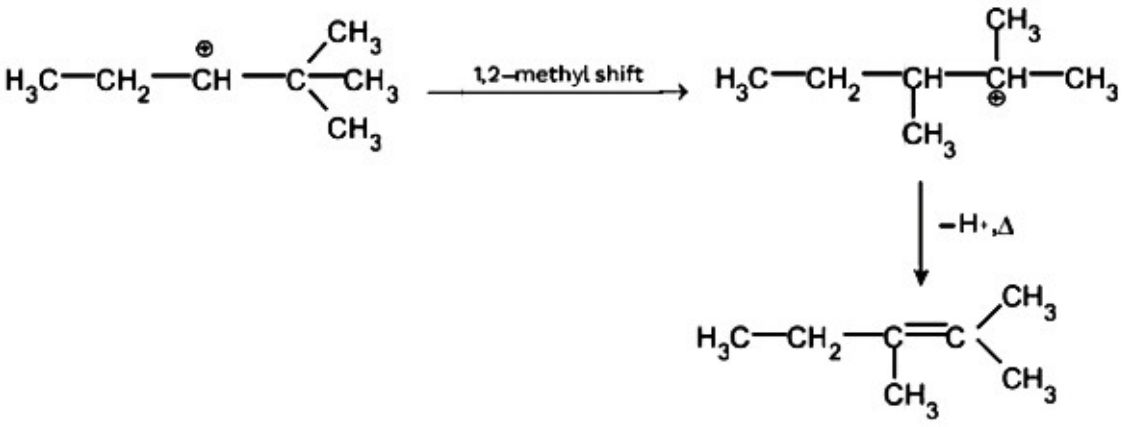
A170



Q171



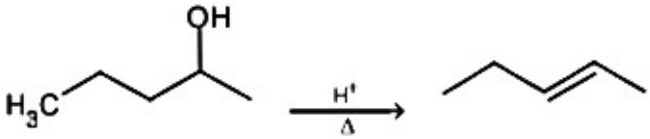
A171



Predict the major product of acid-catalysed dehydration of alcohols

Q172 **2-pentanol**

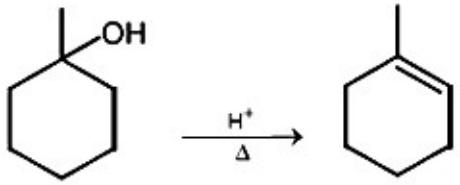
A172



(2-pentanol)

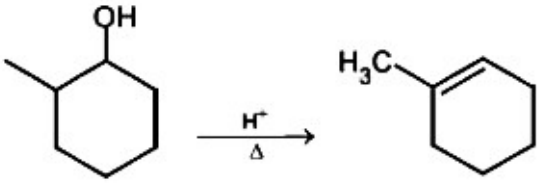
Q173 **1-methylcyclopentanol**

A173



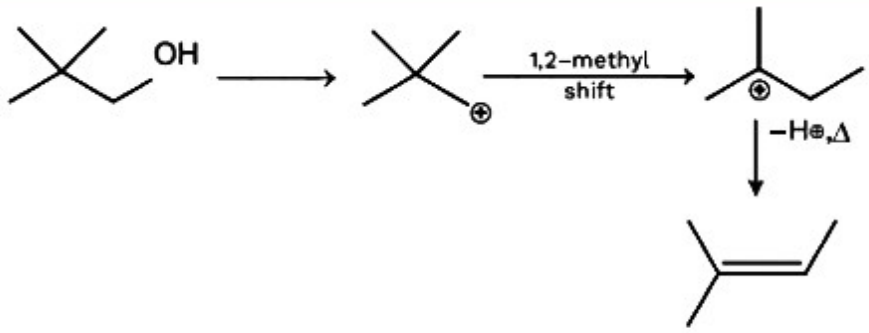
Q174 **2-methyl cyclohexanol**

A174

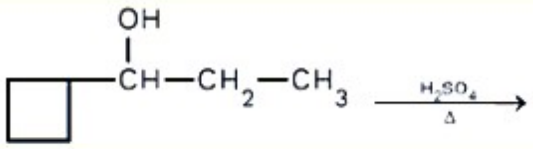


Q175 2,2-dimenthyl-1-propanol

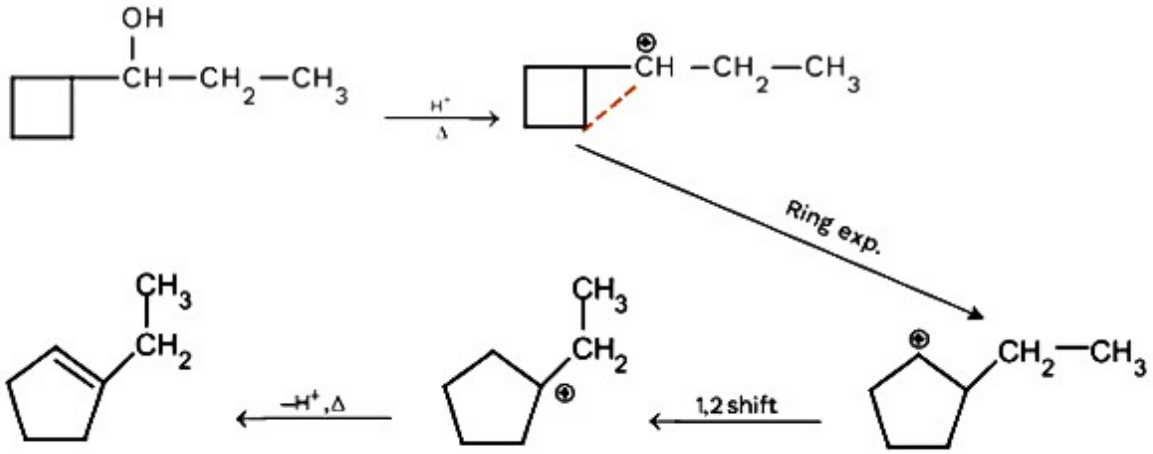
A175



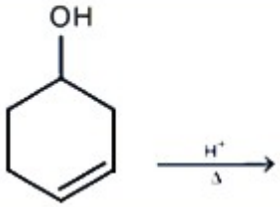
Q176



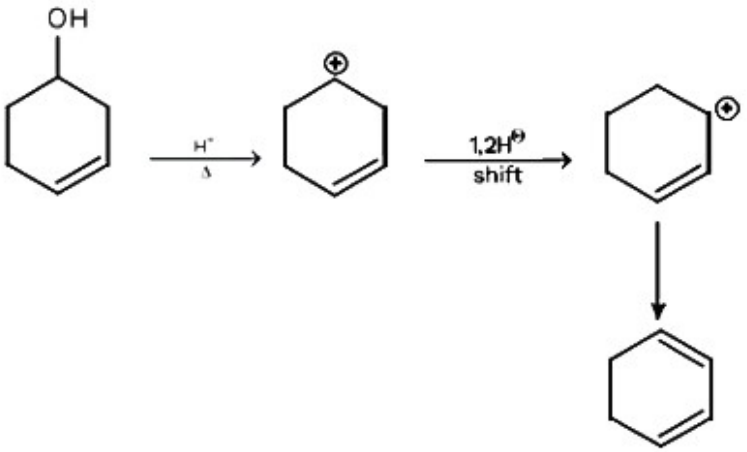
A176



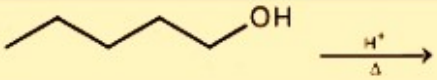
Q177



A178



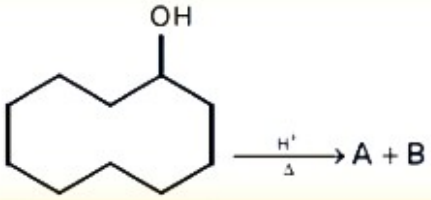
Q179



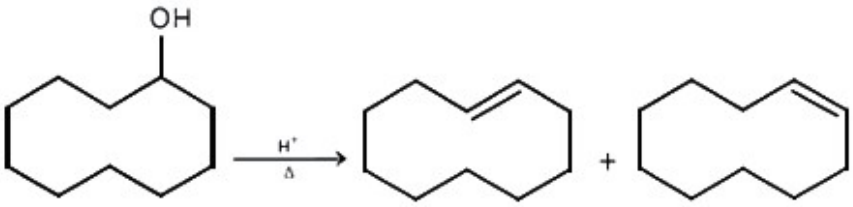
A179



Q180



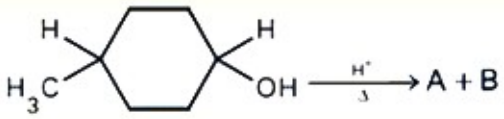
A180



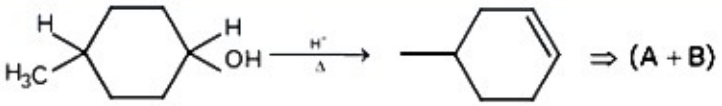
A
(trans)

B
(cis)

Q181

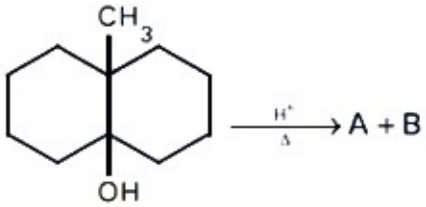


A181

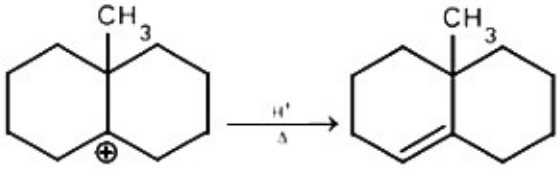


Single product

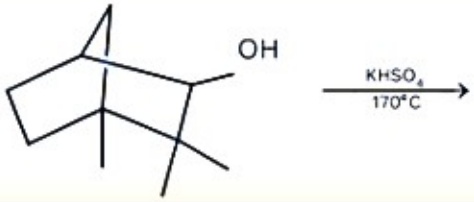
Q182



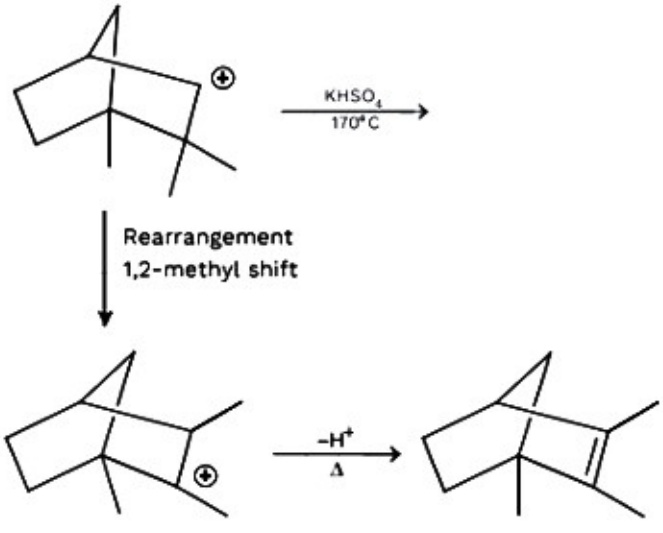
A182



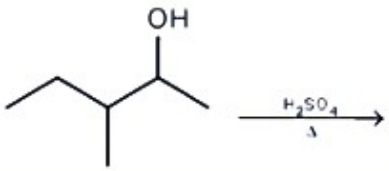
Q183



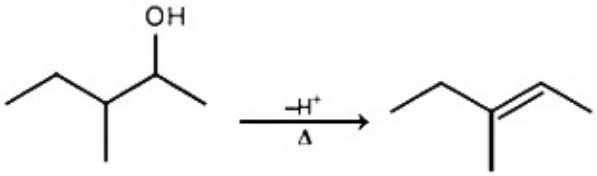
A183



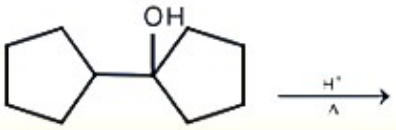
Q184



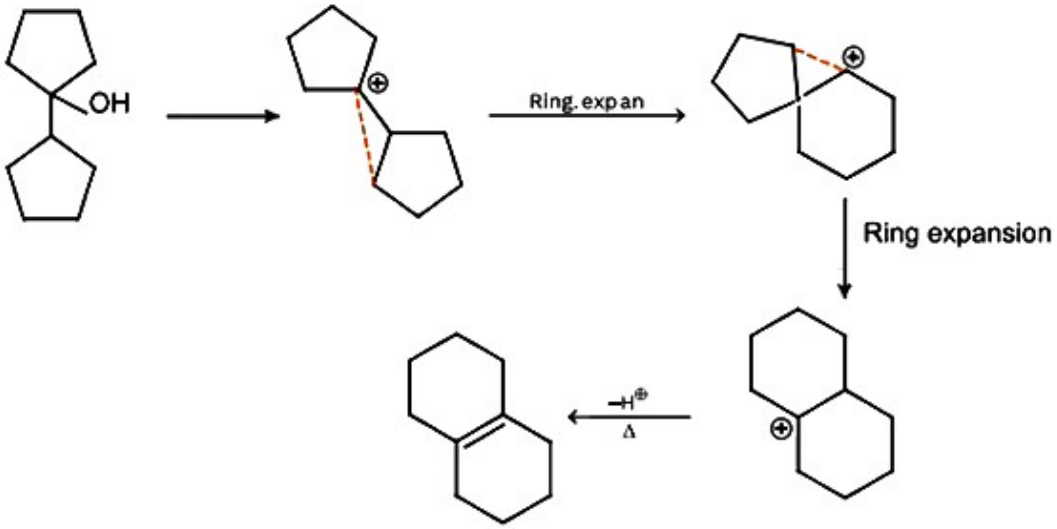
A184



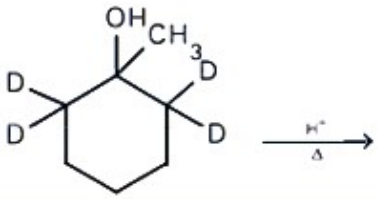
Q185



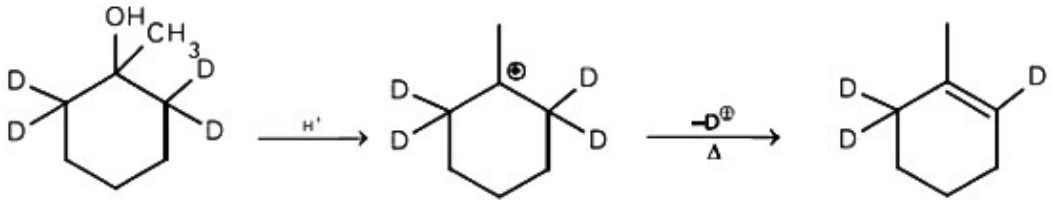
A185



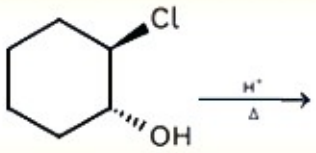
Q186



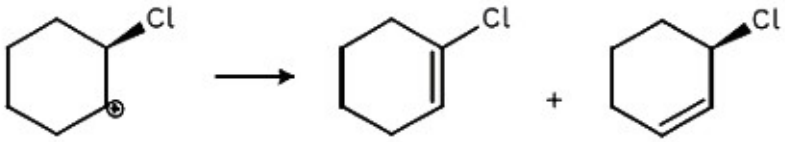
A186



Q187



A187



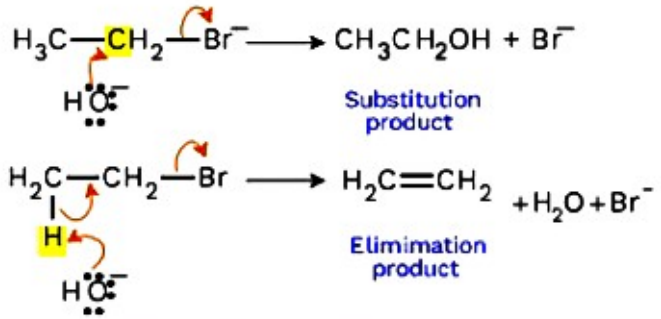
Points to be followed :

- We must decide whether the reaction conditions favour S_N2/E2 or SN1/E1 reactions. Conditions that favour an S_N2 reaction also favours an E2 reaction and the conditions that favour an S_N1 reaction also favour an E1 reaction
- If the reactant is a primary alkyl halide-it undergoes only S_N2/E2 reactions. Primary carbocations are highly unstable to be formed. so primary alkyl halides cannot undergo S_N1/E1 reactions. (Except some special cases)
- If the reactant is a secondary or a tert alkyl halide, it may undergo either S_N2/E2 or S_N1/E1 reactions, depending on the reaction conditions.
- S_N2/E2 reactions are preferred by a high concentration of a good nucleophilic/ strong base,
- S_N1/E1 reactions are favored by a poor nucleophile/weak base.
- The solvent in which the reaction is processed out can influence the mechanism.
- The comparative amounts of substitution and elimination output will depend on whether the alkyl halide is primary. Secondary, or tertiary, and on the nature of the nucleophile/base.

Conditions that lead to S_N2/E2 reactions:

A high con. of a efficient nucleophile/strong base.
For Substitution: The negatively charged species can act as a nucleophile and attack the back side of the a-carbon to form the substitution product.
For Elimination: It can act as a base and remove a b-hydrogen, leading to the elimination product. Thus, the two reactions compete with each other.

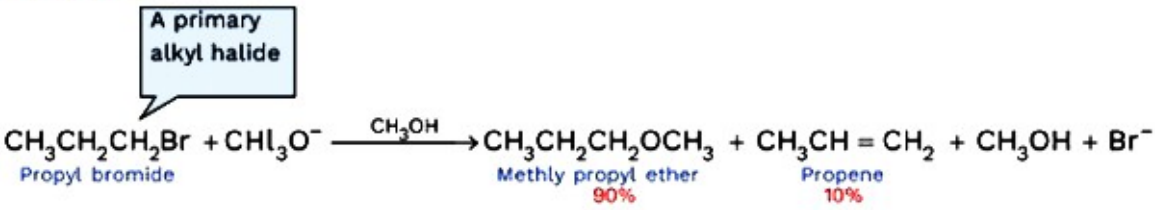
Notice that both reactions occur for the same reason–the electron-withdrawing halogen gives the carbon to which it is bonded a partial positive charge.



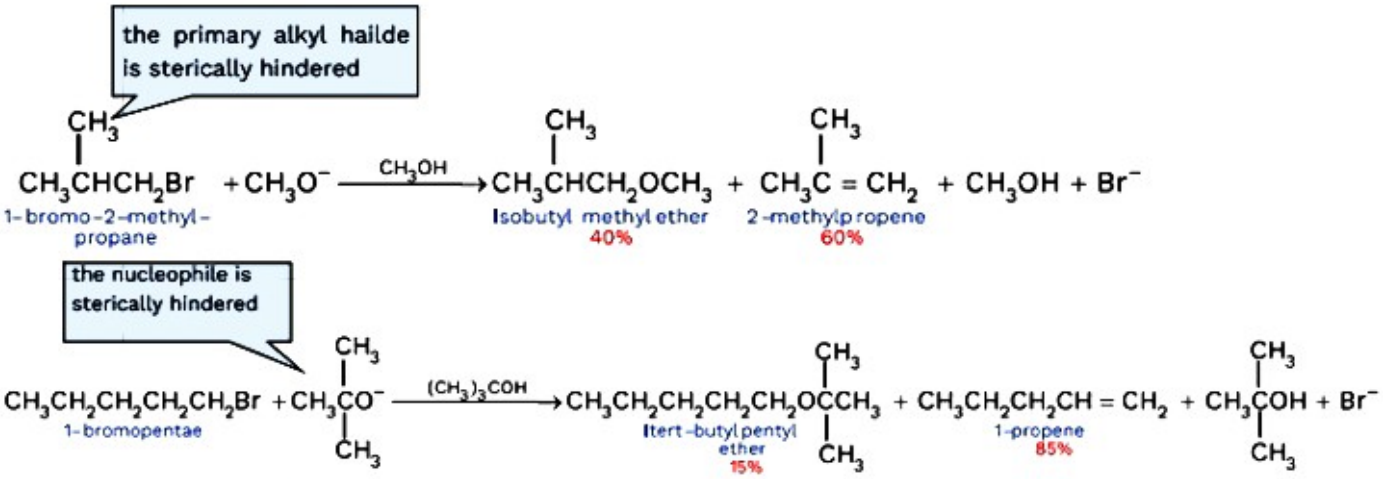
Relative Reactivities of Alkyl Halides

In an S_N2 reaction : 1° > 2° > 3°	In an S_N1 reaction : 3° > 2° > 1°
In an E2 reaction : 3° > 2° > 1°	In an E1 reaction : 3° > 2° > 1°

Example :



However, if either the primary alkyl halide or the nucleophile/base is sterically hindered, the nucleophile will have difficulty getting to the back side of the α-carbon. As a result, elimination will win the competition, so the elimination product will predominate.



A secondary alkyl halide can form both substitution and elimination products under

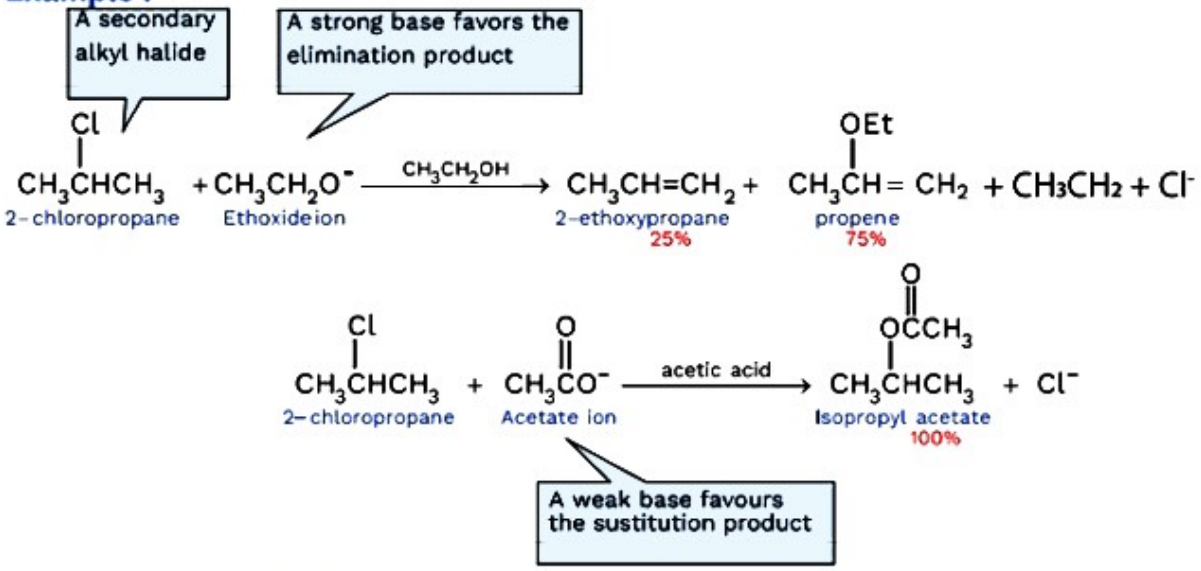
SN2/E2 conditions.

The relative amounts of the two products depend on the base strength and the bulk of the nucleophile/base.

The stronger and bulkier the base, the greater the percentage of the elimination product.

For example, acetic acid is a stronger acid (pK_a = 4.76) than ethanol (pK_a = 15.9), which means that acetate ion is a weaker base than ethoxide ion. The elimination product is the main product formed from the reaction of 2-chloropropane with the strongly basic ethoxide ion. whereas no elimination product is formed with the weakly basic acetate ion.

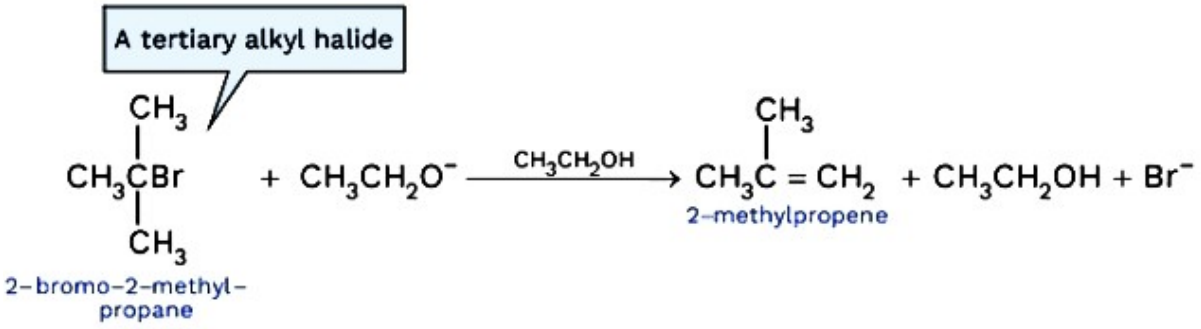
Example :



The percentage of elimination product produced would be increased father if the bulky tert-butoxide ion were used instead of ethoxide ion.

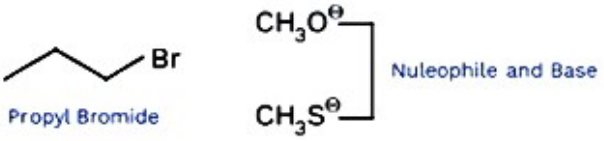
A tertiary alkyl halide is the least reactive of the alkyl halides in an S_N2 reaction and the most reactive in an E2 reaction.

Only the elimination product is formed when a tertiary alkyl halide reacts with a nucleophile/base under S_N2/E2 conditions.

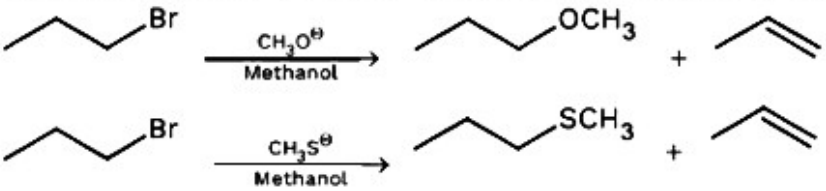


Q188 **How would you suppose the ratio of substitution product to elimination product formed from the reaction of propyl bromide with CH_3O in methanol to change when the nucleophile is changed to CH_3S^- ?**

A188

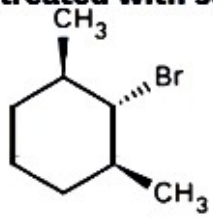


- In methanol CH_3S^- is better nucleophile compare to CH_3O^- due to solvation.



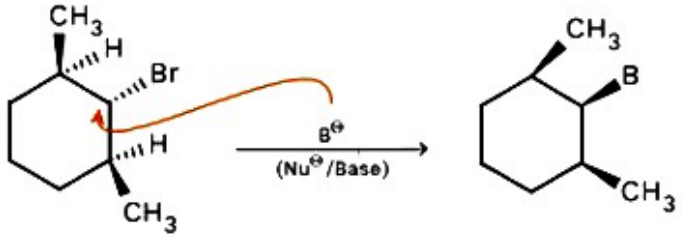
- Better nucleophile/weak base CH_3S^- leads more substitution than elimination.

Q189 **(a) Only a substitution product is obtained when the following compound is treated with sodium methoxide:**

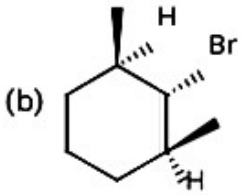


(b) Explain why an elimination product is not obtained.

A189 (a)



- –Br and –H could not participate in E2 elimination due to not a anit peri-palaner mechanism.
- Due to absence of polar protic medium formation of carbocation not position.
- Only substitution is possible.

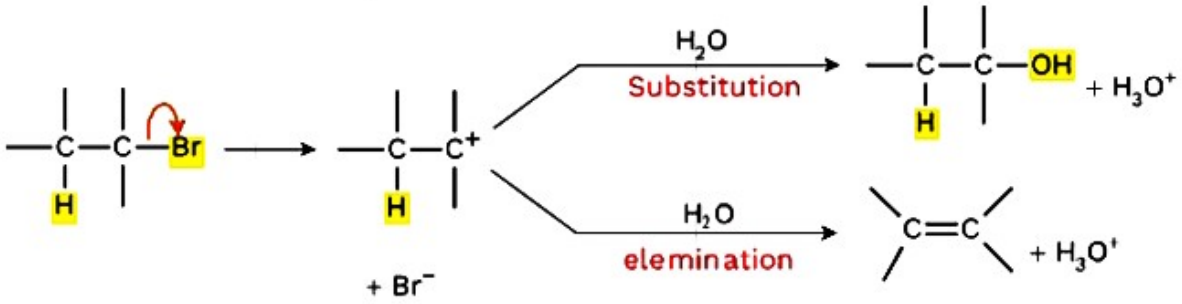
(b) 

Same side / not anti peri planar
E2 not possible

- E1 not possible due to lack of solvation/carbocation formation.

SN1/E1 Conditions: A poor nucleophile/weak base

In S_N1/E1 reactions, the alkyl halide dissociates to form a carbocation, which can then either combine with the nucleophile to form the substitution product or lose a proton to form the elimination product.



Why Alkyl halides have the same order of reactivity in SN1 reactions as they do in E1 reactions?

Reason : Because both reactions have the same rate-determining step dissociation of the alkyl halide. This means that all alkyl halides that react under S_N1/E1 conditions will give both substitution and elimination products.

Remember that primary alkyl halides do not undergo S_N1/El reactions because primary carbocations are too unstable to be formed.

Summary of the Products Expected in Substitution and Elimination Reactions		
Class of alkyl halide	**S_N2 versus E2**	**S_N1 versus E1**
Primary alkyl halide	Primarily substitution, unless there is steric hindrance in the alkyl halide or nucleophile, in which case elimination is favored	Cannot undergo S_N1/ E1 reactions

Secondary alkyl halide	Both substitution and elimination: the stronger and bulkier the base, the greater is the percentage of elimination	Both substitution and elimination
Tertiary alkyl halide	Only elimination	Both substitution and elimination

Q190 **Indicate whether the alkyl halides listed will give mostly substitution products, mostly elimination products, or about equal amounts of substitution and elimination products when they react with the following:**

(a) methanol under S_N1/E1 conditions

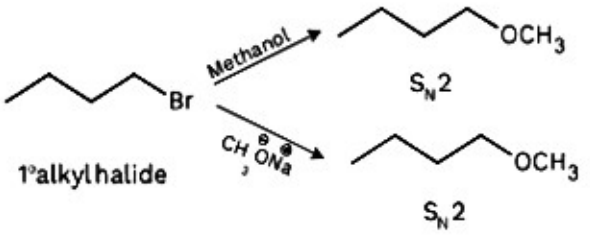
(b) sodium methoxide under S_N2/E2 conditions

(1) 1-bromobutane

(2) 1-bromo-2-methylpropane

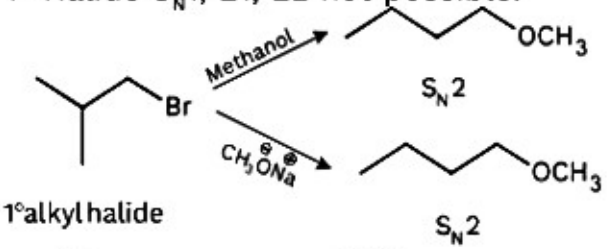
(3) 2-bromo-2-methylpropane

A190 (1)

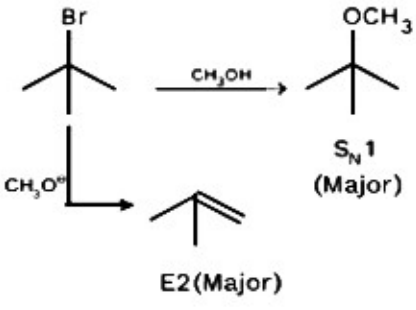


1° Halide S_N1, E1, E2 not possible.

(2)

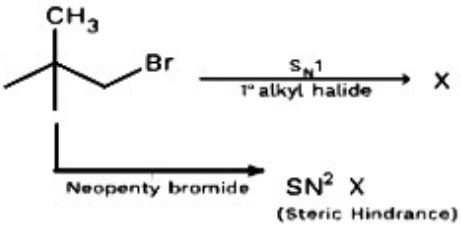


(3)



Q191 **1-Bromo-2.2-dimethylpropane has difficulty undergoing either S_N2 or S_N1 re actions.**
a. Explain why
b. Can it undergo E2 and El reactions?

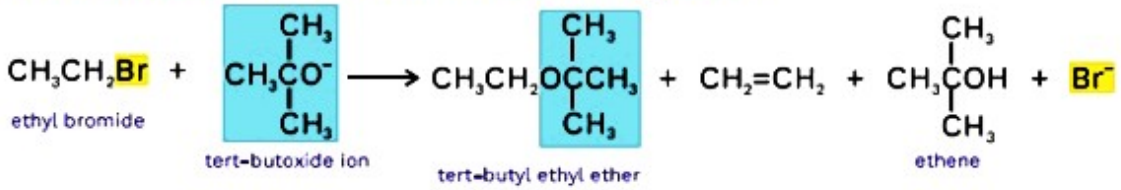
A191 (a)

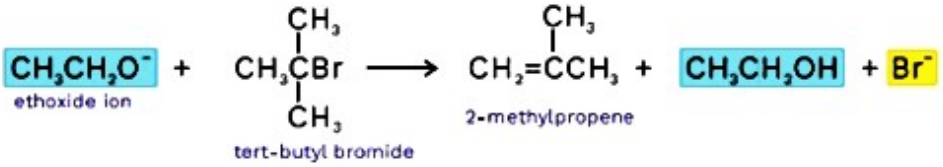


- Steric Hindrance

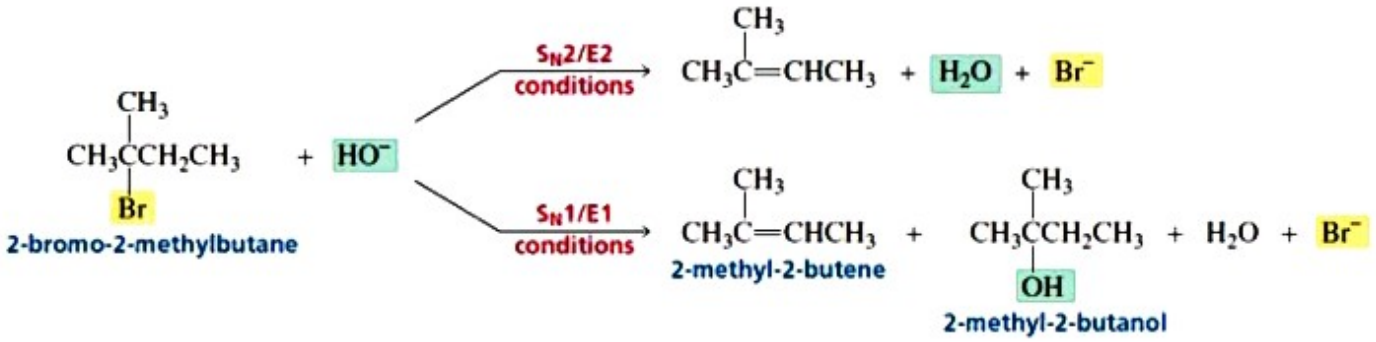
(b) Not even E1 and E2 due to absence of α-H

Substitution and Elimination Reactions in Synthesis





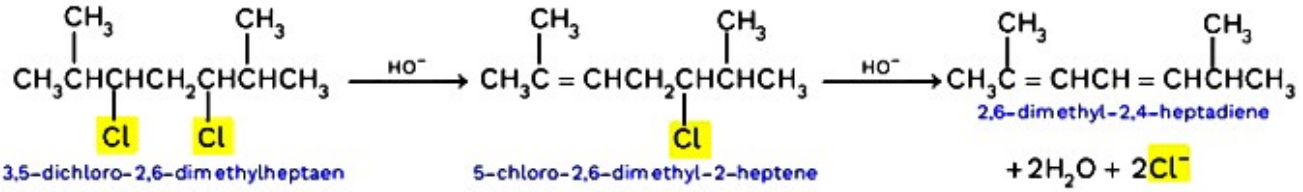
S_N2/E2 conditions (a high concentration of HO^- in an aprotic polar solvent) because a tertiary alkyl halide gives only the elimination product under those conditions. If S_N1/E1 conditions were used (a low concentration of HO^- in water), both elimination and substitution products would be obtained.



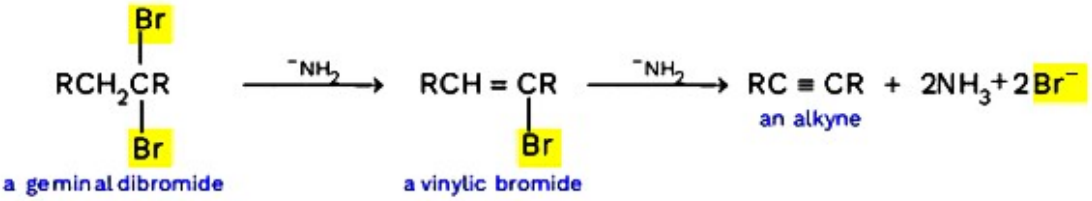
Consecutive E2 Elimination Reactions :

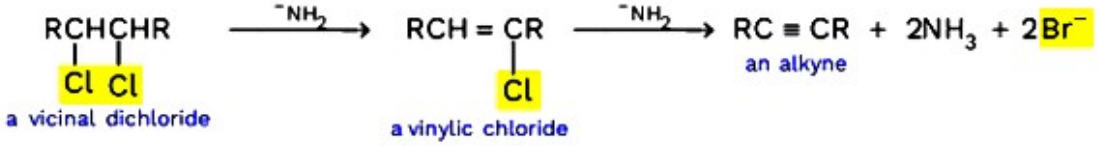
Alkyl dihalides can undergo two consecutive dehydrohalogenations, giving products that

contain two double bonds. In the following example. Zaitsev's rule predicts the most stable product of the first dehydrohalogenation, but not the most stable product of the second. The reason Zaitsey's rule fails in the second reaction is that a conjugated diene is more stable than an isolated diene.



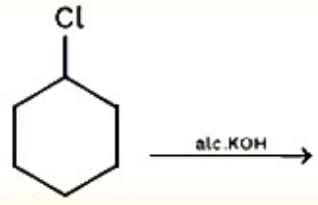
If the two halogens are on the same carbon (geminal dihalides) or on adjacent carbons (vicinal dihalides), the two consecutive E2 dehydrohalogenations can result in the formation of a triple bond. This is how alkynes are commonly synthesized.



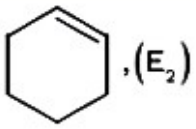


Identify major product and type of reaction (SN1/SN2/E1/E2).

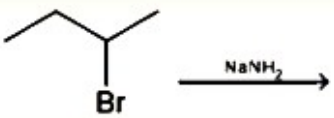
Q192



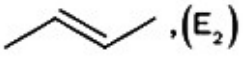
A192



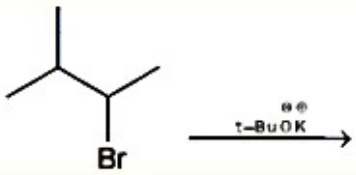
Q193



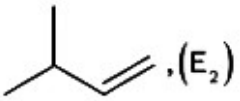
A193



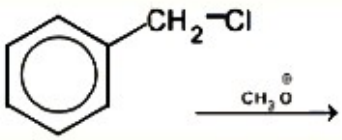
Q194



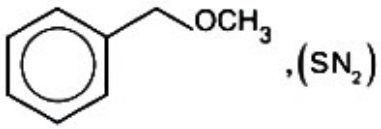
A194



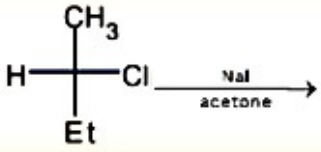
Q195



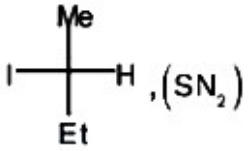
A195



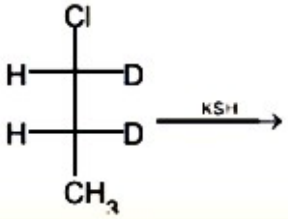
Q196



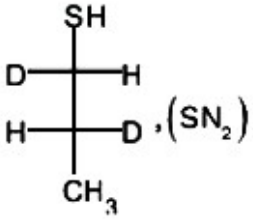
A196



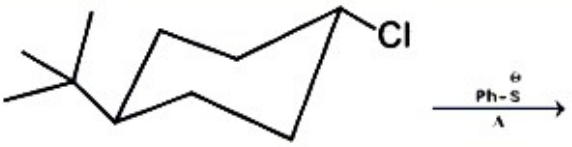
Q197



A197



Q198

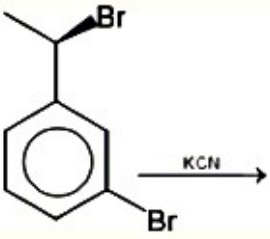


A198

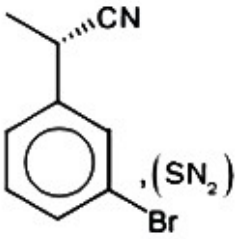


$,(SN_2)$

Q199

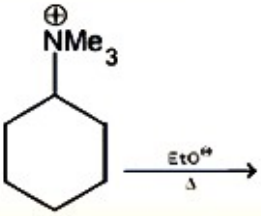


A199

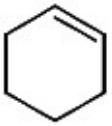


$,(SN_2)$

Q200

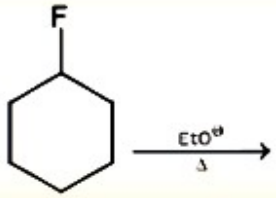


A200

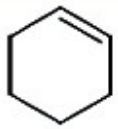


$,(E_2)$

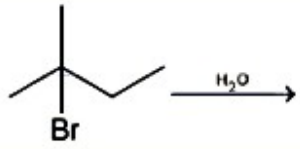
Q201



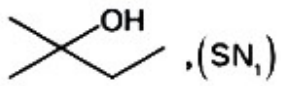
A201



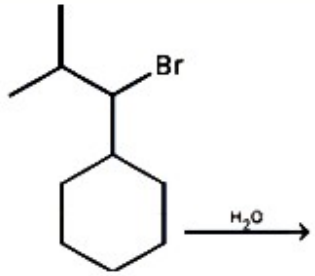
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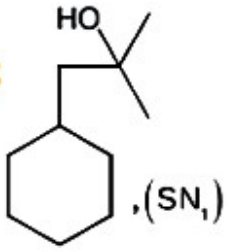
A202



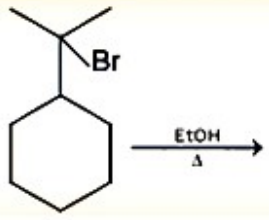
Q203



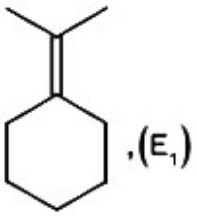
A203



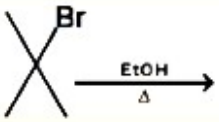
Q204



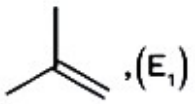
A204



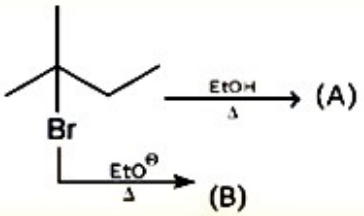
Q205



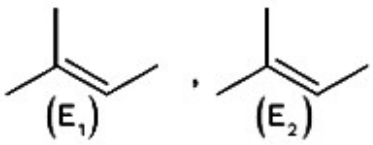
A205



Q206

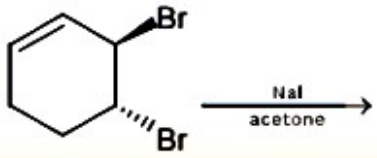


A206

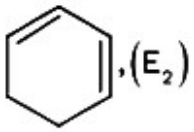


(A) (B)

Q207

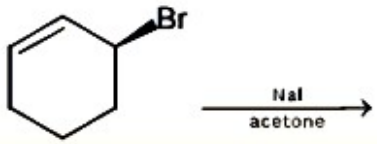


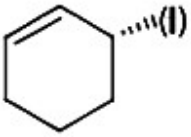
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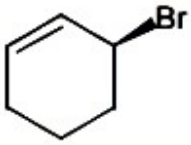
Anti periplanar mechanism

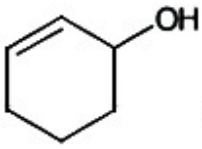
Q208



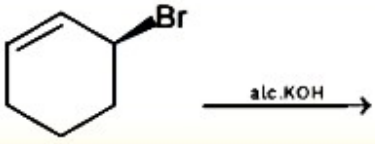
A208 

, (SN_2)

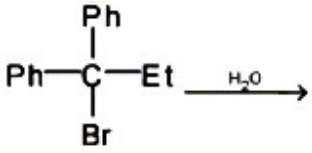
Q209 

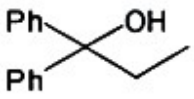
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, (SN_2)

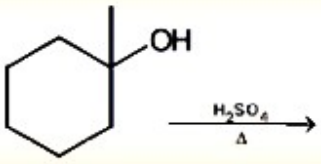
Q210 

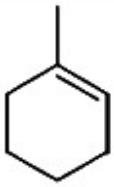
A210 
, (E_2)

Q211 

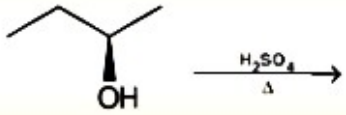
A211 

, (SN_1)

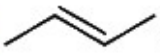
Q212 

A212 
, (E_1)

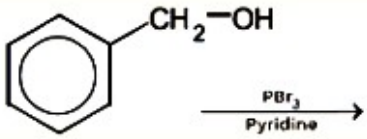
Q213



A213



Q214



A214

